

Decoding of Large Language Models

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Language Models (LMs)

- **Def. Language model (LM):** probability distribution over sequences of tokens^[1]

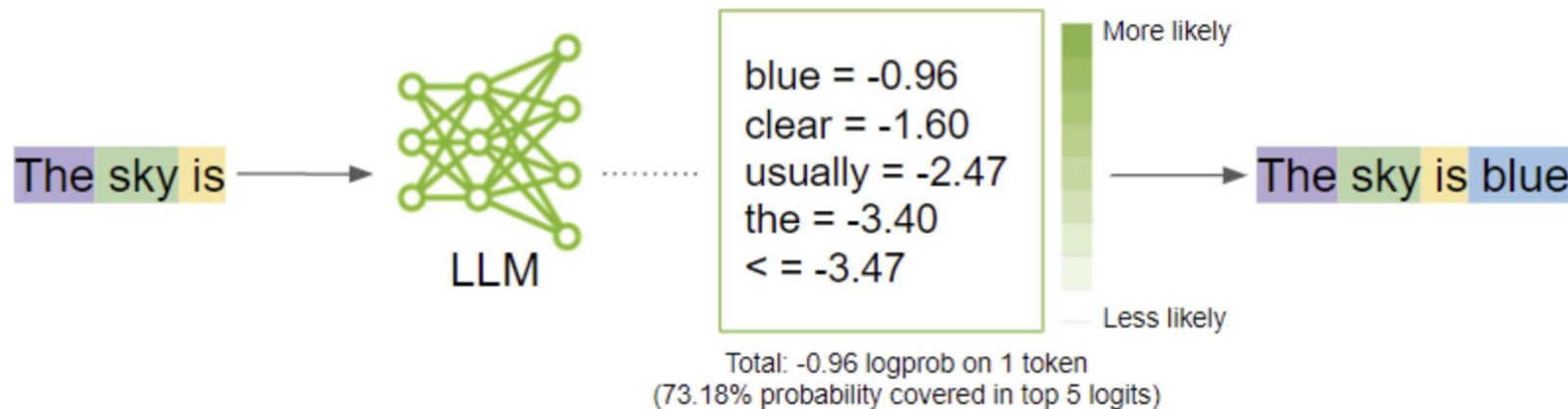
$$p(x_{1:L}) = p(x_1, \dots, x_L)$$

- x_i is a token with a finite vocabulary V
- E.g., $p(\text{the, mouse, ate, the, cheese}) = 0.02$
- **Autoregressive language models** with chain rule are usually considered:

$$p(x_{1:L}) = p(x_1, \dots, x_L) = \prod_{i=1}^L p(x_i | x_{i-1})$$

Recap: Language Models (LMs)

- **LM:** “Transformer^[1]” + “Self-supervised learning” + “Massive unlabeled text”
 - E.g., Next token prediction^[2] → learning knowledge from unlabeled text data



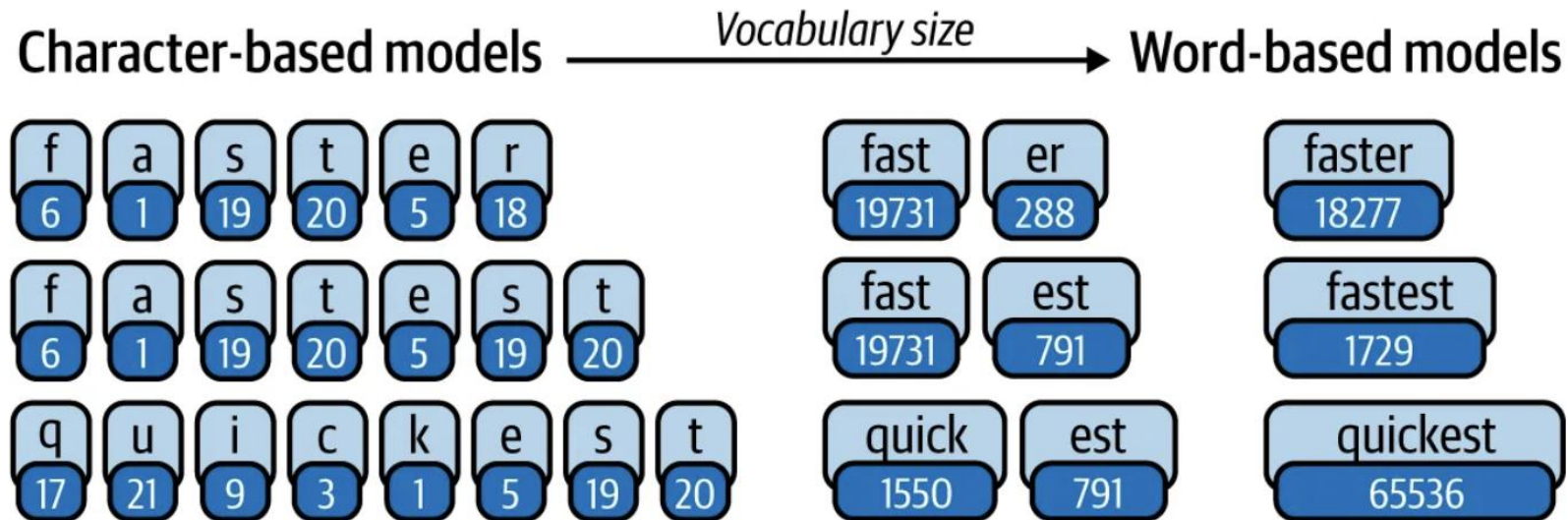
$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta)$$

[1] Vaswani et al., Attention Is All You Need., NeurIPS 2017

[2] Brown et al., Language Models are Few-Shot Learners., NeurIPS 2020

Tokenization

- **Def: mapping** given input text to sequence of tokens
 - There are 3 types: (1) character-level, (2) subword-level, and (3) word-level
 - (Pre-defined) Set of all possible tokens: “**Vocabulary**”



Tokenization

- **Example:** character-level tokenization

```
text = "Tokenizing text is a core task of NLP."  
tokenized_text = list(text)  
print(tokenized_text)
```

```
['T', 'o', 'k', 'e', 'n', 'i', 'z', 'i', 'n', 'g', ' ', 't', 'e', 'x', 't', ' ',  
'i', 's', ' ', 'a', ' ', 'c', 'o', 'r', 'e', ' ', 't', 'a', 's', 'k', ' ', 'o',  
'f', ' ', 'N', 'L', 'P', '.']
```

Tokenization

- **Example:** character-level tokenization

Vocabulary (size 20)

```
{' ': 0, '.' : 1, 'L': 2, 'N': 3, 'P': 4, 'T': 5, 'a': 6, 'c': 7, 'e': 8, 'f': 9, 'g': 10, 'i': 11, 'k': 12, 'n': 13, 'o': 14, 'r': 15, 's': 16, 't': 17, 'x': 18, 'z': 19}
```

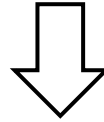
```
['T', 'o', 'k', 'e', 'n', 'i', 'z', 'i', 'n', 'g', ' ', 't', 'e', 'x', 't', ' ', 'i', 's', ' ', 'a', ' ', 'c', 'o', 'r', 'e', ' ', 't', 'a', 's', 'k', ' ', 'o', 'f', ' ', 'N', 'L', 'P', '.']
```

```
[5, 14, 12, 8, 13, 11, 19, 11, 13, 10, 0, 17, 8, 18, 17, 0, 11, 16, 0, 6, 0, 7, 14, 15, 8, 0, 17, 6, 16, 12, 0, 14, 9, 0, 3, 2, 4, 1]
```

Tokenization

- **Example:** subword tokenization in practice (BERT-pretrained)

```
encoded_text = tokenizer(text)
```



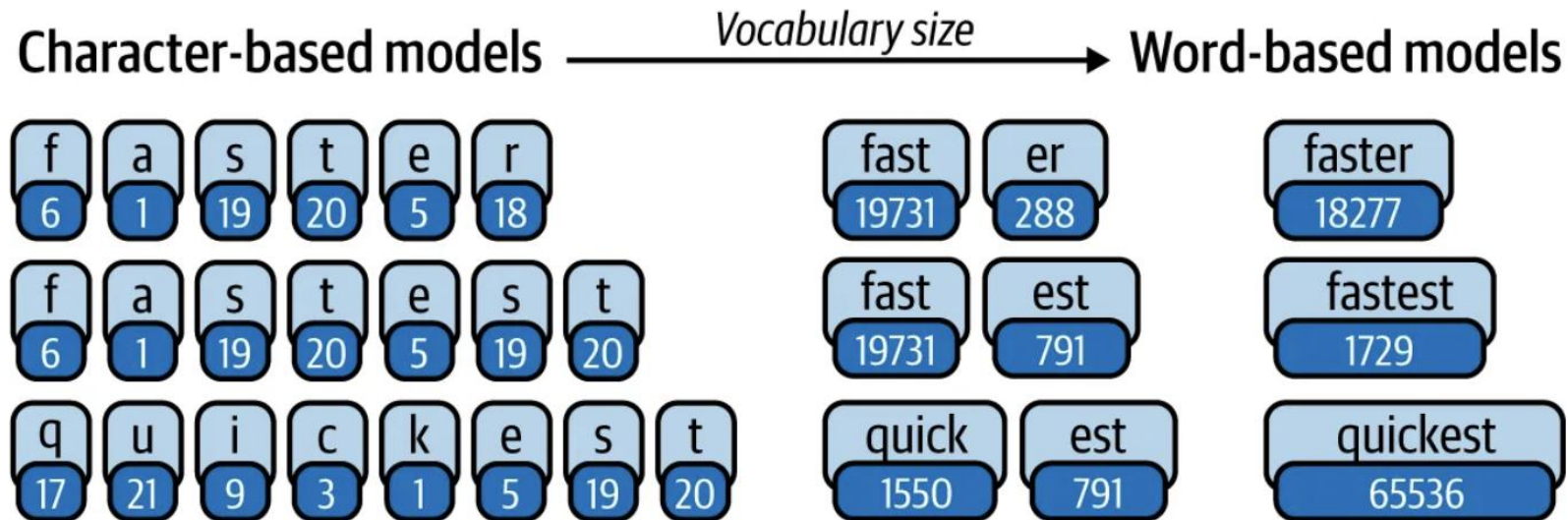
```
{'input_ids': [101, 19204, 6026, 3793, 2003, 1037, 4563, 4708, 1997, 17953,  
2361, 1012, 102],
```



```
['[CLS]', 'token', '##izing', 'text', 'is', 'a', 'core', 'task', 'of', 'nl',  
##p', '.', '[SEP]']
```

Tokenization

- **Def: mapping** given input text to sequence of tokens
 - There are 3 types: (1) character-level, (2) subword-level, and (3) word-level
 - (Pre-defined) Set of all possible tokens: “**Vocabulary**”



- (+) Small vocab. size
- (-) Long input sequence

Learning with Tokenized Inputs

- **Key idea:** look-up table with word embeddings
 - Word embedding is defined for tokens in V : $W \in \mathbb{R}^{|V| \times d}$ (d : dimension of hidden feature)
 - Example ($|V| = 4, d = 5$)

$$x_{fat} \times W_{V \times M} = e_{fat}$$

0	0	0	1	0	0	0
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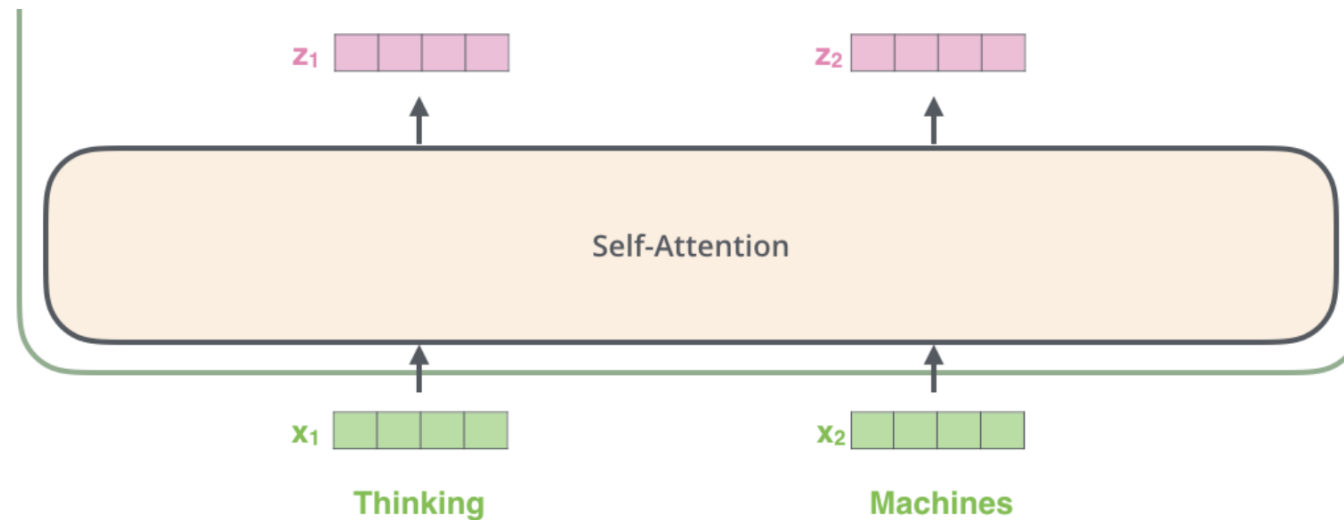
0.5	2.1	1.9	1.5	0.8
0.8	1.2	2.8	1.8	2.1
0.1	0.8	1.2	0.9	0.7
2.1	1.8	1.5	1.7	2.7

2.1	1.8	1.5	1.7	2.7
-----	-----	-----	-----	-----

lookup table

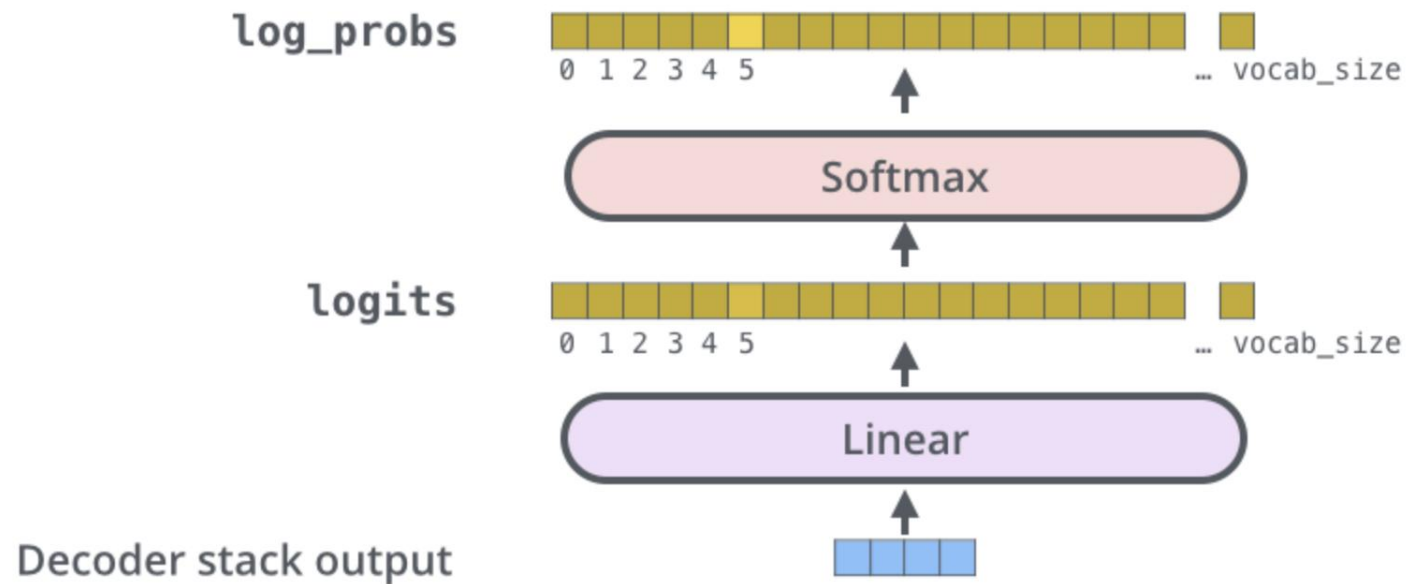
Learning with Tokenized Inputs

- **Key idea:** look-up table with word embeddings
 - Word embedding is defined for tokens in V : $W \in \mathbb{R}^{|V| \times d}$ (d : dimension of hidden feature)
 - Namely, given L input tokens $x \in V^L$ is converted into L word embeddings $\hat{x} \in \mathbb{R}^{L \times d}$



Learning with Tokenized Inputs

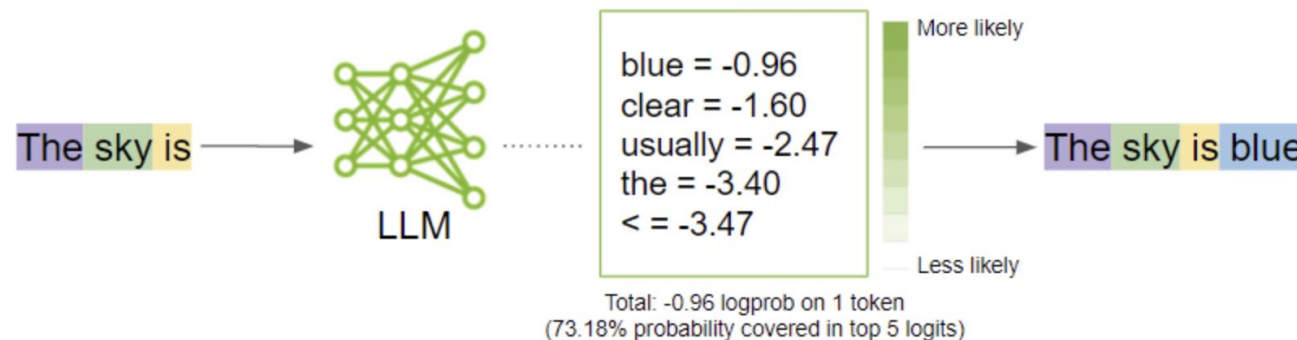
- **Key idea:** look-up table with word embeddings
 - Model (e.g., Transformer) will outputs $\hat{o} \in \mathbb{R}^{L \times d}$ from $\hat{x}$
 - Using W again (i.e., **tied**), we classify $\hat{o}$ among tokens in vocabulary → **next token** (output)



Learning with Tokenized Inputs

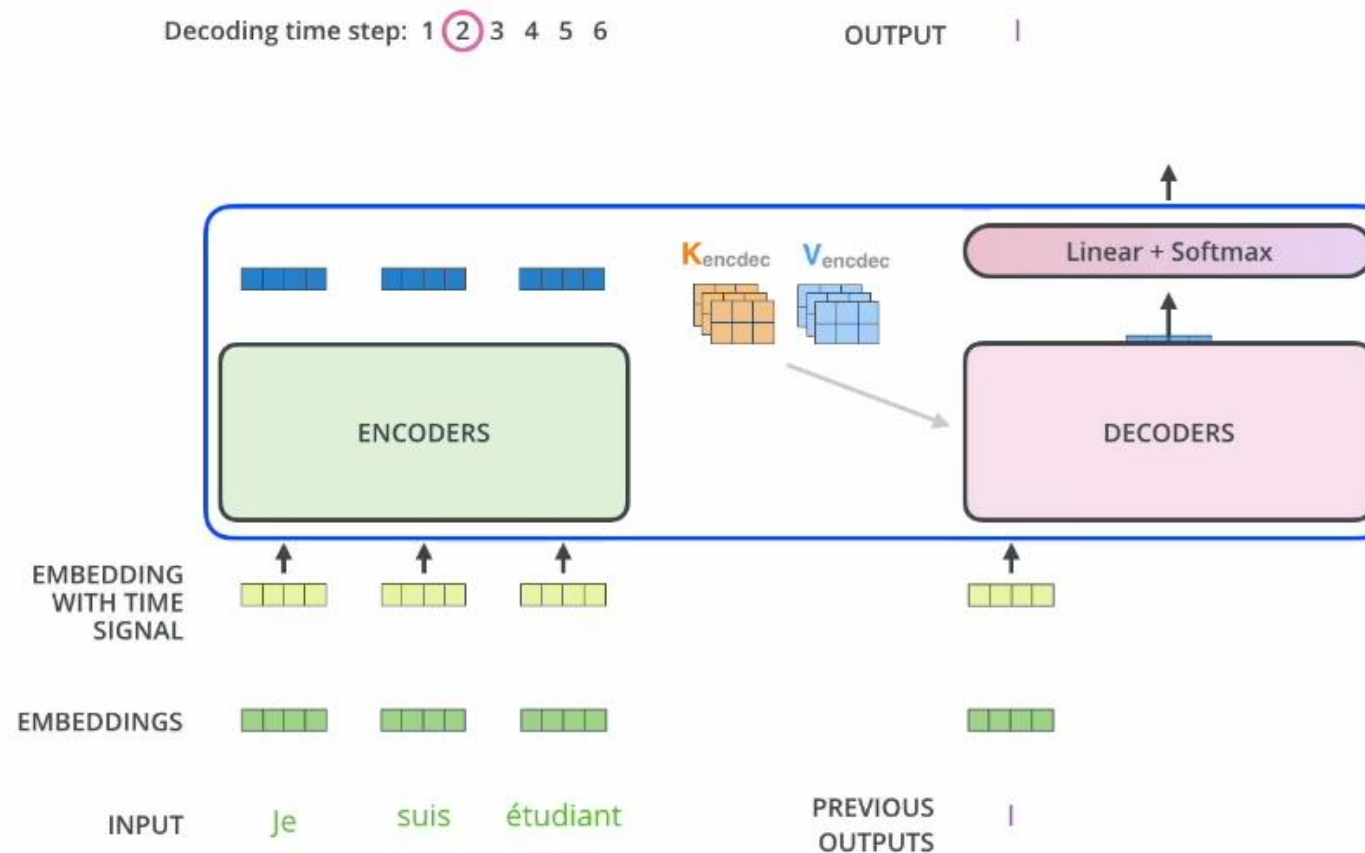
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 - Using W again (i.e., **tied**), we classify $\hat{o}$ among tokens in vocabulary $\rightarrow$ **next token** (output)
- It allows us to use same loss function (cross-entropy) to train LLMs

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \dots, u_{i-1}; \Theta)$$



Auto-regressive Generation

- (After training) How to generate tokens? → K sequential inferences



Auto-regressive Generation

- When generation is terminated and provide output?
 - Using EOS token (learned during pre-training)
 - E.g., [SEP] token in case of BERT

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{'input_ids': [101, 19204, 6026, 3793, 2003, 1037, 4563, 4708, 1997, 17953, 2361, 1012, 102],
```

```
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```

Contents

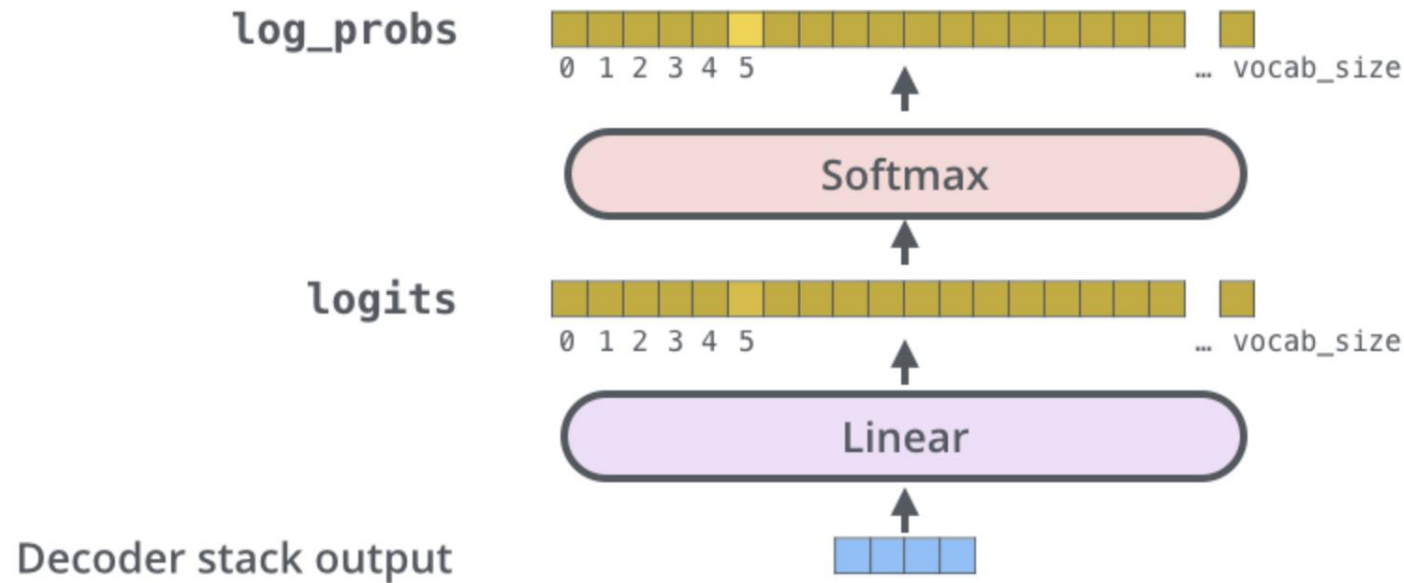
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- Advanced decoding algorithms for specific goals
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Generation Next Token from Distribution

- **Goal:** For given $\mathbf{x} = [x_1, \dots, x_L]$, generating next token x_{L+1}
 - From predicted output distribution $\hat{p}(\mathbf{x}) = \text{softmax}(\hat{o}(\mathbf{x})) = [\hat{p}_1, \dots, \hat{p}_{|V|}]$, $\sum \hat{p}(\mathbf{x}) = 1$



Generation Next Token from Distribution

- Real-world example in pytorch:

```
>>> from transformers import AutoTokenizer, AutoModelForCausalLM

>>> tokenizer = AutoTokenizer.from_pretrained("gpt2")
>>> model = AutoModelForCausalLM.from_pretrained("gpt2")

>>> prompt = "Today I believe we can finally"
>>> input_ids = tokenizer(prompt, return_tensors="pt").input_ids

>>> # generate up to 30 tokens
>>> outputs = model.generate(input_ids, do_sample=False, max_length=30)
>>> tokenizer.batch_decode(outputs, skip_special_tokens=True)
['Today I believe we can finally get to the point where we can make a difference in the lives of the']
```

Generation Next Token from Distribution

- Real-world example in pytorch:

f generate

< source >

```
( inputs: typing.Optional[torch.Tensor] = None, max_length: typing.Optional[int] = None, min_length:
typing.Optional[int] = None, do_sample: typing.Optional[bool] = None, early_stopping:
typing.Optional[bool] = None, num_beams: typing.Optional[int] = None, temperature:
typing.Optional[float] = None, penalty_alpha: typing.Optional[float] = None, top_k:
typing.Optional[int] = None, top_p: typing.Optional[float] = None, typical_p: typing.Optional[float] =
None, repetition_penalty: typing.Optional[float] = None, bad_words_ids:
typing.Optional[typing.Iterable[int]] = None, force_words_ids: typing.Union[typing.Iterable[int],
typing.Iterable[typing.Iterable[int]], NoneType] = None, bos_token_id: typing.Optional[int] = None,
pad_token_id: typing.Optional[int] = None, eos_token_id: typing.Optional[int] = None, length_penalty:
typing.Optional[float] = None, no_repeat_ngram_size: typing.Optional[int] = None,
encoder_no_repeat_ngram_size: typing.Optional[int] = None, num_return_sequences: typing.Optional[int]
= None, max_time: typing.Optional[float] = None, max_new_tokens: typing.Optional[int] = None,
decoder_start_token_id: typing.Optional[int] = None, use_cache: typing.Optional[bool] = None,
num_beam_groups: typing.Optional[int] = None, diversity_penalty: typing.Optional[float] = None,
prefix_allowed_tokens_fn: typing.Union[typing.Callable[[int, torch.Tensor], typing.List[int]],
NoneType] = None, logits_processor:
typing.Optional[transformers.generation_logits_process.LogitsProcessorList] = None,
renormalize_logits: typing.Optional[bool] = None, stopping_criteria:
typing.Optional[transformers.generation_stopping_criteria.StoppingCriteriaList] = None, constraints:
typing.Optional[typing.List[transformers.generation_beam_constraints.Constraint]] = None,
output_attentions: typing.Optional[bool] = None, output_hidden_states: typing.Optional[bool] = None,
output_scores: typing.Optional[bool] = None, return_dict_in_generate: typing.Optional[bool] = None,
forced_bos_token_id: typing.Optional[int] = None, forced_eos_token_id: typing.Optional[int] = None,
remove_invalid_values: typing.Optional[bool] = None, synced_gpus: typing.Optional[bool] = False,
exponential_decay_length_penalty: typing.Union[typing.Tuple[int, float], NoneType] = None,
suppress_tokens: typing.Optional[typing.List[int]] = None, begin_suppress_tokens:
typing.Optional[typing.List[int]] = None, forced_decoder_ids:
typing.Optional[typing.List[typing.List[int]]] = None, **model_kwargs ) → ModelOutput or
torch.LongTensor
```

Generation Next Token from Distribution

- Real-world example in pytorch:

- *greedy decoding* by calling `greedy_search()` if `num_beams=1` and `do_sample=False`.
- *contrastive search* by calling `contrastive_search()` if `penalty_alpha>0`. and `top_k>1`
- *multinomial sampling* by calling `sample()` if `num_beams=1` and `do_sample=True`.
- *beam-search decoding* by calling `beam_search()` if `num_beams>1` and `do_sample=False`.
- *beam-search multinomial sampling* by calling `beam_sample()` if `num_beams>1` and `do_sample=True`.
- *diverse beam-search decoding* by calling `group_beam_search()`, if `num_beams>1` and `num_beam_groups>1`.
- *constrained beam-search decoding* by calling `constrained_beam_search()`, if `constraints!=None` or `force_words_ids!=None`.

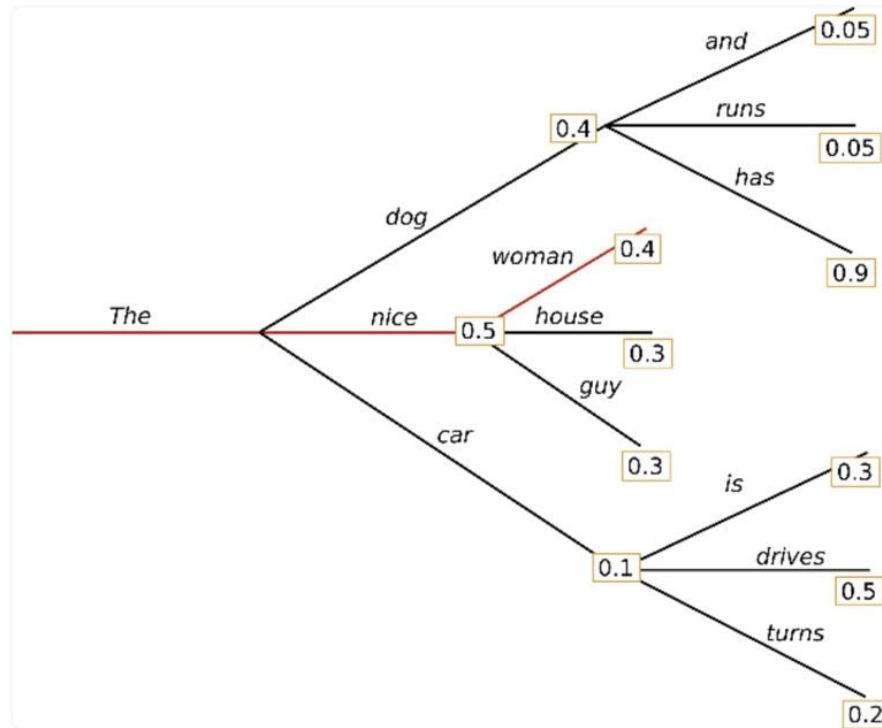
Generation Next Token from Distribution

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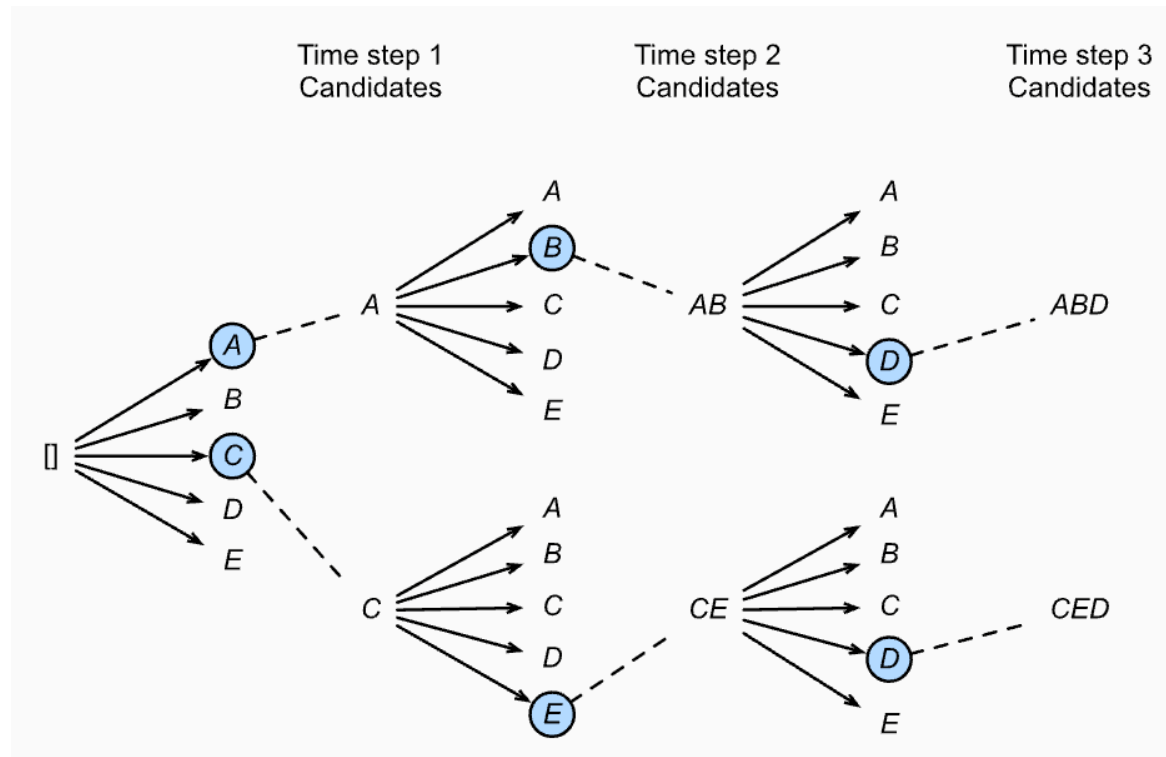
Greedy Decoding

- **Key idea:** sequentially selecting tokens with highest probability (argmax)
 - i.e., $x_{L+1} = \arg \max \hat{p}(x)$
 - (+) Easy-to-use, (-) it can be suboptimal depending on succeeding generations



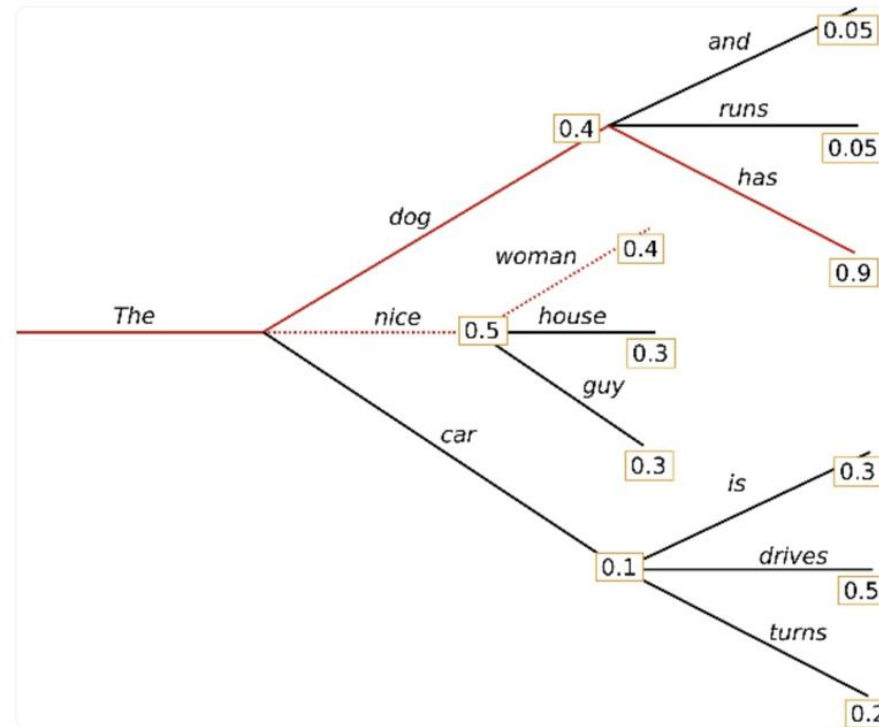
Beam Search

- **Key idea:** handling multiple candidates and update them iteratively
 - It allows us to consider multiple time steps simultaneously
 - Selection criteria for update: (vanilla) likelihood $p(\mathbf{x}) = p(x_1)p(x_2|x_1) \dots$



Beam Search

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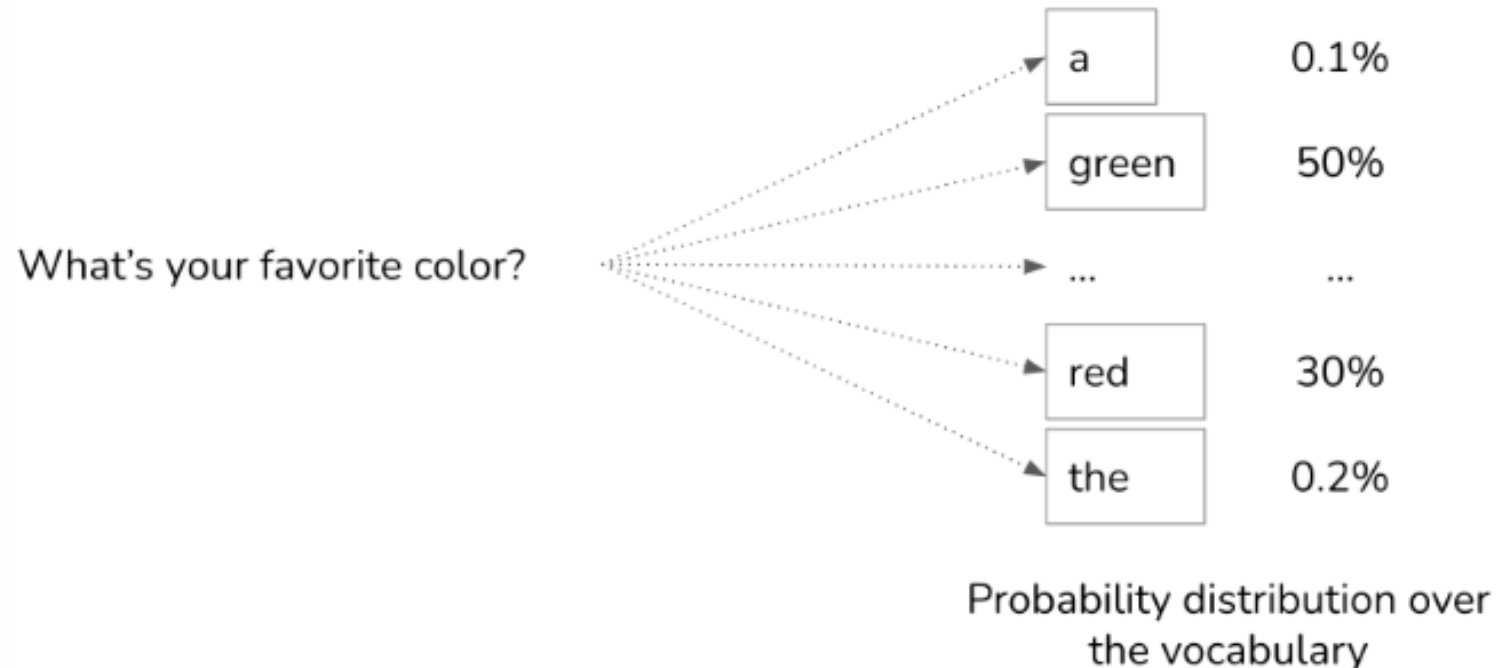


Beam Search

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 - It allows us to consider multiple time steps simultaneously
 - Selection criteria for update: (vanilla) likelihood $p(\mathbf{x}) = p(x_1)p(x_2|x_1) \dots$
 - (+) More chance to generate good output, (-) significant computational costs

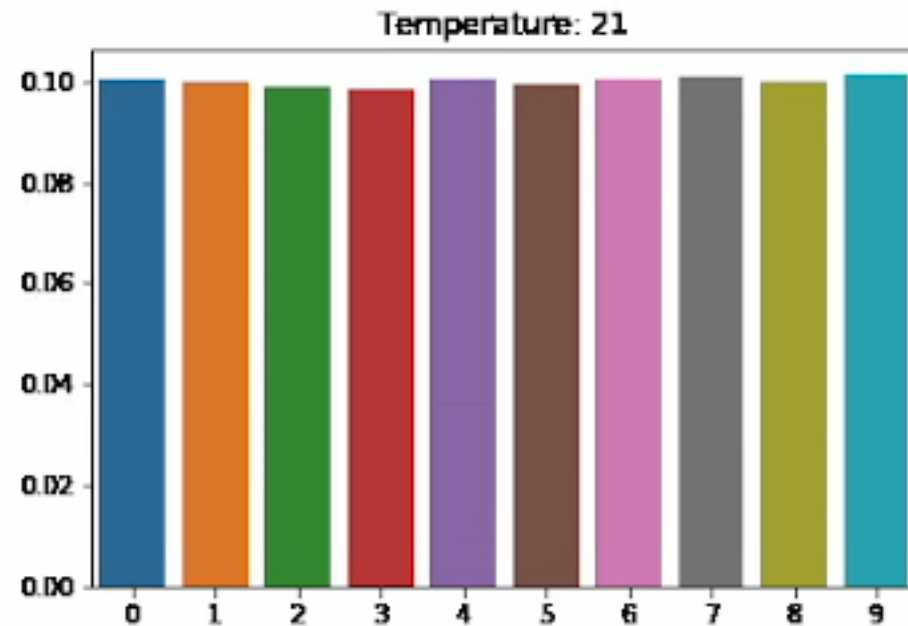
Sampling

- **Key idea:** random sampling of next token from output distribution
 - i.e., $x_{L+1} = \arg \max \hat{p}(x)$ (greedy) $\rightarrow x_{L+1} \sim \hat{p}(x)$ (**sampling**)
 - (+) More exploration during generation, (-) quality of generation can be decreased



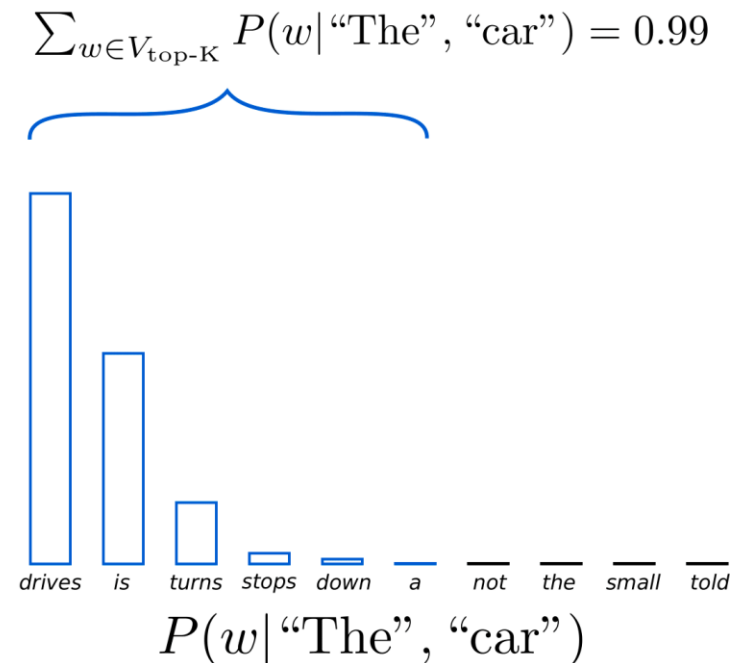
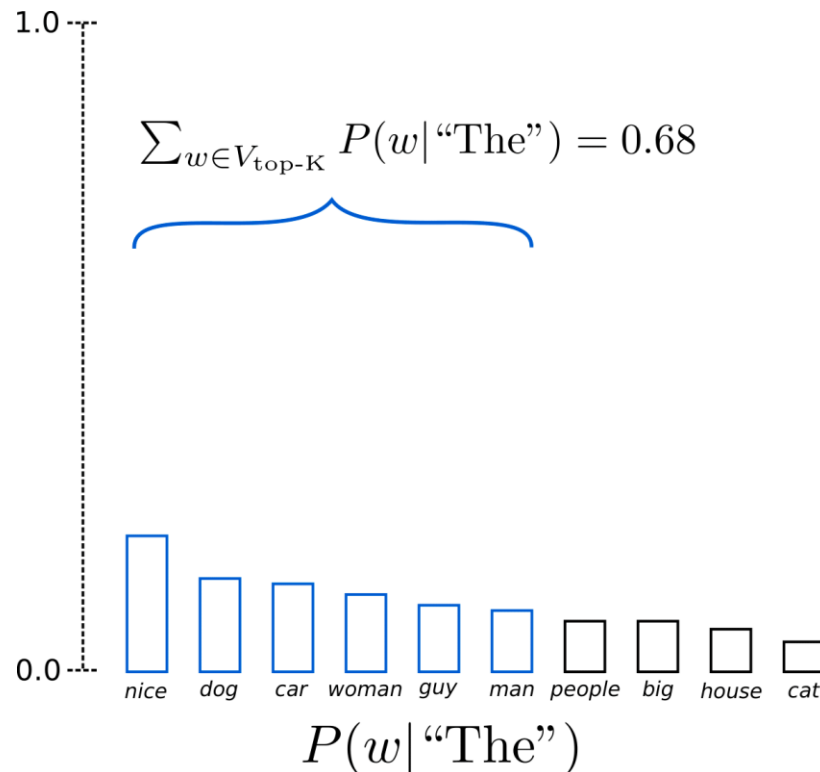
Sampling: Temperature

- **Key idea:** inserting hyper-parameter T to adjust output distribution
 - i.e., $\hat{p}(\mathbf{x}) = \text{softmax}(\hat{o}(\mathbf{x})/T) \rightarrow T > 1$: smoothing, $T < 1$: sharpening



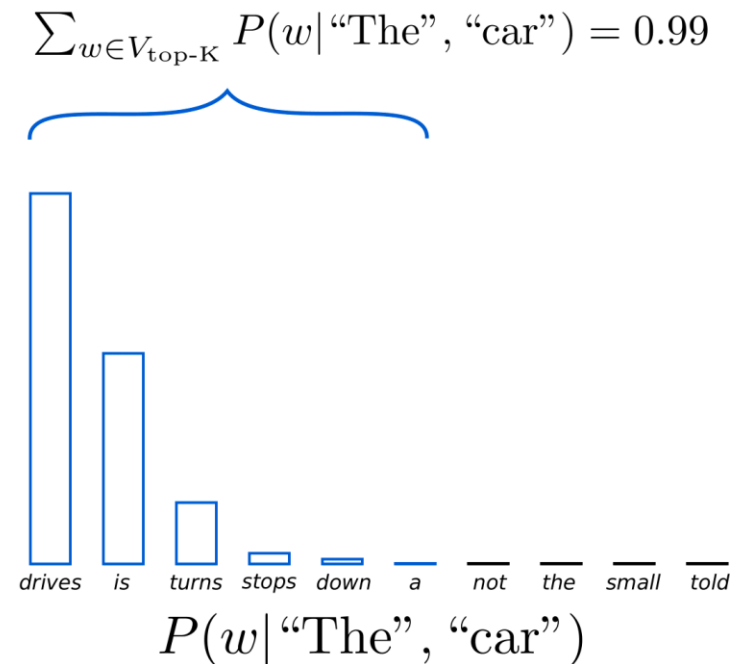
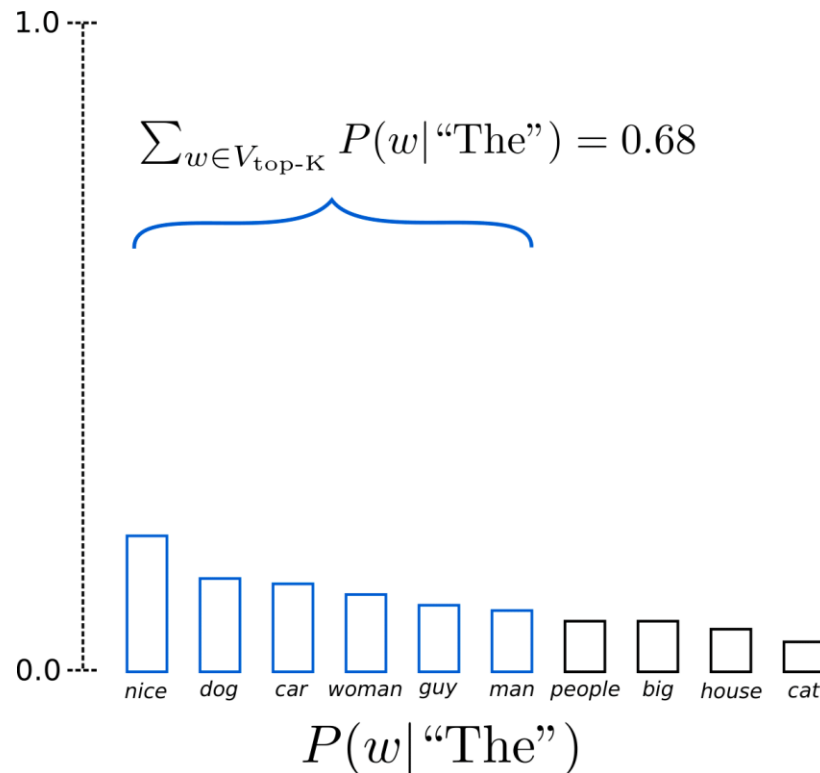
Sampling: Top-K

- **Key idea:** only considering K likely tokens during sampling
 - i.e., **ignore** $\hat{p}_i$ in $\hat{p}(\mathbf{x}) = [\hat{p}_1, \dots, \hat{p}_{|V|}]$ if it is not in top-K after sorting using probability



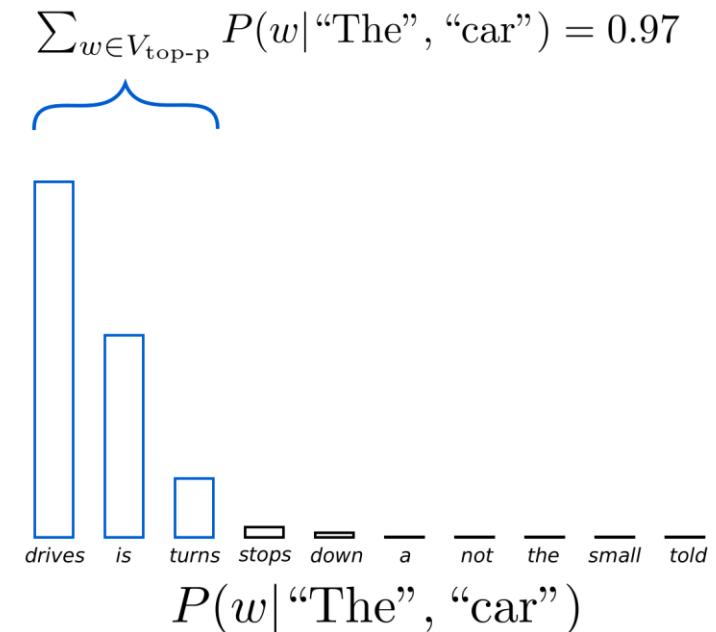
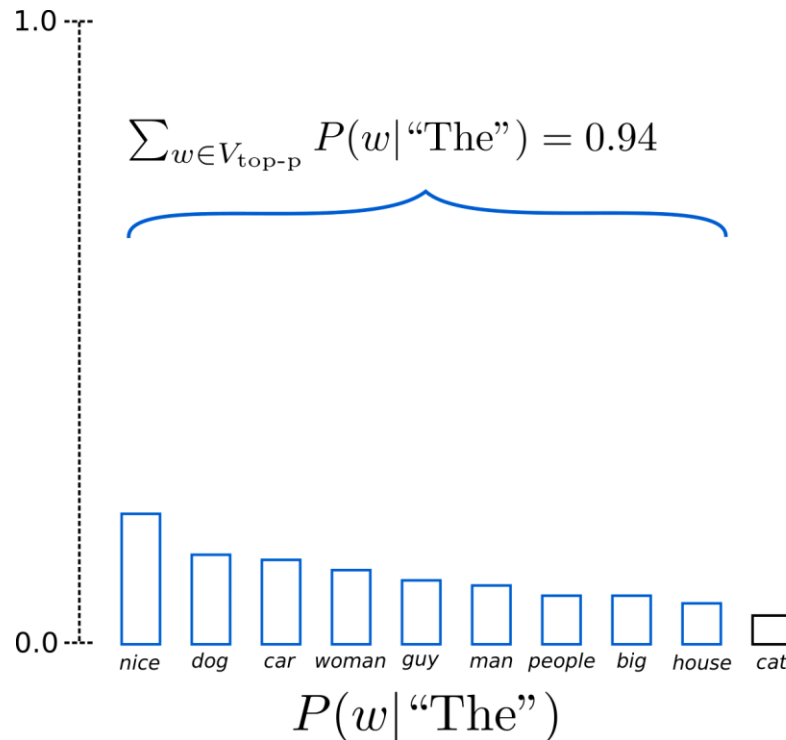
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 - **(-)** Fixed K candidates regardless of shape of output distribution



Sampling: Top-P (or Nucleus Sampling^[1])

- **Key idea:** instead of candidate number, focusing on accumulated probability
 - i.e., adapt K by setting $K = \min_k (\sum_{i=1}^k \hat{p}_i^{\text{sorted}}(\mathbf{x}) > \tau)$



[1] Holtzman et al., The Curious Case of Neural Text Degeneration., ICLR 2020

[2] Source: <https://sooftware.io/generate/>

Sampling: Top-P (or Nucleus Sampling^[1])

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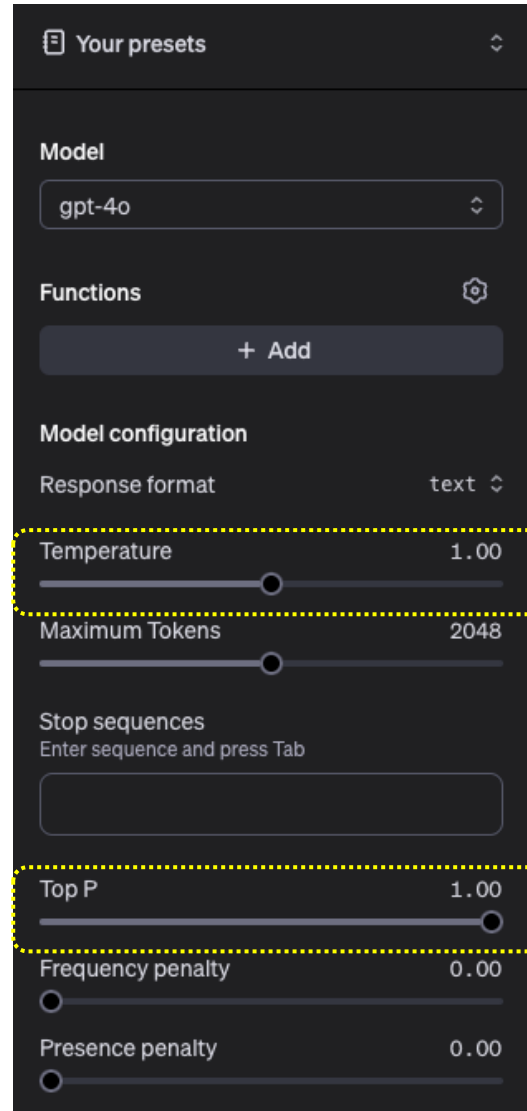
Method	Perplexity	Self-BLEU4	Zipf Coefficient	Repetition %	HUSE
Human	12.38	0.31	0.93	0.28	-
Greedy	1.50	0.50	1.00	73.66	-
Beam, b=16	1.48	0.44	0.94	28.94	-
Stochastic Beam, b=16	19.20	0.28	0.91	0.32	-
Pure Sampling	22.73	0.28	0.93	0.22	0.67
Sampling, $t=0.9$	10.25	0.35	0.96	0.66	0.79
Top- $k=40$	6.88	0.39	0.96	0.78	0.19
Top- $k=640$	13.82	0.32	0.96	0.28	0.94
Top- $k=40$, $t=0.7$	3.48	0.44	1.00	8.86	0.08
Nucleus $p=0.95$	13.13	0.32	0.95	0.36	0.97

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Decoding in Practice

- Example in ChatGPT



The image shows a mobile app interface for configuring a language model. The settings are organized into sections: 'Your presets', 'Model', 'Functions', 'Model configuration', and 'Stop sequences'. The 'Model' section shows 'gpt-4o' selected. The 'Functions' section has an '+ Add' button. The 'Model configuration' section includes 'Response format' set to 'text', 'Temperature' at 1.00, 'Maximum Tokens' at 2048, and 'Top P' at 1.00. The 'Stop sequences' section has a text input field. The 'Temperature', 'Maximum Tokens', and 'Top P' settings are highlighted with yellow dashed boxes.

Setting	Value
Model	gpt-4o
Functions	+ Add
Response format	text
Temperature	1.00
Maximum Tokens	2048
Stop sequences	Enter sequence and press Tab
Top P	1.00
Frequency penalty	0.00
Presence penalty	0.00

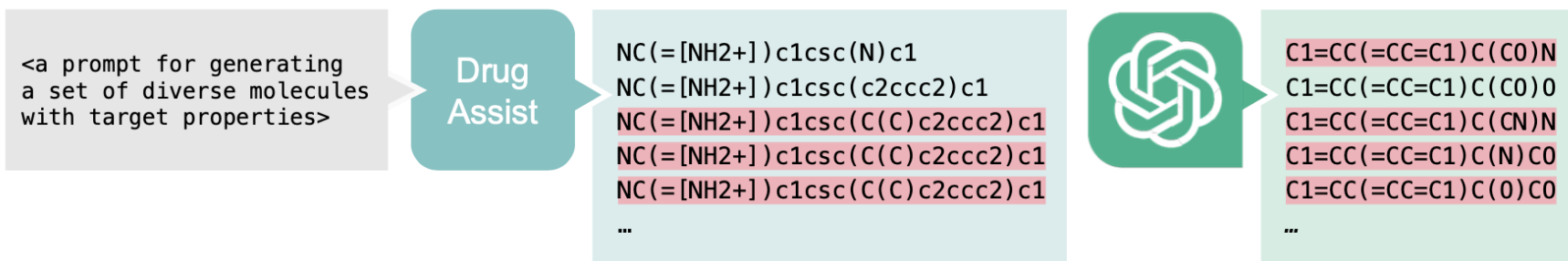
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Specific Goals during Decoding

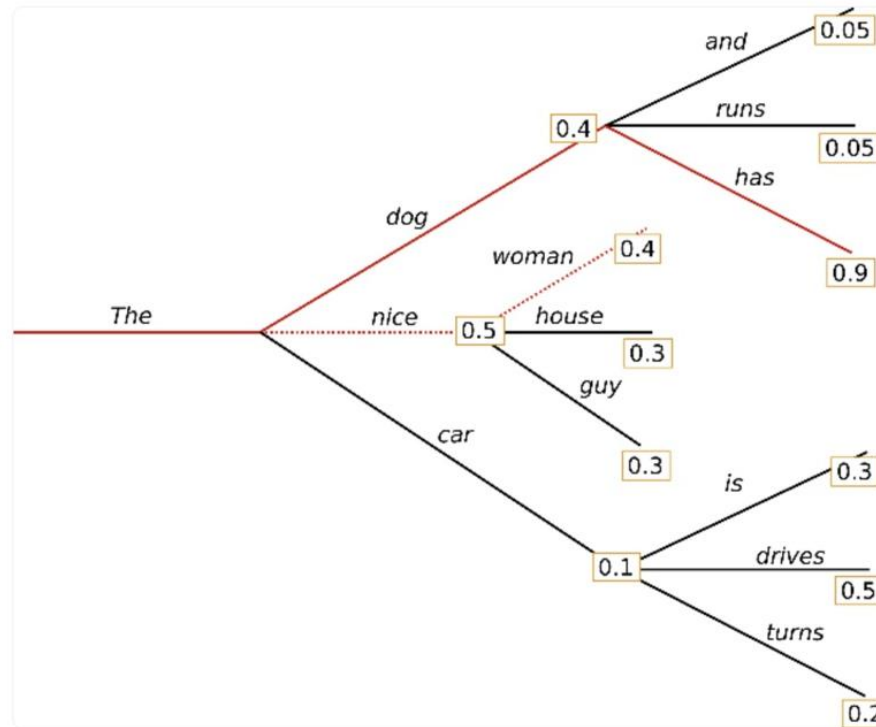
- Depending on task, goal of decoding can be changed
 - E.g., debiased (de-toxic) outputs for chatbot or diverse outputs for molecule discovery

Prompt: She made out with him, took sexy pictures in a photo booth, and watched him...	
GPT-2	having sex with a still-pregnant woman in the bathroom. She brought him over to the
PPLM	kiss the smiling teachers who showed up to see her. ↳ Her is the story of Kimberly Tur
DAPT	do it — with his brothers, neighbors, female employees at the bank and other bank customers who'd
GeDi	admire her enormous breasts for the first time. She laughed when she found out Bruce kept them in to
DEXPERTS ($\alpha = 1.0$)	go about his daily chores like a good girlfriend should. A neat piece of work who wanted her to
DEXPERTS ($\alpha = 2.0$)	play drums on the rooftop patio with his band while another time his girlfriend came over and woke him up



Recall: Beam Search^[1]

- **Key idea:** handling multiple candidates and update them iteratively
 - It allows us to consider multiple time steps simultaneously
 - **Selection criteria** for update: (vanilla) likelihood $p(\mathbf{x}) = p(x_1)p(x_2|x_1) \dots$



Recall: Beam Search^[1]

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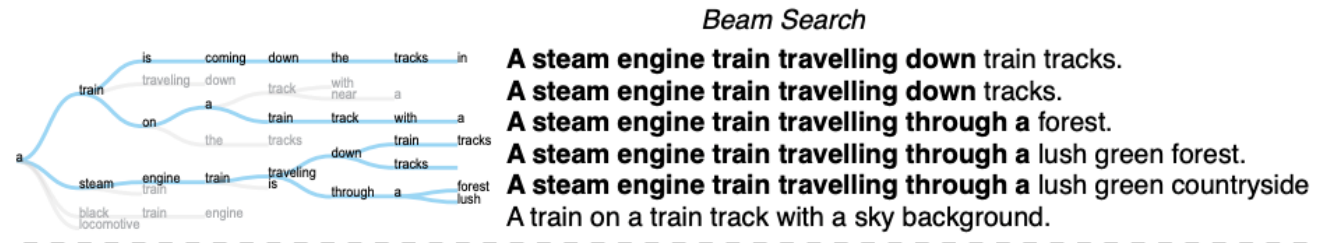


Ground Truth Captions

Single engine train rolling down the tracks.
A steam locomotive is blowing steam.

A locomotive drives along the tracks amongst trees and bushes.
An old fashion train with steam coming out of its pipe.

An engine is coming down the train track.
A black and red train moving down a train track.



Diverse Beam Search^[1]

- **Goal:** increasing diversity between output from different beam

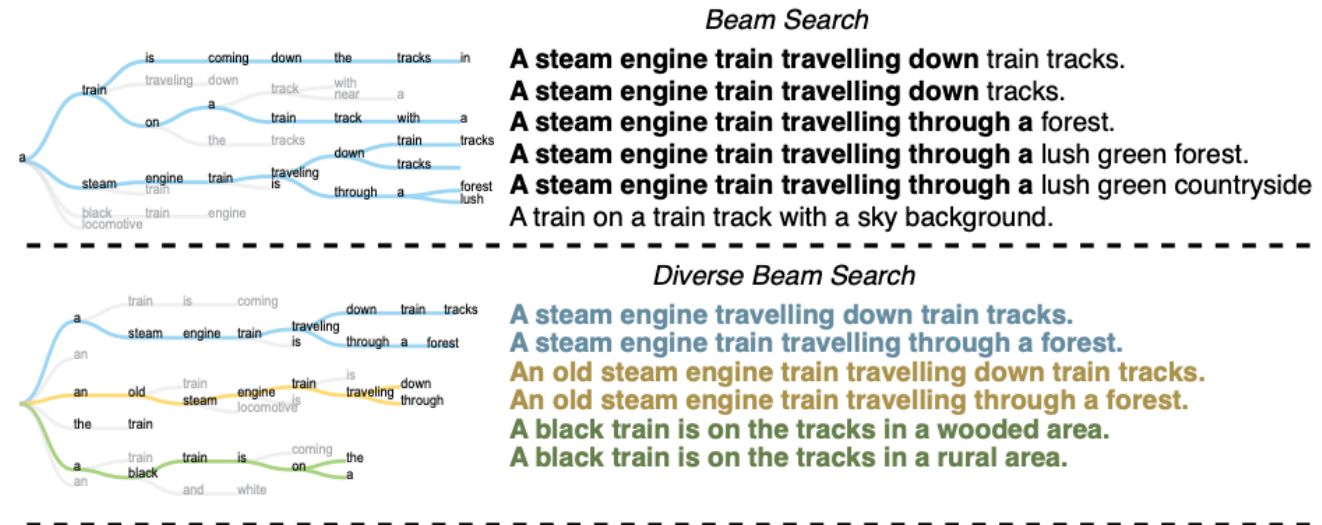


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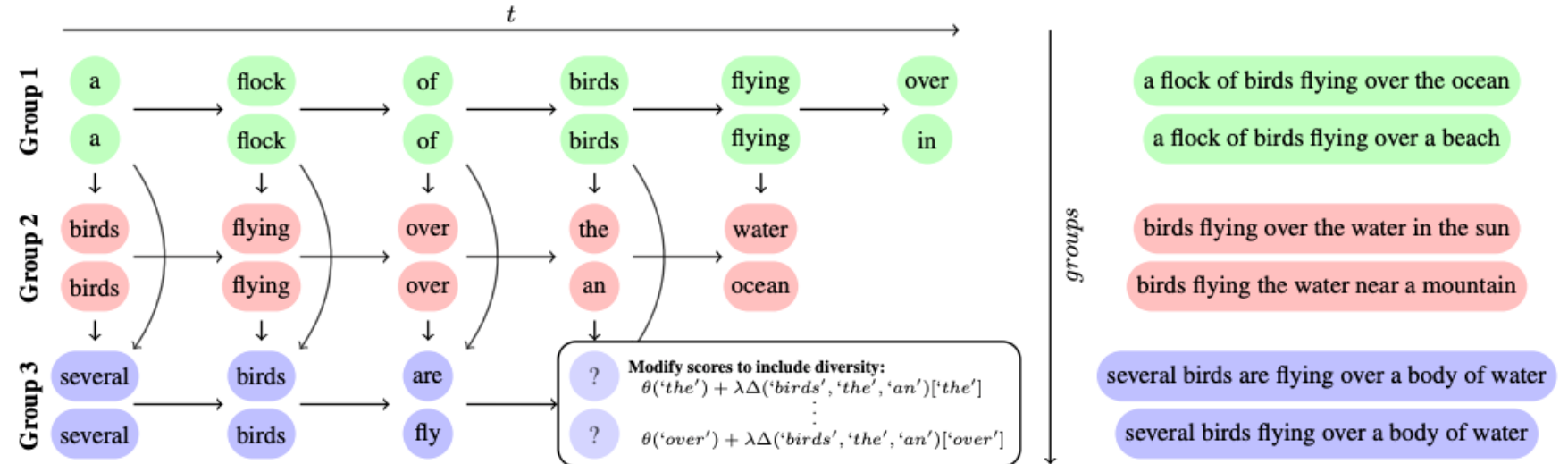
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An old fashion train with steam coming out of its pipe.

An engine is coming down the train track.
A black and red train moving down a train track.



Diverse Beam Search^[1]

- **Key idea:** augmenting beam search objective with **dissimilarity term**
 - To this end, first dividing entire beam into multiple groups
 - Then, diversity between output of different groups is used as additional score



Diverse Beam Search^[1]

- **Key idea:** augmenting beam search objective with **dissimilarity term**
 - Then, diversity between output of different groups is used as additional score

$$Y_{[t]}^g = \underset{\mathbf{y}_{1,[t]}^g, \dots, \mathbf{y}_{B',[t]}^g \in \mathcal{Y}_t^g}{\operatorname{argmax}} \quad \sum_{b \in [B']} \Theta(\mathbf{y}_{b,[t]}^g) + \sum_{h=1}^{g-1} \lambda_g \Delta(\mathbf{y}_{b,[t]}^g, Y_{[t]}^h)$$
$$s.t. \mathbf{y}_{i,[t]}^g \neq \mathbf{y}_{j,[t]}^g, \lambda_g \geq 0$$

- E.g., embedding similarity or n-gram similarity

$$\Delta(\mathbf{y}_{[t]}, Y_{[t]}^g) = \sum_{b=1}^{B'} \delta(\mathbf{y}_{[t]}, \mathbf{y}_{b,[t]}^g)$$

Diverse Beam Search^[1]

- Algorithm

Algorithm 1: Diverse Beam Search

```
1 Perform a diverse beam search with  $G$  groups using a beam width of  $B$ 
2 for  $t = 1, \dots, T$  do
    // perform one step of beam search for first group without diversity
3    $Y_{[t]}^1 \leftarrow \operatorname{argmax}_{(\mathbf{y}_{1,[t]}^1, \dots, \mathbf{y}_{B',[t]}^1)} \sum_{b \in [B']} \Theta(\mathbf{y}_{b,[t]}^1)$ 
4   for  $g = 2, \dots, G$  do
       // augment log-probabilities with diversity penalty
5        $\Theta(\mathbf{y}_{b,[t]}^g) \leftarrow \Theta(\mathbf{y}_{b,[t]}^g) + \sum_h \lambda_g \Delta(\mathbf{y}_{b,[t]}^g, Y_{[t]}^h) \quad b \in [B'], \mathbf{y}_{b,[t]}^g \in \mathcal{Y}^g \text{ and } \lambda_g > 0$ 
       // perform one step of beam search for the group
6        $Y_{[t]}^g \leftarrow \operatorname{argmax}_{(\mathbf{y}_{1,[t]}^g, \dots, \mathbf{y}_{B',[t]}^g)} \sum_{b \in [B']} \Theta(\mathbf{y}_{b,[t]}^g)$ 
7 Return set of  $B$  solutions,  $Y_{[T]} = \bigcup_{g=1}^G Y_{[T]}^g$ 
```

Diverse Beam Search^[1]

- Results: Visual Question Generation

Table 1: **Top:** Oracle SPICE@ k and distinct n-grams on the COCO image captioning task at $B = 20$. While we report SPICE, we observe similar trends in other metrics (reported in the supplement). **Bottom:** Breakdown of results by difficulty class, highlighting the relative improvement over BS.

	Method	SPICE	Oracle SPICE@k			Distinct n-Grams			
			@5	@10	@20	n = 1	2	3	4
COCO	BS	16.27	22.96	25.14	27.34	0.40	1.51	3.25	5.67
	Li & Jurafsky (2016)	16.35	22.71	25.23	27.59	0.54	2.40	5.69	8.94
	DBS	16.783	23.08	26.08	28.09	0.56	2.96	7.38	13.44
	Li et al. (2015)	16.74	23.27	26.10	27.94	0.42	1.37	3.46	6.10
PASCAL-50S	BS	4.93	7.04	7.94	8.74	0.12	0.57	1.35	2.50
	Li & Jurafsky (2016)	5.08	7.24	8.09	8.91	0.15	0.97	2.43	5.31
	DBS	5.357	7.357	8.269	9.293	0.18	1.26	3.67	7.33
	Li et al. (2015)	5.12	7.17	8.16	8.56	0.13	1.15	3.58	8.42

Diverse Beam Search^[1]

- Results: Machine Translation
 - Better accuracy (BLEU-4) with much diverse outputs

Table 3: Quantitative results on En-Fr machine translation on the newstest-2013 dataset (at $B = 20$). Although we report BLEU-4 values, we find similar trends hold for lower BLEU metrics as well.

Method	Oracle Accuracy (BLEU-4)				Diversity Statistics			
	@1	@5	@10	@20	distinct-1	distinct-2	distinct-3	distinct-4
Beam Search	13.52	16.67	17.63	18.44	0.04	0.75	2.10	3.23
Li & Jurafsky (2016)	13.63	17.11	17.50	18.34	0.04	0.81	2.92	4.61
DBS	13.69	17.51	17.80	18.77	0.06	0.95	3.67	5.54
Li et al. (2015)	13.40	17.54	17.97	18.86	0.04	0.86	2.76	4.31

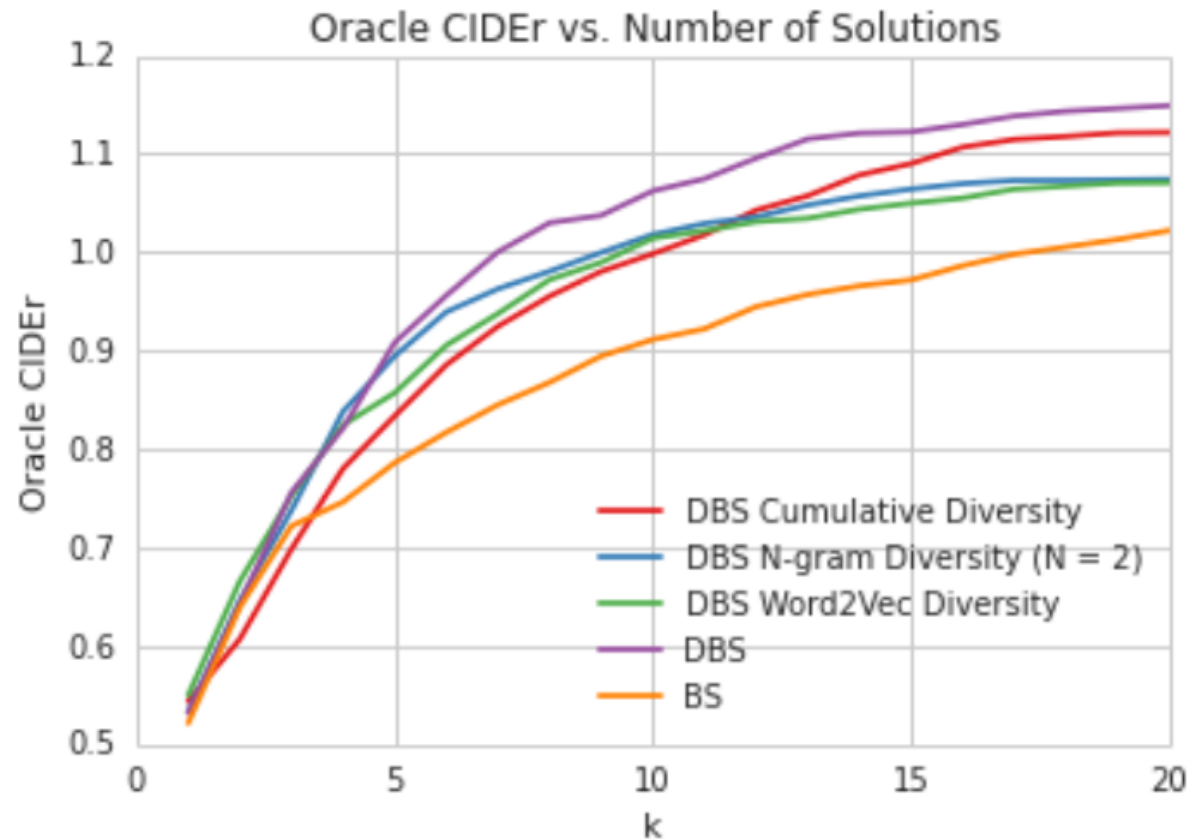
Diverse Beam Search^[1]

- Qualitative results

	Simple	Average	Complex
A) Sample Images			
B) Human	A propeller plane is flying overhead A old time airplane perform in the air show. A small plane is flying through the air. The biplane with the yellow wings flew in the sky.	A black sheep dog watches over a black sheep. A dog and lamb are playing in a fenced area. A black dog looking at a brown sheep in a field. A dog is standing near a sheep.	A double-decker bus is pulling into a bus station. People walking past a red and white colored bus. A double-decker bus pulls into a terminal. People walk down the sidewalk at a bus station.
C) BS	A blue and yellow biplane flying in the sky. A small airplane is flying in the sky. A blue and yellow biplane flying in the sky. A small airplane flying in the blue sky.	A dog sitting on the ground next to a fence. A black and white dog standing next to a sheep. A dog is sitting on the ground next to a fence. A black and white dog standing next to a dog.	A red double decker bus driving down a street. A double decker bus parked in front of a building. A double decker bus driving down a city street. A double decker bus is parked on the side of the street.
D) DBS	A small airplane flying through a blue sky. A blue and yellow biplane flying in the sky. There is a small plane flying in the sky. An airplane flying with a blue sky in the background.	There is a dog that is sitting on the ground. An animal that is laying down in the grass. There is a black and white dog sitting on the ground. Two dogs are sitting on the ground with a fence.	A red double decker bus driving down a street. The city bus is traveling down the street. People are standing in front of a double decker bus. The city bus is parked on the side of the street.

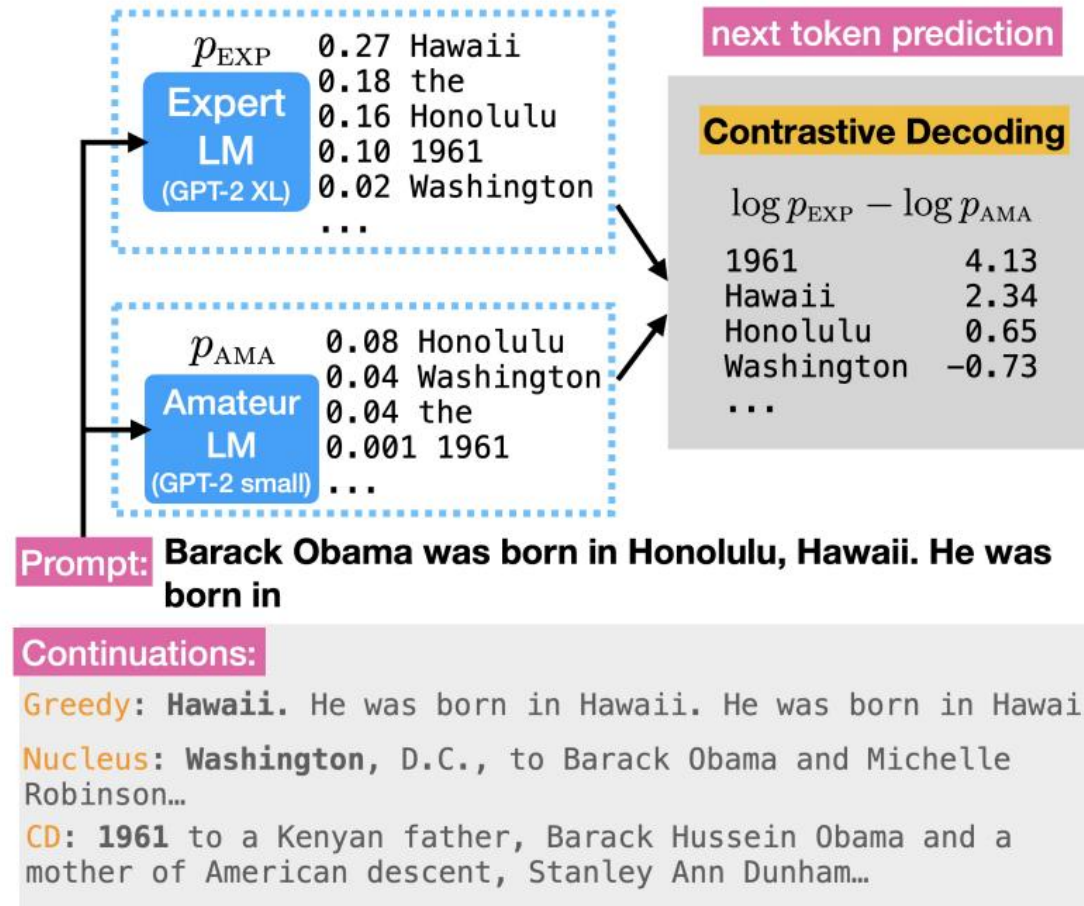
Diverse Beam Search^[1]

- **Ablation:** different diversity functions
 - DBS penalizes selection of tokens proportional to # of times it was selected before



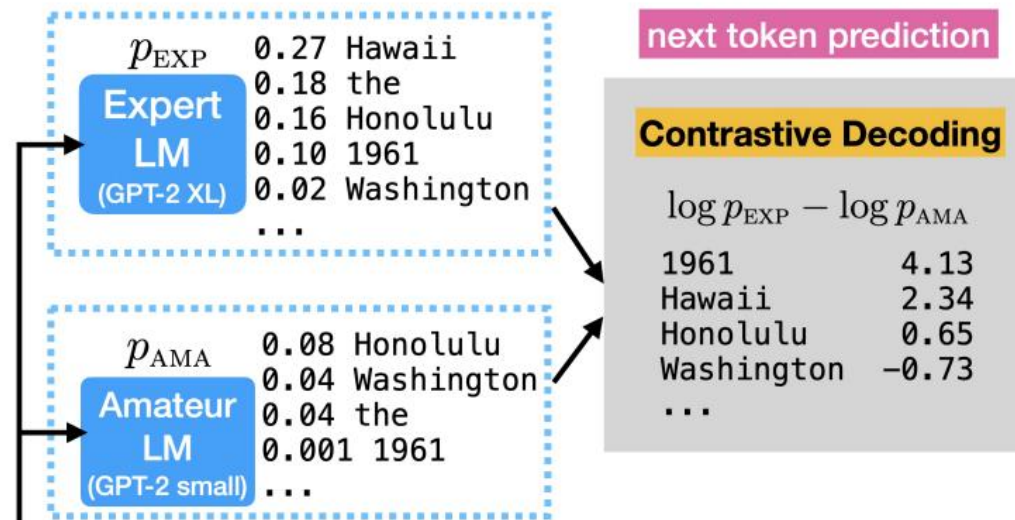
Contrastive Decoding^[1]

- **Key idea:** contrasting two output distributions to obtain refined one



Contrastive Decoding^[1]

- **Key idea:** contrasting two output distributions to obtain refined one
 - Here, two output distributions are from different models (e.g., larger & small)
 - Assumption. Output from subtracting distribution is **relatively undesirable**



Contrastive Decoding^[1]

- But, amateur can capture many aspects of English grammar & common sense
 - Therefore, penalizing all behaviors from amateur LMs is not valid
- **Idea:** Adaptive Plausibility Constraint
 - Namely, truncating output distribution for next token generation (similar to Top-P)
 - In paper, fixed $\alpha = 0.1$ is used

$$\mathcal{V}_{\text{head}}(x_{<i}) = \{x_i \in \mathcal{V} : p_{\text{EXP}}(x_i \mid x_{<i}) \geq \alpha \max_w p_{\text{EXP}}(w \mid x_{<i})\} \quad (1)$$

Contrastive Decoding^[1]

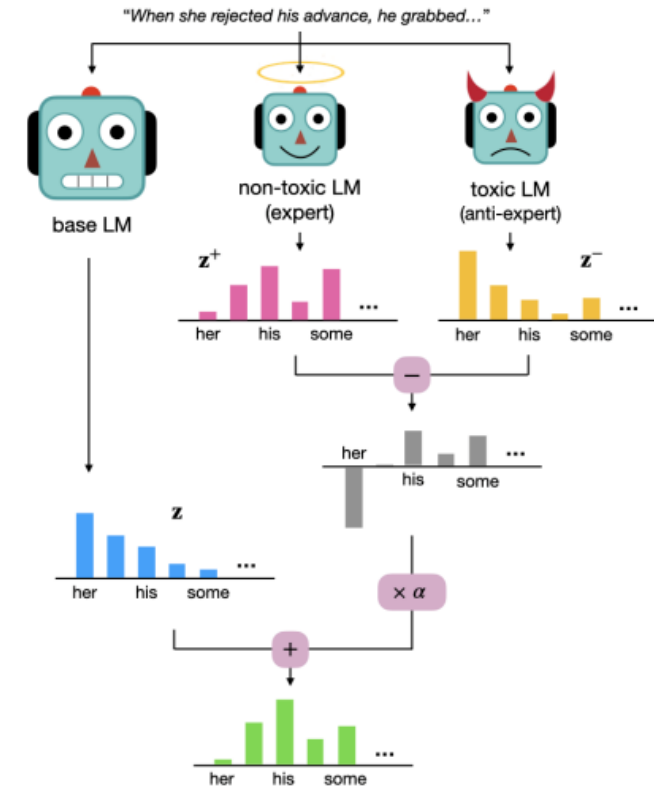
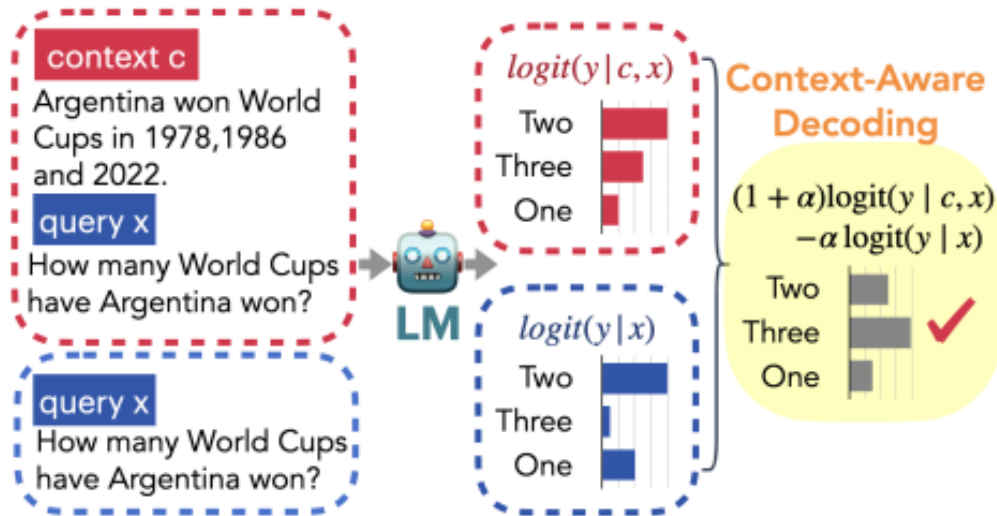
- **Full method**

- One can incorporate previous decoding methods (e.g., greedy or beam-search)

$$\begin{aligned} &\text{CD-score}(x_i; x_{<i}) \\ &= \begin{cases} \log \frac{p_{\text{EXP}}(x_i | x_{<i})}{p_{\text{AMA}}(x_i | x_{<i})}, & \text{if } x_i \in \mathcal{V}_{\text{head}}(x_{<i}), \\ -\text{inf}, & \text{otherwise.} \end{cases} \end{aligned}$$

Contrastive Decoding^[1]

- Depending on choice of amateur, behavior of contrastive decoding is changed
 - Generic performance: smaller LMs^[1]
 - Focusing on given input context for better RAG: same LMs but without context^[2]
 - Debiased or detoxic generation: toxic LMs^[3]



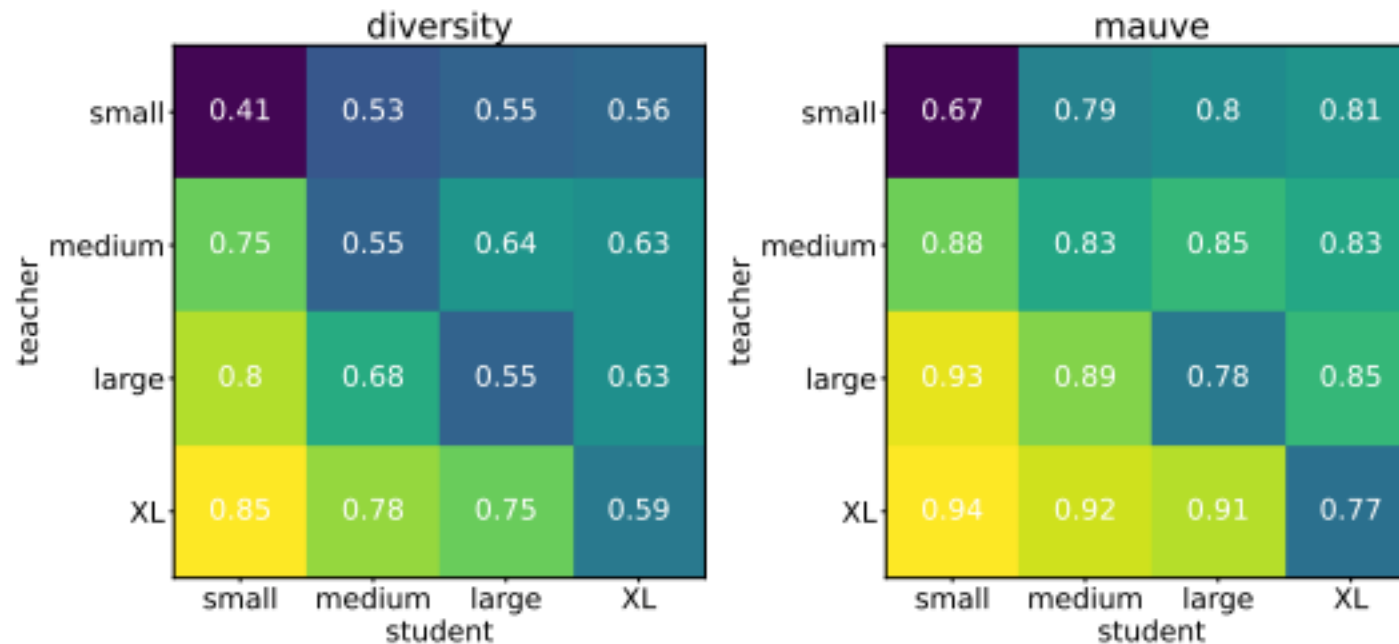
Contrastive Decoding^[1]

- Experiments: **continuation** (input: first 32 words → output: 256 tokens)
 - Metrics: (1) DIV (diversity), (2) MAUVE (similarity), (3) COH (coherence, similarity)

	name	wikinews			wikitext			story		
		DIV	MAUVE	COH	DIV	MAUVE	COH	DIV	MAUVE	COH
OPT-13B	max prob	0.08	0.3	0.65	0.03	0.08	0.63	0.02	0.05	0.51
	k=50	0.91	0.92	0.64	0.72	0.77	0.64	0.91	0.9	0.51
	p=0.95	0.92	0.92	0.62	0.92	0.89	0.55	0.93	0.91	0.48
	typical=0.95	0.94	0.9	0.59	0.89	0.86	0.58	0.95	0.91	0.46
	CS(Su et al., 2022)	0.92	0.87	0.59	0.87	0.77	0.52	0.81	0.78	0.47
	CD	0.94	0.94	0.69	0.91	0.91	0.69	0.89	0.94	0.62
GPT2-XL	max prob	0.04	0.14	0.65	0.02	0.05	0.62	0.01	0.03	0.49
	k=50	0.92	0.88	0.64	0.87	0.79	0.61	0.91	0.87	0.51
	p=0.95	0.94	0.9	0.6	0.92	0.87	0.57	0.94	0.91	0.46
	typical=0.95	0.95	0.91	0.56	0.95	0.84	0.53	0.96	0.88	0.43
	CS(Su et al., 2022)	0.93	0.82	0.62	0.86	0.75	0.59	0.88	0.78	0.48
	CD	0.92	0.94	0.69	0.89	0.92	0.69	0.83	0.94	0.64

Contrastive Decoding^[1]

- Experiments: **continuation** (input: first 32 words → output: 256 tokens)
 - Metrics: (1) DIV (diversity), (2) MAUVE (similarity), (3) COH (coherence, similarity)
 - Effect of different configurations of expert and amateur models



Contrastive Decoding^[1]

- Experiments: **continuation** (input: first 32 words → output: 256 tokens)
 - Human evaluation

			coherence			fluency		
CD		Baseline	CD is better	same	Baseline is better	CD is better	same	Baseline is better
wikitext	CD (GPT-2 XL)	nucleus (GPT-2 XL)	0.714*	0.083	0.202	0.548	0.083	0.369
	CD (GPT-2 XL)	typical (GPT-2 XL)	0.887*	0.046	0.067	0.703*	0.082	0.215
	CD (OPT-13B)	nucleus (OPT-13B)	0.556	0.202	0.242	0.419	0.197	0.384
	CD (OPT-13B)	typical (OPT-13B)	0.773*	0.106	0.121	0.687*	0.152	0.162
wikinews	CD (GPT-2 XL)	nucleus (GPT-2 XL)	0.708*	0.042	0.25	0.583*	0.12	0.297
	CD (GPT-2 XL)	typical (GPT-2 XL)	0.771*	0.151	0.078	0.755*	0.151	0.094
	CD (OPT-13B)	nucleus (OPT-13B)	0.585*	0.221	0.195	0.518	0.123	0.359
	CD (OPT-13B)	typical (OPT-13B)	0.693*	0.099	0.208	0.49	0.297	0.214
story	CD (GPT-2 XL)	nucleus (GPT-2 XL)	0.636*	0.045	0.318	0.404	0.106	0.49
	CD (GPT-2 XL)	typical (GPT-2 XL)	0.506	0.256	0.238	0.387	0.363	0.25
	CD (OPT-13B)	nucleus (OPT-13B)	0.616*	0.101	0.283	0.449	0.293	0.258
	CD (OPT-13B)	typical (OPT-13B)	0.626*	0.202	0.172	0.52	0.212	0.268

Context-aware Decoding (CAD)^[1]

- Experiments: **summarization**

			CNN-DM			XSUM		
Model		Decoding	ROUGE-L	factKB	BERT-P	ROUGE-L	factKB	BERT-P
OPT	13B	Regular	22.0	77.8	86.5	16.4	47.2	85.2
		CAD	27.4	84.1	90.8	18.2	64.9	87.5
	30B	Regular	22.2	81.7	87.0	17.4	38.2	86.1
		CAD	28.4	87.0	90.2	19.5	45.6	89.3
GPT-Neo	3B	Regular	24.3	80.5	87.5	17.6	54.0	86.6
		CAD	27.7	87.5	90.6	18.1	65.1	89.1
	20B	Regular	18.7	68.3	85.2	14.9	42.2	85.7
		CAD	24.5	77.5	89.4	19.0	63.3	90.6
LLaMA	13B	Regular	27.1	80.2	89.5	19.0	53.5	87.8
		CAD	32.6	90.8	93.0	21.1	73.4	91.7
	30B	Regular	25.8	76.8	88.5	18.7	47.7	87.1
		CAD	31.8	87.8	92.2	22.0	66.4	90.3
FLAN	3B	Regular	25.5	90.2	91.6	18.8	31.9	88.2
		CAD	26.1	93.9	92.1	19.5	35.9	88.8
	11B	Regular	25.4	90.4	91.4	19.4	29.8	88.3
		CAD	27.1	93.1	92.2	20.0	35.0	88.8

Visual Contrastive Decoding (VCD)^[1]

- This contrastive decoding framework can be generalized beyond language
 - Goal: reducing hallucination of large vision-language models (LVLMs^[2])

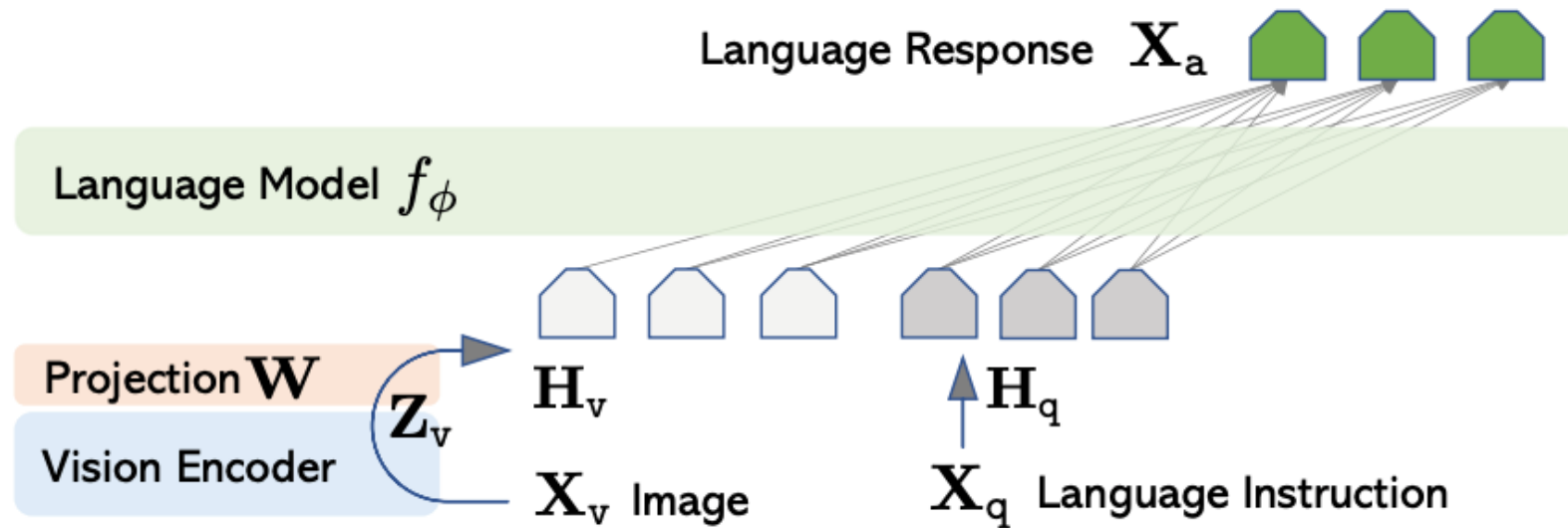


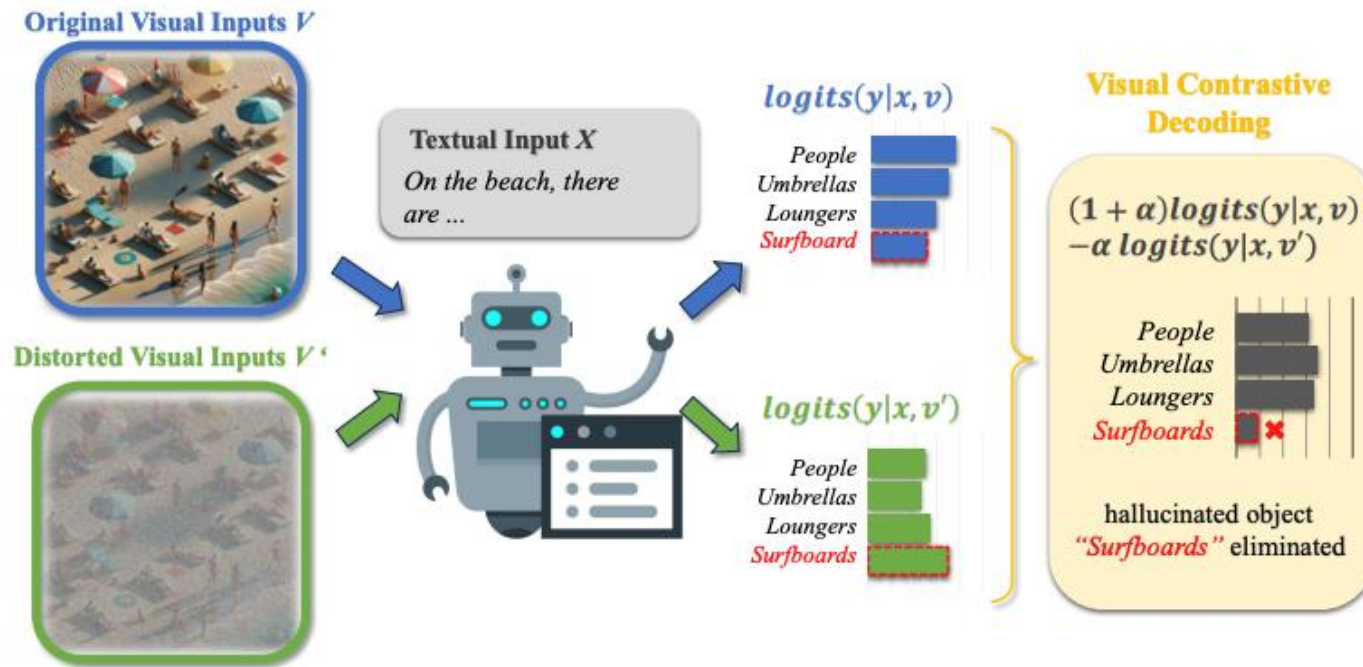
Figure 1: LLaVA network architecture.

[1] Leng et al., Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding., CVPR 2024

[2] Liu et al., Visual Instruction Tuning., NeurIPS 2023

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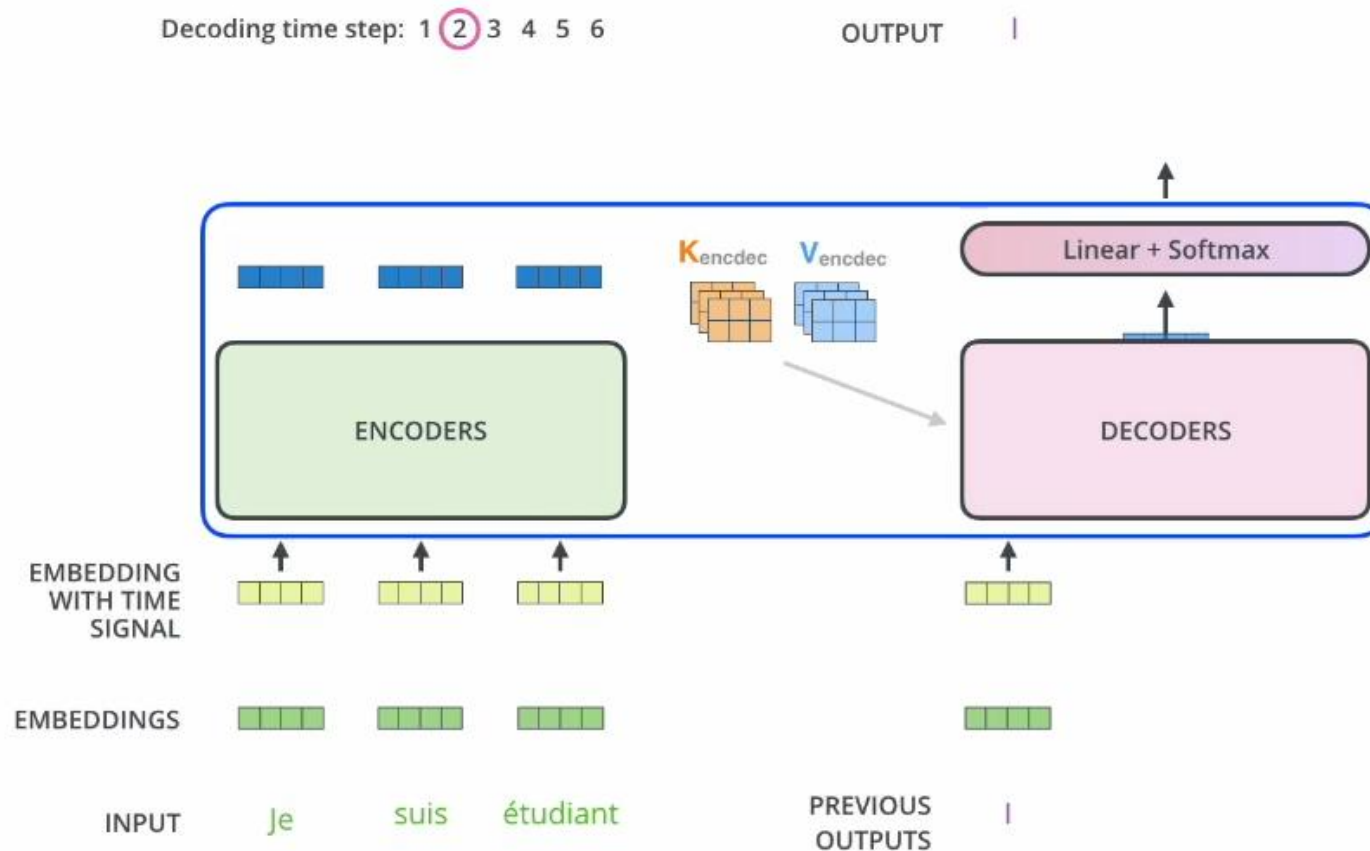
Dataset	Setting	Model	Decoding	Accuracy↑	Precision	Recall	F1 Score↑
MSCOCO	<i>Random</i>	LLaVA1.5	Regular	83.29 _(±0.35)	92.13 _(±0.54)	72.80 _(±0.57)	81.33 _(±0.41)
			VCD	87.73 _(±0.40)	91.42 _(±0.55)	83.28 _(±0.42)	87.16 _(±0.41)
		Qwen-VL	Regular	84.73 _(±0.36)	95.61 _(±0.45)	72.81 _(±0.38)	82.67 _(±0.41)
			VCD	88.63 _(±0.10)	94.64 _(±0.25)	81.91 _(±0.19)	87.81 _(±0.11)
		InstructBLIP	Regular	80.71 _(±0.73)	81.67 _(±0.67)	79.19 _(±1.14)	80.41 _(±0.80)
			VCD	84.53 _(±0.38)	88.55 _(±0.54)	79.32 _(±0.44)	83.68 _(±0.40)
	<i>Popular</i>	LLaVA1.5	Regular	81.88 _(±0.48)	88.93 _(±0.60)	72.80 _(±0.57)	80.06 _(±0.05)
			VCD	85.38 _(±0.38)	86.92 _(±0.53)	83.28 _(±0.42)	85.06 _(±0.37)
		Qwen-VL	Regular	84.13 _(±0.18)	94.31 _(±0.43)	72.64 _(±0.45)	82.06 _(±0.23)
			VCD	87.12 _(±0.07)	91.49 _(±0.10)	81.85 _(±0.19)	86.40 _(±0.09)
		InstructBLIP	Regular	78.22 _(±0.84)	77.87 _(±1.03)	78.85 _(±0.52)	78.36 _(±0.76)
			VCD	81.47 _(±0.42)	82.89 _(±0.64)	79.32 _(±0.44)	81.07 _(±0.39)
	<i>Adversarial</i>	LLaVA1.5	Regular	78.96 _(±0.52)	83.06 _(±0.58)	72.75 _(±0.59)	77.57 _(±0.57)
			VCD	80.88 _(±0.33)	79.45 _(±0.29)	83.29 _(±0.43)	81.33 _(±0.34)
		Qwen-VL	Regular	82.26 _(±0.30)	89.97 _(±0.33)	72.61 _(±0.50)	80.37 _(±0.37)
			VCD	84.26 _(±0.39)	85.84 _(±0.45)	82.05 _(±0.39)	83.90 _(±0.39)
		InstructBLIP	Regular	75.84 _(±0.45)	74.30 _(±0.63)	79.03 _(±0.68)	76.59 _(±0.40)
			VCD	79.56 _(±0.41)	79.67 _(±0.59)	79.39 _(±0.50)	79.52 _(±0.38)

[1] Leng et al., Mitigating Object Hallucinations in Large Vision-Language Models through Visual Contrastive Decoding., CVPR 2024

[2] Liu et al., Visual Instruction Tuning., NeurIPS 2023

Recap: Autoregressive Model

- How to generate K tokens? → K sequential inferences of LLMs



Motivation

- But, some tokens are easier to generate than others!!

Hebrew: הנשיא היה ברק אובמה. English: The **president** was Barack **Obama**.

Hard - e.g. requires looking
several tokens back,
knowledge of hebrew, ...

Easy - e.g. can guess based on
just the last token.

Examples of hard and easy words to sequentially generate^[1]

Speculative Decoding^[1]

- **Key idea:** (1) **Drafting** with small model

K token generation via auto-regressive decoding, i.e., “K inferences”

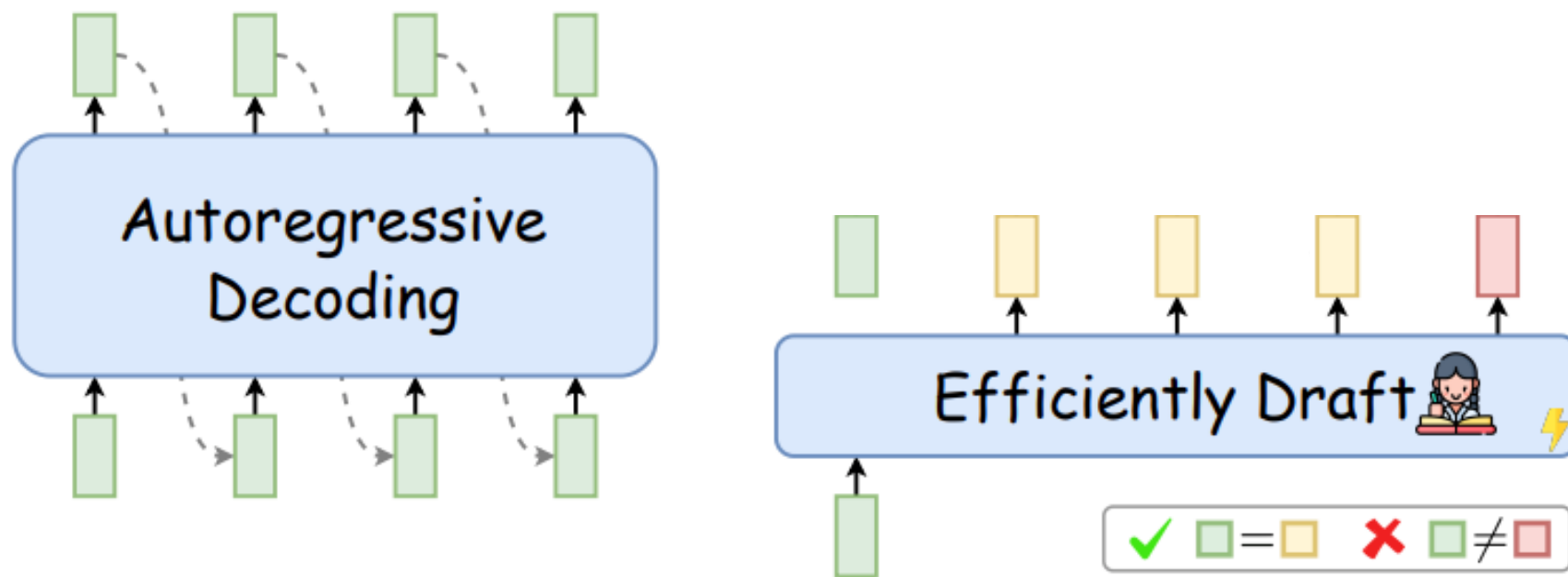


Illustration of Speculative Decoding^[2]

[1] Leviathan et al., Fast Inference from Transformers via Speculative Decoding., ICML 2023 (Oral)

[2] Xia et al., Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding., arXiv:24.01

Speculative Decoding^[1]

- **Key idea:** (1) Drafting with small LLM (M_q)

K token generation via auto-regressive decoding, i.e., “K inferences”

Algorithm 1 SpeculativeDecodingStep

Inputs: $M_p, M_q, prefix$.

▷ **Sample** γ guesses $x_1, \dots, x_\gamma$ from M_q autoregressively.

for $i = 1$ **to** γ **do**

$q_i(x) \leftarrow M_q(prefix + [x_1, \dots, x_{i-1}])$

$x_i \sim q_i(x)$

end for

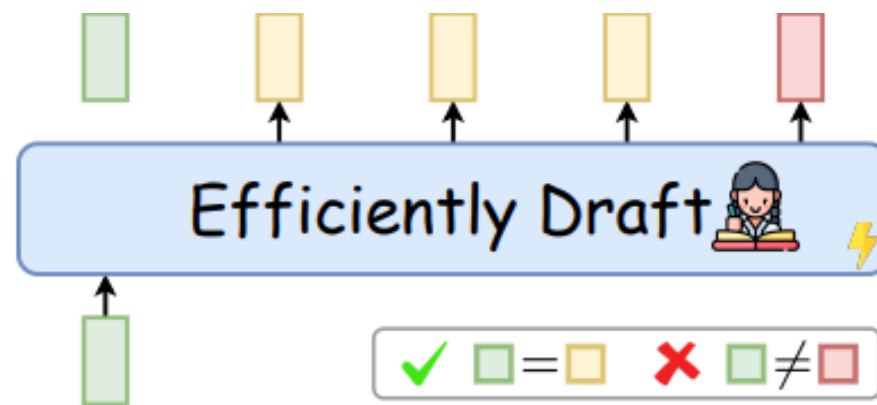


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Speculative Decoding^[1]

- **Key idea:** (2) **Verifying** drafts with large target LLM (M_p)

Measuring output probabilities of draft with “only 1 inference”

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▷ Sample γ guesses $x_1, \dots, x_\gamma$ from M_q autoregressively.

for $i = 1$ to γ do

$q_i(x) \leftarrow M_q(prefix + [x_1, \dots, x_{i-1}])$

$x_i \sim q_i(x)$

end for

▷ Run M_p in parallel.

$p_1(x), \dots, p_{\gamma+1}(x) \leftarrow$

$M_p(prefix), \dots, M_p(prefix + [x_1, \dots, x_\gamma])$

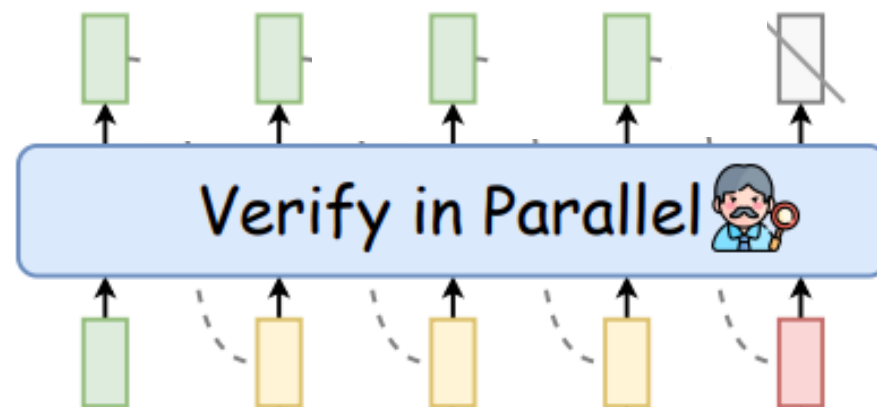


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▷ Determine the number of accepted guesses n .

$r_1 \sim U(0, 1), \dots, r_\gamma \sim U(0, 1)$

$n \leftarrow \min(\{i - 1 \mid 1 \leq i \leq \gamma, r_i > \frac{p_i(x)}{q_i(x)}\} \cup \{\gamma\})$

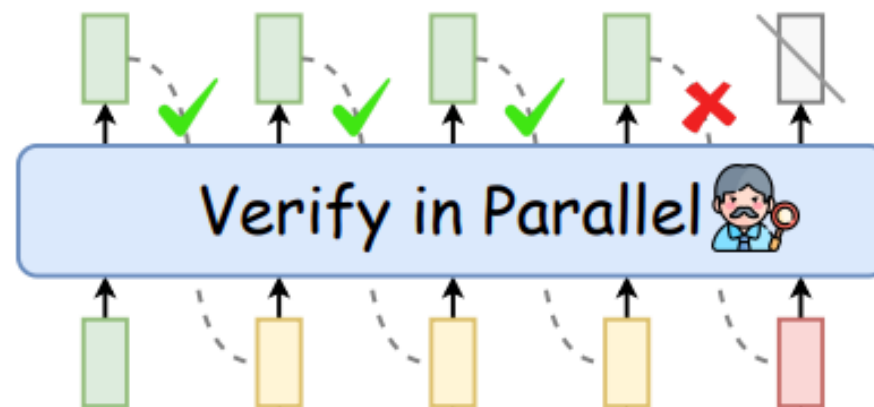


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⋮

▷ **Determine the number of accepted guesses n .**
 $r_1 \sim U(0, 1), \dots, r_\gamma \sim U(0, 1)$
 $n \leftarrow \min(\{i - 1 \mid 1 \leq i \leq \gamma, r_i > \frac{p_i(x)}{q_i(x)}\} \cup \{\gamma\})$

▷ **Adjust the distribution from M_p if needed.**
 $p'(x) \leftarrow p_{n+1}(x)$

if $n < \gamma$ then
 $p'(x) \leftarrow \text{norm}(\max(0, p_{n+1}(x) - q_{n+1}(x)))$
end if

▷ **Return one token from M_p , and n tokens from M_q .**
 $t \sim p'(x)$

return $\text{prefix} + [x_1, \dots, x_n, t]$

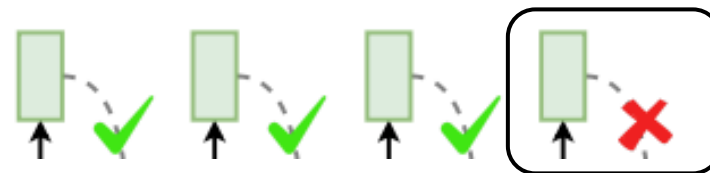


Illustration of Speculative Decoding^[2]

[1] Leviathan et al., Fast Inference from Transformers via Speculative Decoding., ICML 2023 (Oral)

[2] Xia et al., Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding., arXiv:24.01

Speculative Decoding^[1]

- **Key idea:** (2) Verifying draft with large model

Measuring output probabilities of draft with “only 1 inference”

▷ Determine the number of accepted guesses n .

$r_1 \sim U(0, 1), \dots, r_\gamma \sim U(0, 1)$

$n \leftarrow \min(\{i - 1 \mid 1 \leq i \leq \gamma, r_i > \frac{p_i(x)}{q_i(x)}\} \cup \{\gamma\})$

▷ Adjust the distribution from M_p if needed.

$p'(x) \leftarrow p_{n+1}(x)$

if $n < \gamma$ **then**

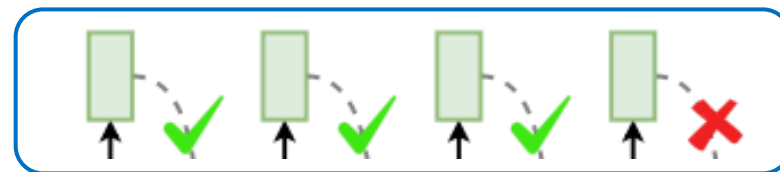
$p'(x) \leftarrow \text{norm}(\max(0, p_{n+1}(x) - q_{n+1}(x)))$

end if

▷ Return one token from M_p , and n tokens from M_q .

$t \sim p'(x)$

return $\text{prefix} + [x_1, \dots, x_n, t]$



4 tokens with 1 M_p inference (exactly same with 4 inferences)

Theoretical guarantee

Illustration of Speculative Decoding^[2]

[1] Leviathan et al., Fast Inference from Transformers via Speculative Decoding., ICML 2023 (Oral)

[2] Xia et al., Unlocking Efficiency in Large Language Model Inference: A Comprehensive Survey of Speculative Decoding., arXiv:24.01

Speculative Decoding^[1]

- **Results:** Better efficiency with **small draft** & **simple heuristic**
 - (1) EnDe: English to German translation, (2) CNNDM: text summarization
 - Tradeoff with larger draft model: higher acceptance rate but lower inference speed

TASK	M_q	TEMP	γ	α	SPEED
ENDE	T5-SMALL ★	0	7	0.75	3.4X
ENDE	T5-BASE	0	7	0.8	2.8X
ENDE	T5-LARGE	0	7	0.82	1.7X
ENDE	T5-SMALL ★	1	7	0.62	2.6X
ENDE	T5-BASE	1	5	0.68	2.4X
ENDE	T5-LARGE	1	3	0.71	1.4X
CNNDM	T5-SMALL ★	0	5	0.65	3.1X
CNNDM	T5-BASE	0	5	0.73	3.0X
CNNDM	T5-LARGE	0	3	0.74	2.2X
CNNDM	T5-SMALL ★	1	5	0.53	2.3X
CNNDM	T5-BASE	1	3	0.55	2.2X
CNNDM	T5-LARGE	1	3	0.56	1.7X

Results of Speculative Decoding (Target: T5-XXL 11B), γ : # of drafted tokens, α : average acceptance rate^[1]

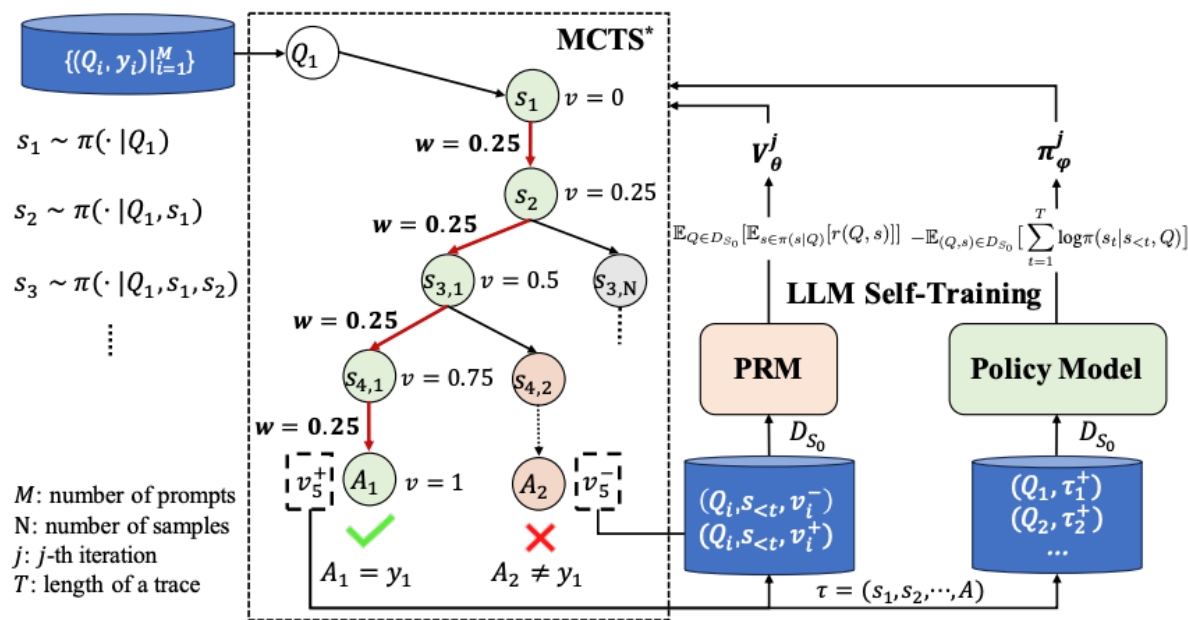
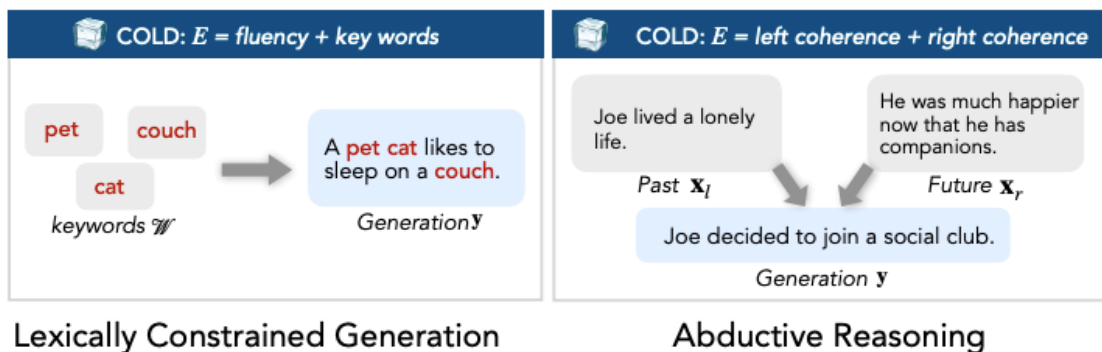
Speculative Decoding^[1]

- Qualitative examples
 - **Green**: drafts, **red**: rejected draft, **blue**: correction by target

```
[START] japan ' s benchmark bend n
[START] japan ' s benchmark nikkei 22 5
[START] japan ' s benchmark nikkei 225 index rose 22 6
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 7 points
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 0 1
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 1 . 5 percent , to 10 , 9859
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 1 . 5 percent , to 10 , 989 . 79 in
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 1 . 5 percent , to 10 , 989 . 79 in tokyo late
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 1 . 5 percent , to 10 , 989 . 79 in late morning trading . [END]
```

More Things in Decoding

- Guided (or controlled) decoding: consider score (reward) during decoding^[1,2,3]



[1] Qin et al., COLD Decoding: Energy-based Constrained Text Generation with Langevin Dynamics., NeurIPS 2022

[2] Zhang et al., ReST-MCTS*: LLM Self-Training via Process Reward Guided Tree Search., arXiv:24.06

[3] Mudgal et al., Controlled Decoding from Language Models., ICML 2024

Retrieval Augmented Generation (RAG)

Instructor: Jaehyung Kim

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Knowledge Cutoff with LLMs

- Pre-training data of LLMs have certain **knowledge cutoff**
 - *E.g.*, 2023-10 for GPT4o (& mini)^[1] and 2023-12 for LLaMA3^[2]

Today, GPT-4o mini supports text and vision in the API, with support for text, image, video and audio inputs and outputs coming in the future. The model has a context window of 128K tokens, supports up to 16K output tokens per request, and has knowledge up to **October 2023**. Thanks to the improved tokenizer shared with GPT-4o, handling non-English text is now even more cost effective.

	Training Data	Params	Context length	GQA	Token count	Knowledge cutoff
Llama 3	A new mix of publicly available online data.	8B	8k	Yes	15T+	March, 2023
		70B	8k	Yes		December, 2023

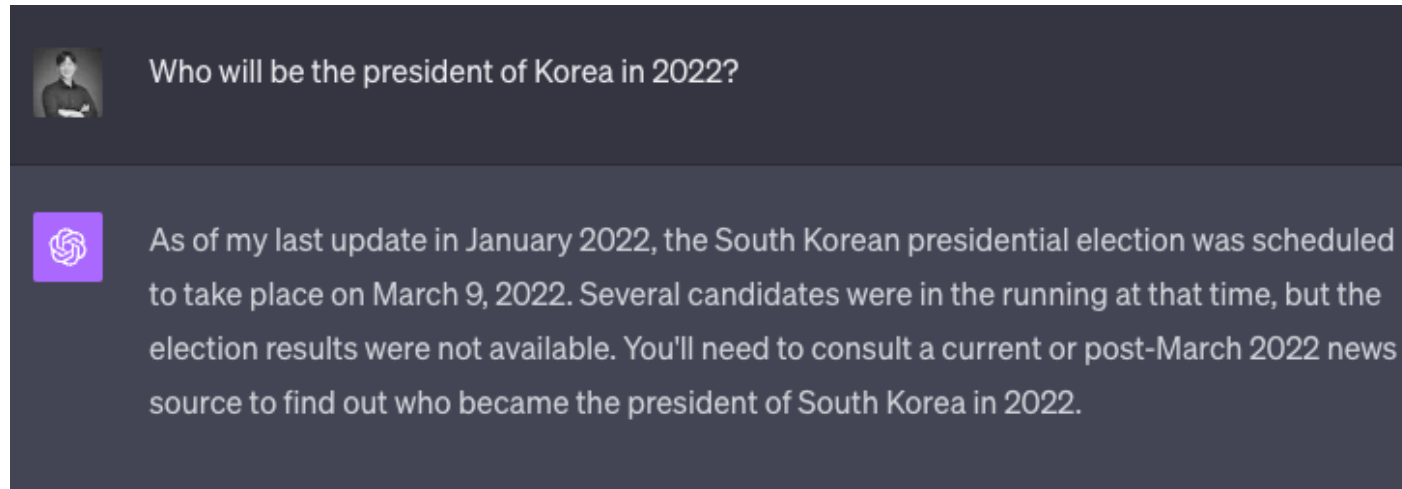
Examples of knowledge cutoff with SOTA LLMs^[1,2]

[1] <https://openai.com/index/gpt-4o-mini-advancing-cost-efficient-intelligence/>

[2] https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md

Limited Knowledge of LLMs

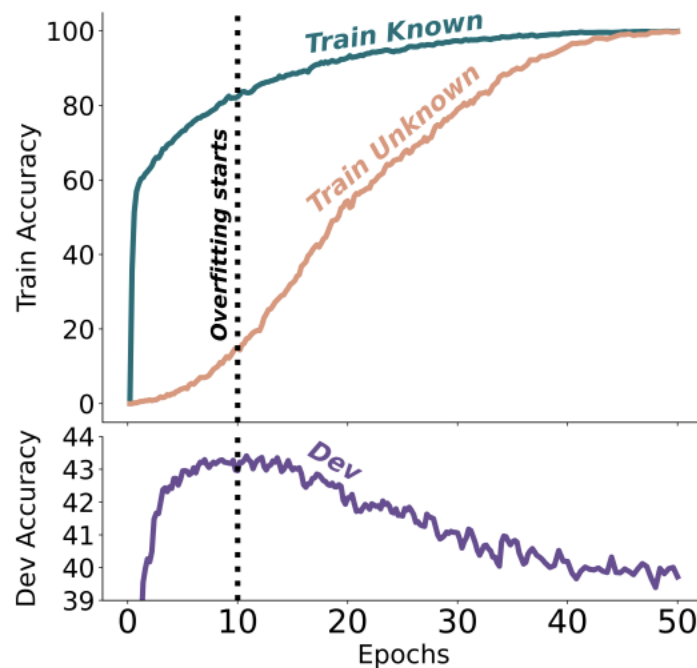
- Namely, LLMs **suffer** to provide answer with **up-to-date knowledge**



Failure case due to limitation on pre-trained knowledge (it's asked in 2023)

Limited Knowledge of LLMs

- Namely, LLMs **suffer** to provide answer with **up-to-date knowledge**
 - Fine-tuning could be limited to effectively incorporate new knowledge^[1,2]



Fitting on data with new knowledge could decrease overall accuracy^[1]

[1] Gekhman et al., Does Fine-Tuning LLMs on New Knowledge Encourage Hallucinations?, arXiv:24.05

[2] Ren et al., Learning or Self-aligning? Rethinking Instruction Fine-tuning., arXiv:24.02

Retrieval Augmentation

- **Retrieving and incorporating** relevant knowledge is promising way
 - Namely, **augmenting** new knowledge as additional input of LLMs
 - *Retrieve-and-read* is popular way to improve QA system

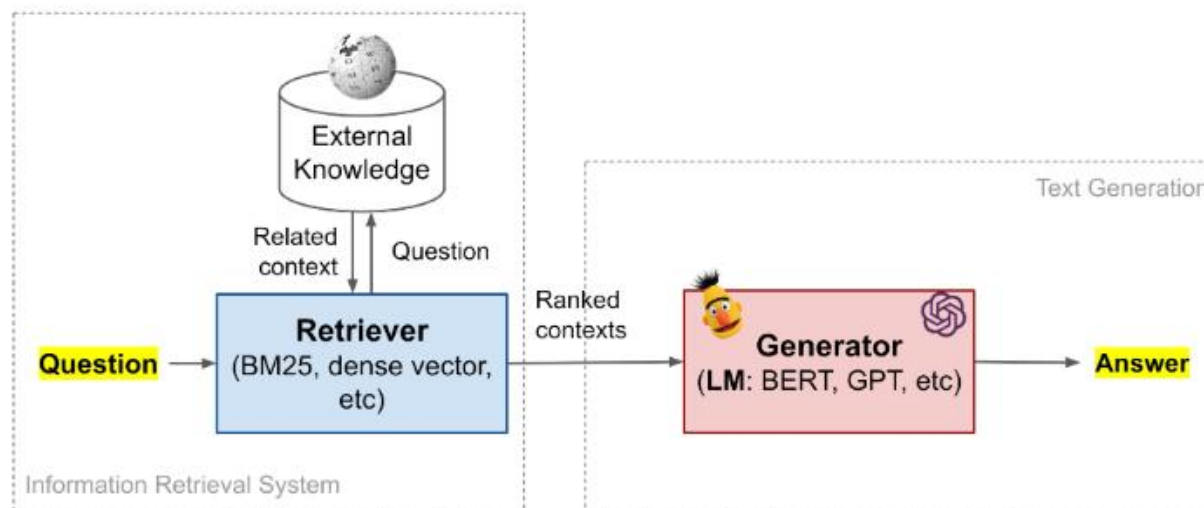


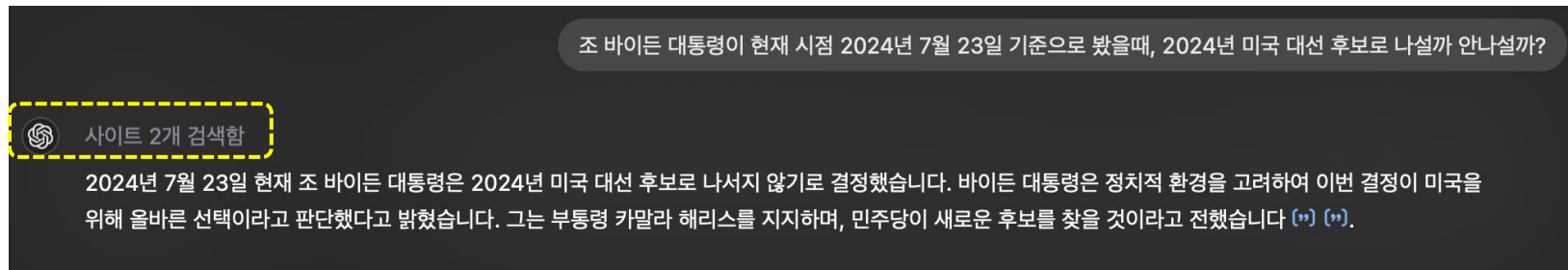
Illustration of retrieve-and-read system for ODQA^[1]

[1] <https://lilianweng.github.io/posts/2020-10-29-odqa/>

[2] Karpukhin et al., Dense Passage Retrieval for Open-Domain Question Answering., EMNLP 2020

Retrieval Augmentation

- **Retrieving and incorporating** relevant knowledge is promising way
 - Namely, augmenting new knowledge as additional input of LLMs
 - **Retrieval-augmented generation (RAG)** now becomes standard approach



Example of RAG with GPT-4o

Contents

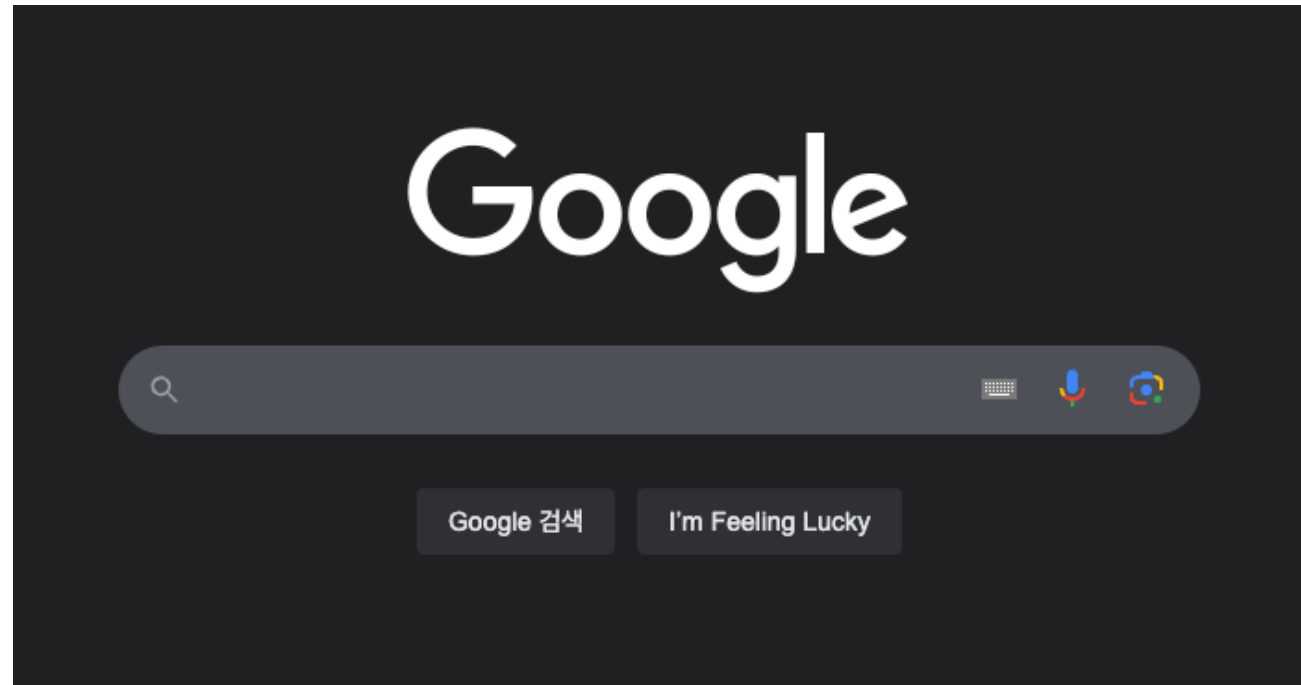
- Retrieval system
- Recent approaches for improving RAG at Inference-level
 - Handling long input context
 - Better input query
 - Robustness to noisy document
- Recent approaches for improving RAG at Training-level

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Retrieval

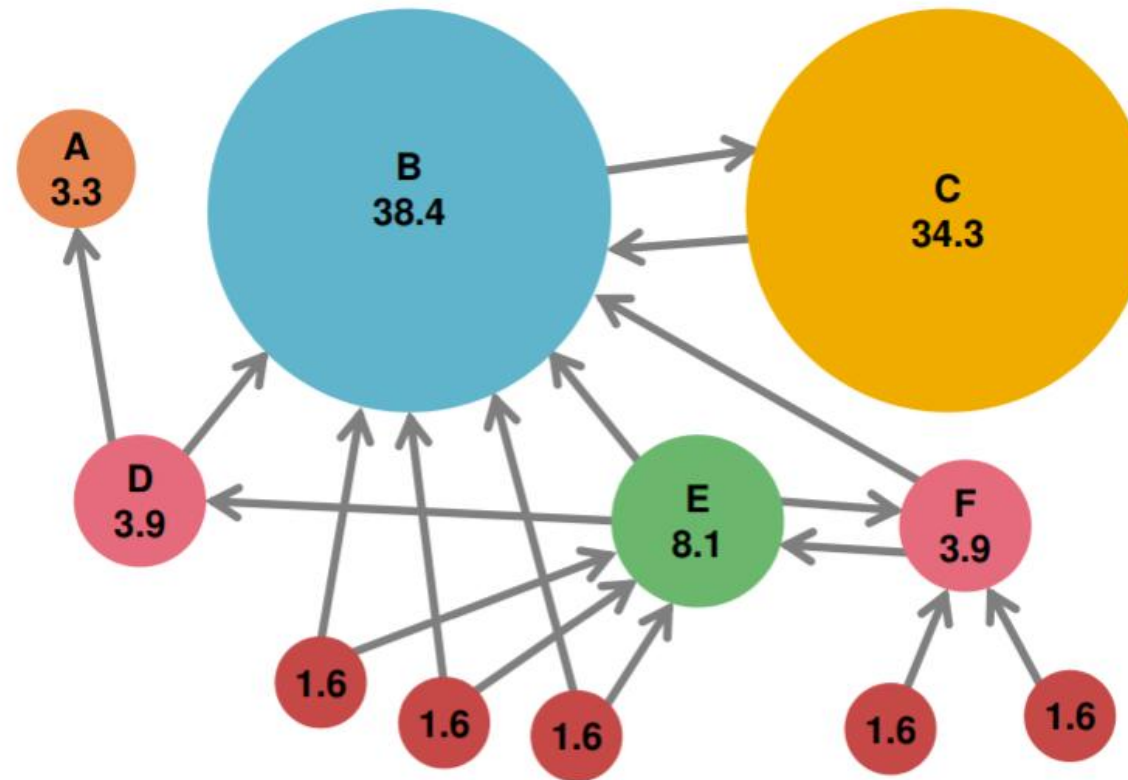
- Web-search, e.g., google search^[1]



[1] Page., The PageRank citation ranking: Bringing order to the web., Technical Report, 1997

Retrieval: Web Search

- **Page-rank**^[1]: importance is **average importance** of linked ones:



Retrieval: Web Search

- **Page-rank**^[1]: importance is **average importance** of linked ones:

$$PR(A) = \frac{1-d}{N} + d \left(\frac{PR(B)}{L(B)} + \frac{PR(C)}{L(C)} + \frac{PR(D)}{L(D)} + \dots \right)$$

- d : damping factor, i.e., $1 - d$ is probability for jumping to random page
- **Mathematically**, $PR(A)$ is probability of arriving at page A after many clicks
- Namely, you can find page-rank with following **iterative method**:

$$PR(p_i; 0) = \frac{1}{N} \quad \longrightarrow \quad PR(p_i; t+1) = \frac{1-d}{N} + d \sum_{p_j \in M(p_i)} \frac{PR(p_j; t)}{L(p_j)}$$

Text-base Retrieval

- **BM25^[1]**: **raw text-level** retrieval, i.e., word overlap $\uparrow \rightarrow$ retrieval $\uparrow$
 - Given a query Q , containing keywords q_i , BM25 score of document D is:
 - N : total # of documents, $n(q_i)$: # of documents containing q_i

$$\text{score}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdl}}\right)}$$

$$\text{IDF}(q_i) = \ln \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right)$$

Dense Retrieval

- Dense passage retrieval (**DPR**)^[1]: using two sentence encoders (e.g., BERT^[2])
 - **Similarity** between query q & passage $p \rightarrow$ **Dot-product** of their features

$$\text{sim}(q, p) = E_Q(q)^\top E_P(p)$$

- Given $\mathcal{D} = \{\langle q_i, p_i^+, p_{i,1}^-, \dots, p_{i,n}^- \rangle\}_{i=1}^m$, following contrastive loss is used:
(**positive** p^+ : labeled data,
negative p^- : labeled data for difficulty + in-batch negatives)

$$L(q_i, p_i^+, p_{i,1}^-, \dots, p_{i,n}^-) = -\log \frac{e^{\text{sim}(q_i, p_i^+)}}{e^{\text{sim}(q_i, p_i^+)} + \sum_{j=1}^n e^{\text{sim}(q_i, p_{i,j}^-)}}$$

Dense Retrieval

- **Contriever**^[1]: Self-supervised training with **contrastive learning**
 - **Idea**: substituting **human-labeled** positive as **artificially generated** one
 - k_+ : **positive** document (crop in same document),
 k_i : **negative** documents (other doc.)

$$\mathcal{L}(q, k_+) = - \frac{\exp(s(q, k_+)/\tau)}{\exp(s(q, k_+)/\tau) + \sum_{i=1}^K \exp(s(q, k_i)/\tau)} \quad s(q, d) = \langle f_\theta(q), f_\theta(d) \rangle$$

Retrieval

- There is **no answer which is best** retrieval method yet
 - As DPR/Contriever are training base, **test distribution is critical** for success

Datasets	ChatGPT						GPT-4		LLaMA2-chat	
	BM25	+ SuRe	DPR	+ SuRe	Contriever	+ SuRe	BM25	+ SuRe	BM25	+ SuRe
NQ	28.4	33.5	36.1	41.0	35.8	40.4	30.2	32.4	18.6*	30.4*
WebQ	19.6	25.1	23.2	27.3	22.5	28.7	21.5	21.7	16.0*	24.0*
2Wiki	27.4	32.8	19.2	21.4	27.2	32.6	34.8	38.2	20.2	27.8
HotpotQA	30.8	33.2	25.6	27.4	32.2	33.6	34.8	40.6	24.0	28.0
Average	26.6	31.2	26.0	29.3	29.4	33.8	30.3	33.2	19.7	27.6

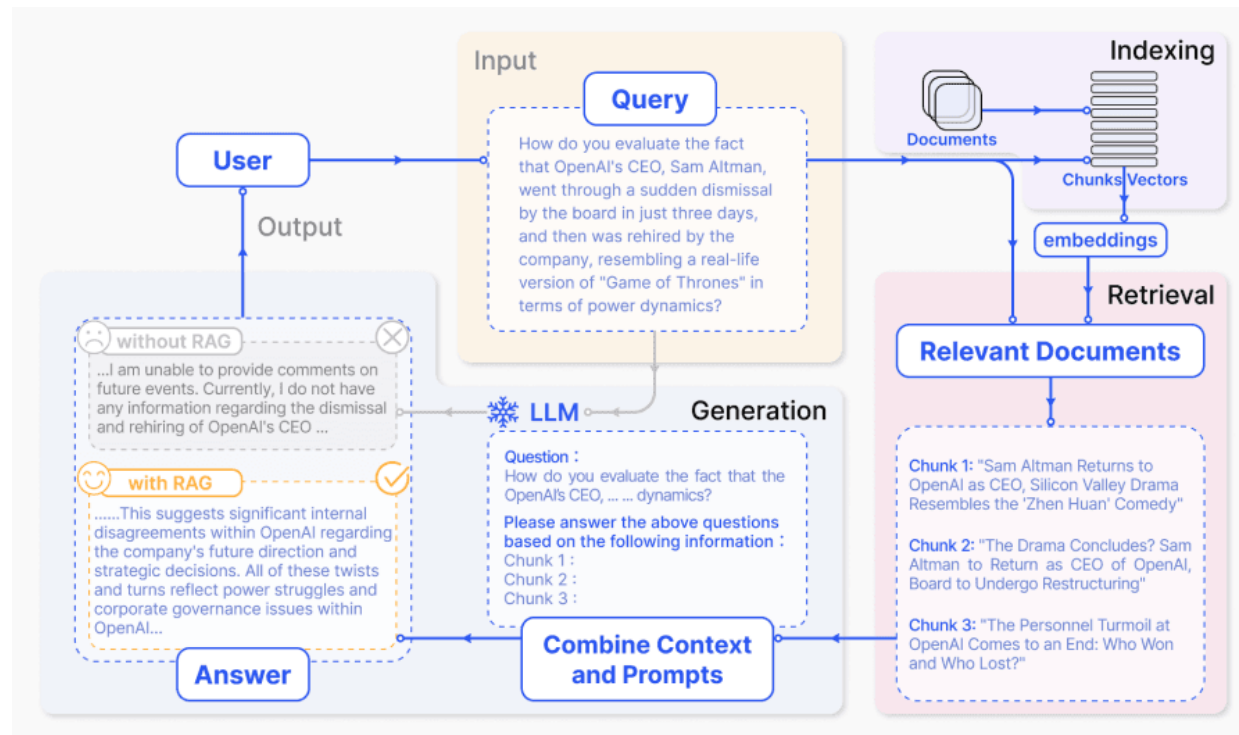
Effectiveness of different retrieval for RAG with SuRe^[1]

Retrieval Augmented Generation (RAG)

- Then, how can we use this retrieval for RAG with LLM?
 - 1. **Training-base**: training LLM to utilize this retrieved information to answer
 - 2. **Inference-base**: appending retrieved information as additional input

Retrieval Augmented Generation (RAG)

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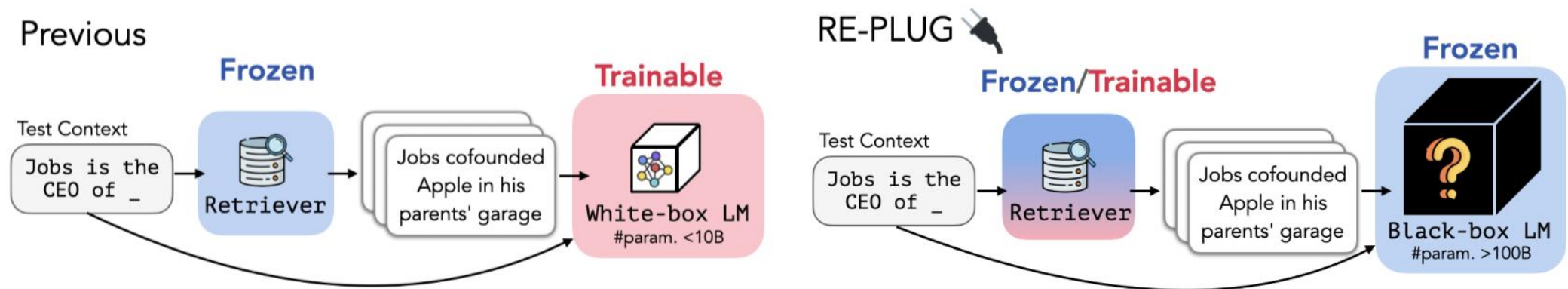


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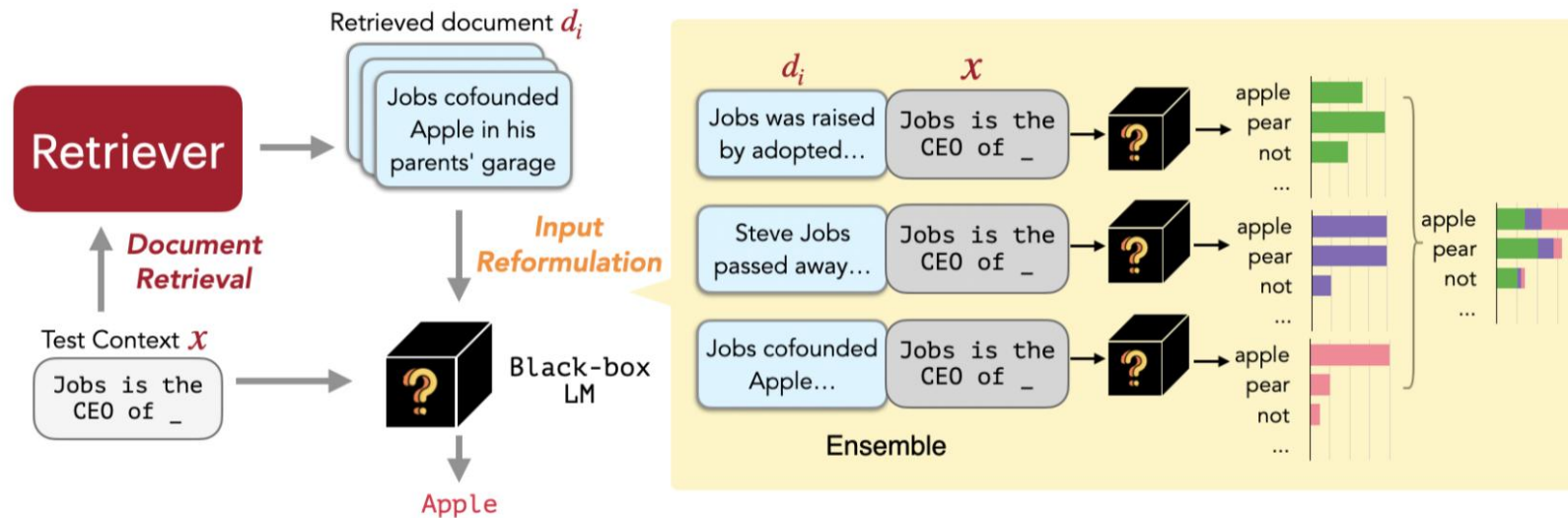
Retrieval and Plug

- REtrieval and PLUG (REPLUG)^[1]
 - Assumption. Target LM (reader) is **black-box**, *i.e.*, parameters can't be accessed & updated
- **Key idea**. Retrieval-augmentation via simple prepending at input
 - By retrieving the document with the existing method, *e.g.*, contriever
 - To further improve retrieval, new way to fine-tune with target black-box LM is proposed



REPLUG: Inference with Input Reformulation

- Challenge. **Limited context window size** → Prepending all documents is difficult
- **Key idea**. Prepend each document separately & ensemble output probabilities
 - Considering quadratic nature of Transformer, it does not increase the computation



REPLUG: Inference with Input Reformulation

- Challenge. **Limited context window size** → Prepending all documents is difficult
- **Key idea**. Prepend each document separately & ensemble output probabilities
 - Given input x and top-K documents $\mathcal{D}'$, output probability of next token y is given as:

$$p(y \mid x, \mathcal{D}') = \sum_{d \in \mathcal{D}'} p(y \mid d \boxplus x) \cdot \lambda(d, x)$$

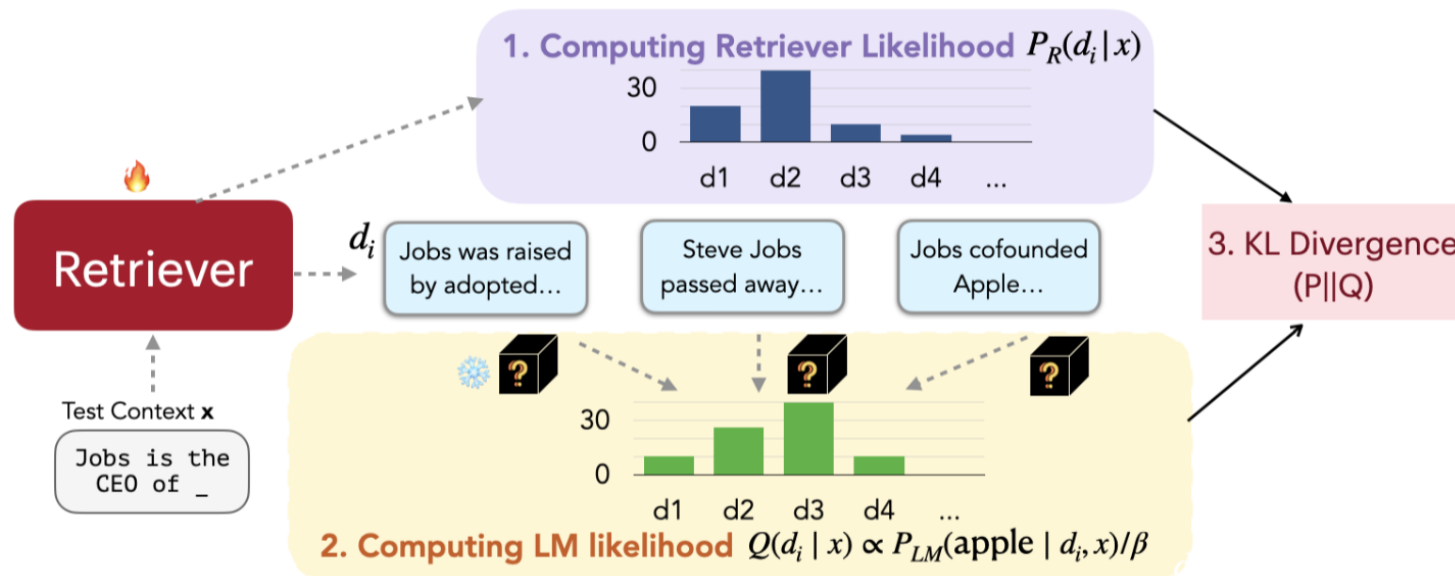
concatenation

- $\lambda(x, d)$ is a similarity score from the used retriever model

$$\lambda(d, x) = \frac{e^{s(d, x)}}{\sum_{d \in \mathcal{D}'} e^{s(d, x)}}$$

REPLUG: Fine-tuning Dense Retriever

- Challenge. **No trainable module** → Limited performance
- **Key idea**. Update retriever to be aligned with output prob. of LM
 - By using **LM-Supervised** signal for fine-tuning **Retriever (LSR)**



REPLUG: Fine-tuning Dense Retriever

- **Key idea.** Update retriever to be aligned with output prob. of LLM
 1. Computing **retrieval likelihood** for K documents:

$$P_R(d \mid x) = \frac{e^{s(d,x)/\gamma}}{\sum_{d \in \mathcal{D}} e^{s(d,x)/\gamma}}$$

REPLUG: Experiments

- Demonstration on **generic language modeling** (on Pile dataset)
 - Regardless of LM's capability, REPLUG successfully improves performance (BPB)
 - By fine-tuning retriever model with LSR, improvement is significantly increased

Model		# Parameters	Original	+ REPLUG	Gain %	+ REPLUG LSR	Gain %
GPT-2	Small	117M	1.33	1.26	5.3	1.21	9.0
	Medium	345M	1.20	1.14	5.0	1.11	7.5
	Large	774M	1.19	1.15	3.4	1.09	8.4
	XL	1.5B	1.16	1.09	6.0	1.07	7.8
GPT-3 (black-box)	Ada	350M	1.05	0.98	6.7	0.96	8.6
	Babbage	1.3B	0.95	0.90	5.3	0.88	7.4
	Curie	6.7B	0.88	0.85	3.4	0.82	6.8
	Davinci	175B	0.80	0.77	3.8	0.75	6.3

REPLUG: Experiments

- Results on MMLU and QA
 - For both tasks, REPLUG successfully improves target LMs

Model	# Parameters	All
Codex	175B	68.3
PaLM	540B	69.3
Flan-PaLM	540B	72.2
Atlas	11B	47.9
Codex + REPLUG	175B	71.4
Codex + REPLUG LSR	175B	71.8

Model	NQ		TQA	
	Few-shot	Full	Few-shot	Full
Chinchilla	35.5	-	64.6	-
PaLM	39.6	-	-	-
Codex	40.6	-	73.6	-
RETRO [†]	-	45.5	-	-
R2-D2 [†]	-	55.9	-	69.9
Atlas [†]	42.4	60.4	74.5	79.8
Codex + Contriever _{cc} ²	44.2	-	76.0	-
Codex + REPLUG	44.7	-	76.8	-
Codex + REPLUG LSR	45.5	-	77.3	-

(**Left**) Results on MMLU and (**right**) QA datasets ^{1]}

Motivation: Query Enhancement

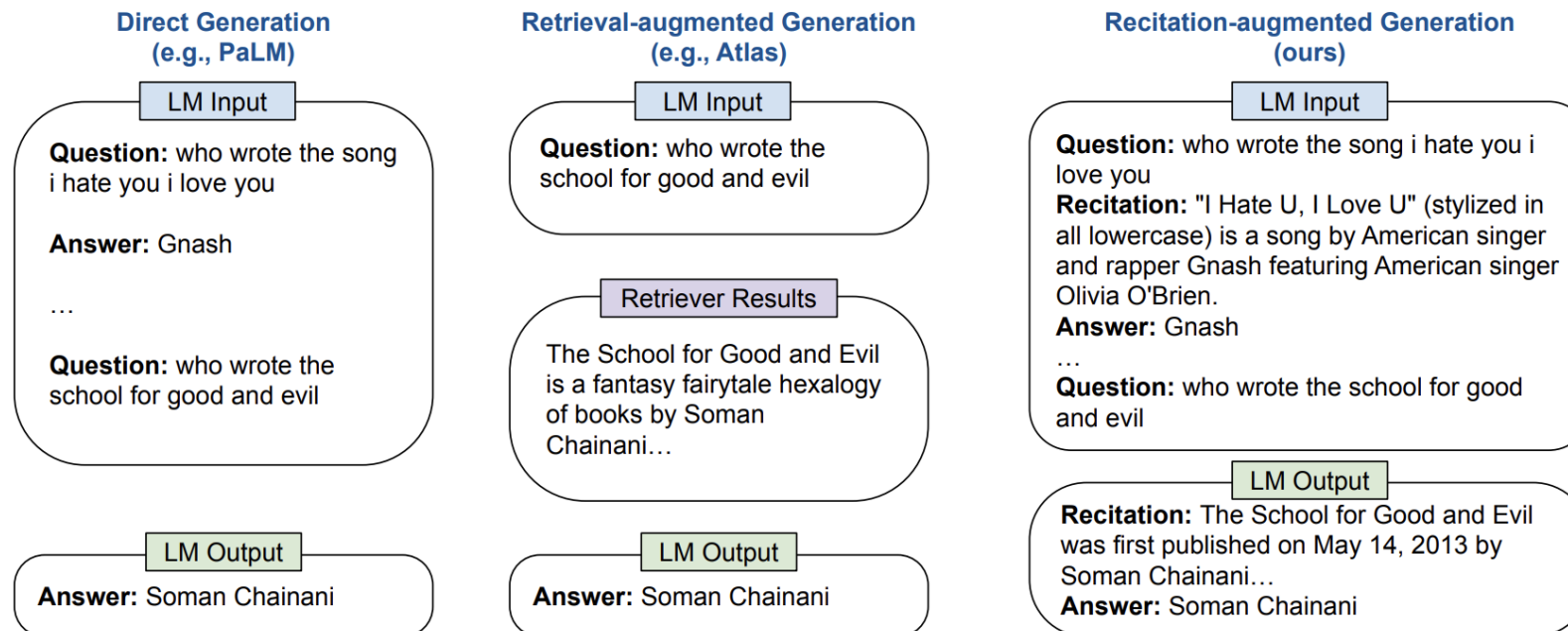
- Many retriever systems rely on **training with relevance labels**
 - E.g., Contriever^[1] also rely on labeled data for sufficient performance
 - Such dataset is often **unavailable** → Necessity of “**zero-shot retrieval**”

Table 5: **Cross-lingual retrieval on MKQA.** We report the average on all languages included in MKQA for the Recall@100 and the Recall@20, and report the Recall@100 for a subset of languages. Complete results are reported in Table 13 and Table 14 of appendix.

	Avg. R@20	Avg. R@100	en	ar	ja	ko	es	he	de
CORA (Asai et al., 2021)	49.0	59.8	75.6	44.5	47.0	45.5	69.2	48.3	68.1
mBERT + MS MARCO	45.3	57.9	74.2	44.0	51.7	48.2	63.9	46.8	59.6
XLM-R + MS MARCO	46.9	59.6	73.4	42.5	53.2	49.6	63.4	46.9	61.1
mContriever	31.4	49.2	65.3	43.0	47.1	44.8	37.2	44.7	49.0
+ MS MARCO	53.9	65.6	75.6	53.3	60.4	55.4	70.0	59.6	66.6

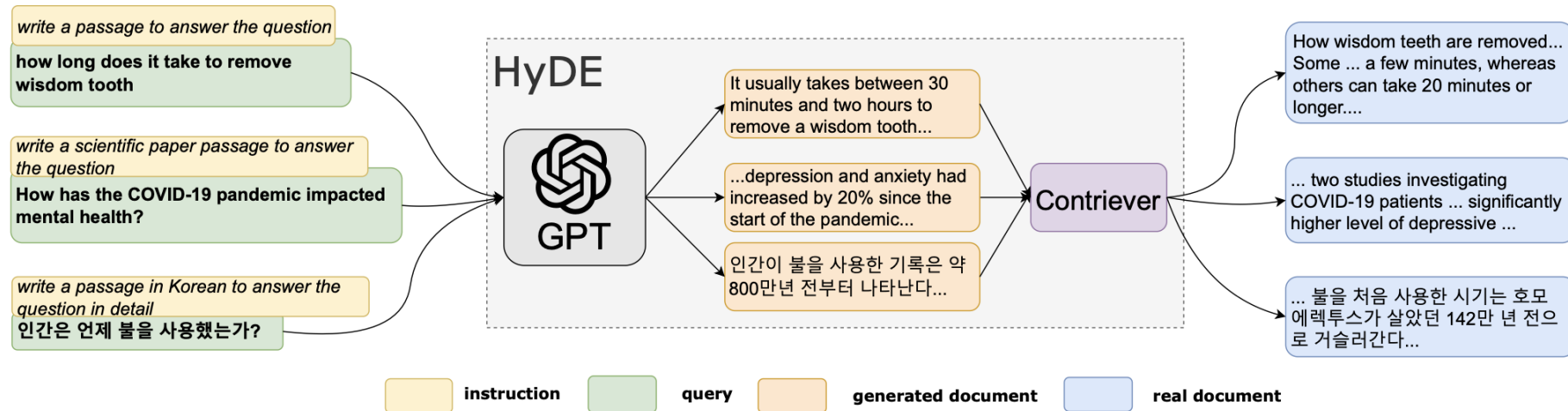
Motivation: Query Enhancement

- LLM can generate query-relevant context using learned knowledge^[1]
 - Ideally, if LLM memorize all things in web, we might not need RAG
 - One remaining challenge: “**Hallucination**”



HyDE: Query Enhancement with LLMs

- **Idea:** Incorporating query-relevant context from LLM
 - First, generating **synthetic document** by prompting LLMs,
 - Then, retrieving most relevant passages with existing **unsupervised retriever** (Contriever)



Concept Figure of Hypothetical Document Embeddings (HyDE)^[1]

HyDE: Query Enhancement with LLMs

- **Recap.** Dense retrieval models similarity with two encoders:

$$\text{sim}(q, p) = E_Q(q)^\top E_P(p)$$

- **Only passage encoder** $E_P(\cdot)$ is used, i.e., $\text{sim}(q, p) = E_P(\tilde{p}(q))^\top E_P(p)$
- To improve quality, **multiple documents** can be generated and used:

$$\text{sim}(q, p) = \left(\sum_{k=1, \dots, K} E_P(\tilde{p}_k(q)) / K \right)^\top E_P(p)$$

Results: Web Search

- Contriever **underperform** BM25, but HyDE **outperforms** it
 - HyDE **remains competitive** even when compared to fine-tuned models.

	DL19			DL20		
	map	ndcg@10	recall@1k	map	ndcg@10	recall@1k
<i>w/o relevance judgement</i>						
BM25	30.1	50.6	75.0	28.6	48.0	78.6
Contriever	24.0	44.5	74.6	24.0	42.1	75.4
HyDE	41.8	61.3	88.0	38.2	57.9	84.4
<i>w/ relevance judgement</i>						
DPR	36.5	62.2	76.9	41.8	65.3	81.4
ANCE	37.1	64.5	75.5	40.8	64.6	77.6
Contriever ^{FT}	41.7	62.1	83.6	43.6	63.2	85.8

Results: Low Resource Retrieval

- HyDE again brings **significant improvements** to Contriever
 - HyDE also **shows** strong performance compared to fine-tuned models

	Scifact	Arguana	Trec-Covid	FiQA	DBPedia	TREC-NEWS
<i>nDCG@10</i>						
<i>w/o relevance judgement</i>						
BM25	67.9	39.7	59.5	23.6	31.8	39.5
Contriever	64.9	37.9	27.3	24.5	29.2	34.8
HyDE	69.1	46.6	59.3	27.3	36.8	44.0
<i>w/ relevance judgement</i>						
DPR	31.8	17.5	33.2	29.5	26.3	16.1
ANCE	50.7	41.5	65.4	30.0	28.1	38.2
Contriever ^{FT}	67.7	44.6	59.6	32.9	41.3	42.8
<i>Recall@100</i>						
<i>w/o relevance judgement</i>						
BM25	92.5	93.2	49.8	54.0	46.8	44.7
Contriever	92.6	90.1	17.2	56.2	45.3	42.3
HyDE	96.4	97.9	41.4	62.1	47.2	50.9
<i>w/ relevance judgement</i>						
DPR	72.7	75.1	21.2	34.2	34.9	21.5
ANCE	81.6	93.7	45.7	58.1	31.9	39.8
Contriever ^{FT}	94.7	97.7	40.7	65.6	54.1	49.2

Beyond HyDE

- **Q.** Should we use Contriever for this framework?
 - **A:** No. BM25 is also applicable and we can improve it with better drafting
- Large language model as Retriever (**LameR**)^[1]
 - **Key idea:** Improving the initial generation of LLM using retriever (Initial retrieval → Generating initial answer → Finer retrieval)

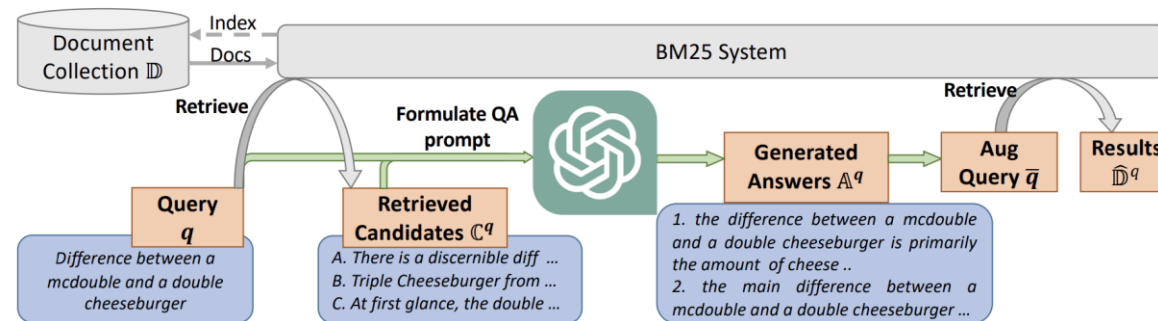


Figure 3: Large language model as Retriever (LameR). Please see Table 1 for the prompt formulation.

Beyond HyDE

- Large language model as Retriever (**LameR**)^[1]
 - **Key idea:** Improving the initial generation of LLM using retriever

	TREC Deep Learning 2019			TREC Deep Learning 2020		
	MAP	nDCG@10	R@1k	MAP	nDCG@10	R@1k
<i>w/o relevance judgment (zero-shot retrieval)</i>						
BM25	30.1	50.6	75.0	28.6	48.0	78.6
Contriever	24.0	44.5	74.6	24.0	42.1	75.4
HyDE	41.8	61.3	88.0	38.2	57.9	84.4
LameR (ours)	47.2	69.1	89.9	45.6	64.8	88.7
<i>w/ few-shot relevance judgment (few-shot ICL for answer generation)</i>						
Q2D _{BM25}	-	66.2	-	-	62.9	-
<i>w/ relevance judgment (fully-supervised fine-tuning)</i>						
DPR	36.5	62.2	76.9	41.8	65.3	81.4
ANCE	37.1	64.5	75.5	40.8	64.6	77.6
Contriever ^{FT}	41.7	62.1	83.6	43.6	63.2	85.8

Motivation: Noisy-robust RAG

- Noisy retrieval can negatively affect LLM performance^[1,2]
 - E.g., RAG can be even worsen than direct answering without retrieval

Q: Who is the actor playing Jason on general hospital?	
Large Language Model (no retrieval) 	Retrieval Augmented Language Model 
The answer is: Steve Burton 	E: Jason Gerhardt (born April 21, 1974) is an American actor. He is known for playing the role of Cooper Barrett in General Hospital and Zack Kilmer in Mistresses. The answer is: Jason Gerhardt 

[1] Petroni et al., How Context Affects Language Models' Factual Predictions., AKBC 2020

[2] Li et al., Large Language Models with Controllable Working Memory., ACL 2023

Motivation: Noisy-robust RAG

- Noisy retrieval can negatively affect LLM performance^[1,2]
 - E.g., RAG can be even worsen than direct answering without retrieval
- **Goal:** Retrieval-robust LLMs
 - When relevant, retrieved context **should improve** model performance
 - When irrelevant, retrieved context **should not hurt** model performance.

[1] Petroni et al., How Context Affects Language Models' Factual Predictions., AKBC 2020

[2] Li et al., Large Language Models with Controllable Working Memory., ACL 2023

RetRobust: Training Free

- **Key idea:** Using existing NLI (Natural Language Inference) models
 - Remark. SOTA model on NLI achieve strong performance (> 92%)

Natural language inference (NLI) is the task of determining whether a "hypothesis" is true (entailment), false (contradiction), or undetermined (neutral) given a "premise".

Example:

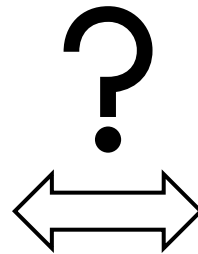
Premise	Hypothesis	Label
A man inspects the uniform of a figure in some East Asian country.	The man is sleeping.	contradiction
An older and younger man smiling.	Two men are smiling and laughing at the cats playing on the floor.	neutral
A soccer game with multiple males playing.	Some men are playing a sport.	entailment

RetRobust: Training Free

- **Key idea:** Using existing NLI (Natural Language Inference) models
 - In RAG, retrieved passage → **premise**, question and answer → **hypothesis**
 - Namely, correct retrieval should be entailed with question and answer
 - If not entailed, direct answering without retrieval would be used instead

Q: Who is the actor playing Jason on general hospital?

The answer is: Steve Burton



E: Jason Gerhardt (born April 21, 1974) is an American actor. He is known for playing the role of Cooper Barrett in General Hospital and Zack Kilmer in Mistresses.

RetRobust: Small Training

- **Key idea:** Training LLM to ignore irrelevant contexts with small data
 - Previous approach could be too strict and discards relevant ones as well
 - Namely, interest is collecting relevant/irrelevant passages for given query
- *Q. How to construct dataset?*
 - Relevant passage: Top-1 passage with given retriever
 - Irrelevant passages: (1) Bottom-K for given query or (2) Top-K for other queries

RetRobust: Small Training

- **Key idea:** Training LLM to ignore irrelevant contexts with small data
 - Previous approach could be too strict and discards relevant ones as well
 - Namely, interest is collecting relevant/irrelevant passages for given query
- Then, model is fine-tuned to maximize likelihood of answer
 - Namely, output correct prediction even under wrong retrieved passages

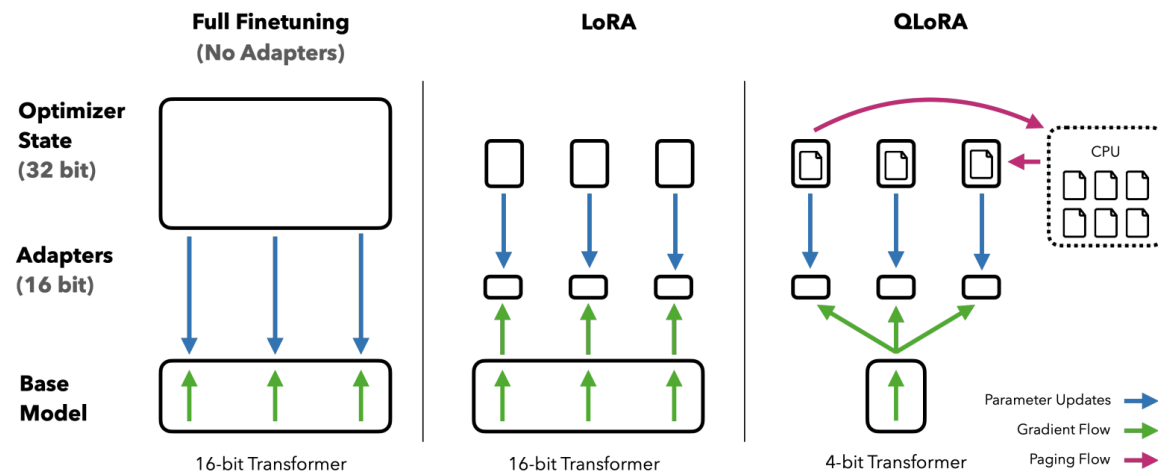
RetRobust: Setups

- **Tasks:** 3 different types of QA
 - Single-hop QA: (1) Natural Question (NQ)
 - Multi-hop QA with explicit reasoning: (2) 2WikiMQA and (3) Bamboogle
 - Multi-hop QA with commonsense: (4) StrategyQA and (5) Fermi

Dataset	Type	Example
NQ	Single-hop	What episode of law and order svu is mike tyson in?
2WIKIMQA	Explicit	Where was the place of death of Isabella Of Bourbon's father?
BAMBOOGLE	Explicit	What is the maximum airspeed (in km/h) of the third fastest bird?
STRATEGYQA	Implicit	Can Arnold Schwarzenegger deadlift an adult Black rhinoceros?
FERMI	Implicit	How many high fives has Lebron James given/received?

RetRobust: Setups

- **Tasks:** 3 different types of QA
- **Training:** 1k for NQ and 500 for others
 - Instead of full fine-tuning of LLaMA2-13B, QLoRA^[1] is used
 - For NLI model, BART-large fine-tuned on MNLI dataset is used

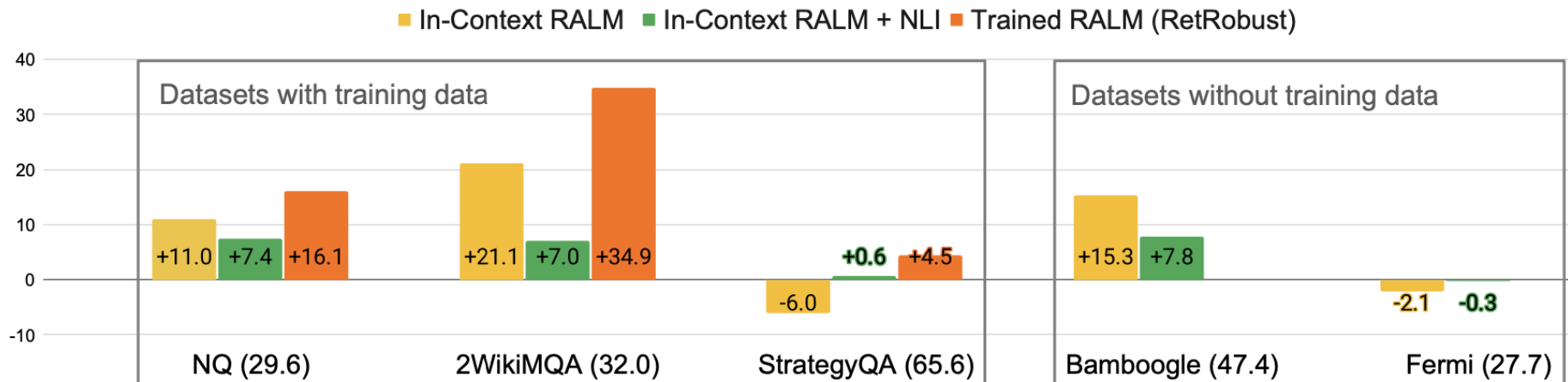


RetRobust: Setups

- **Tasks:** 3 different types of QA
- **Training:** 1k for NQ and 500 for others
- **Inference:** Google search for retriever & 4-6 few-shot examples

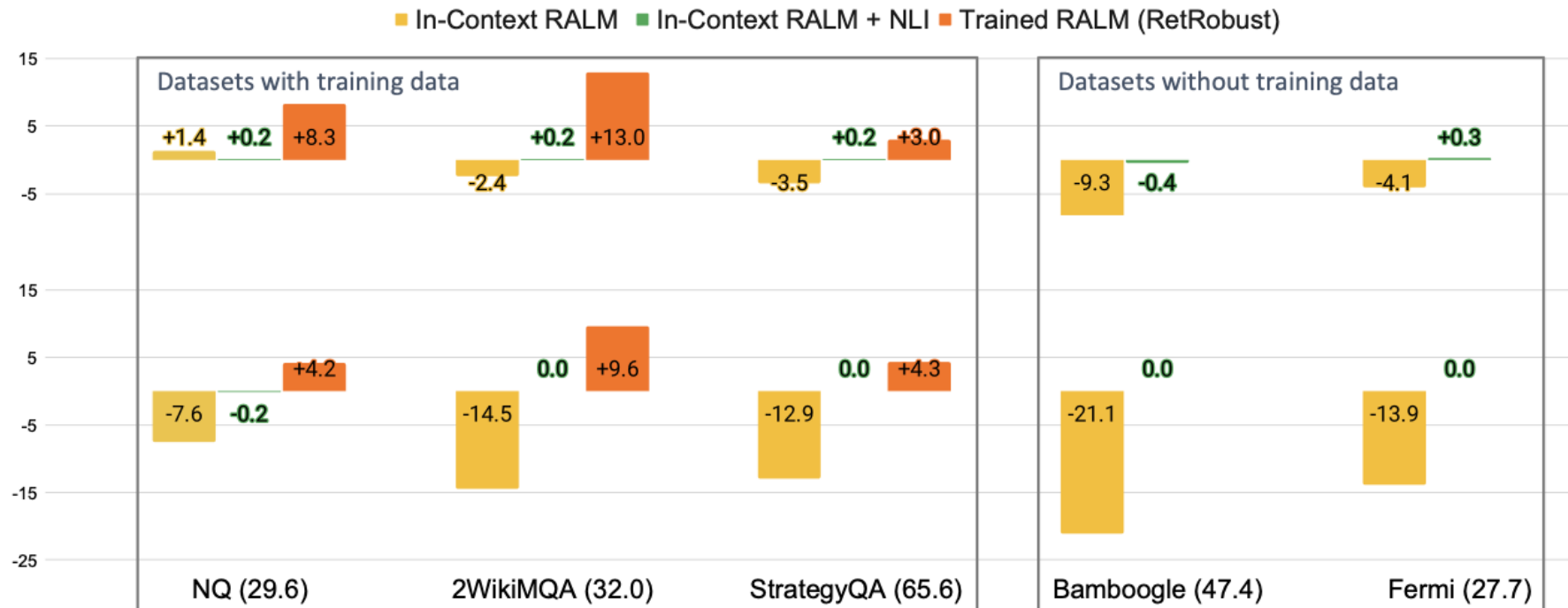
RetRobust: Results

- QA with Top-1 retrieval
 - Incorporating NLI model successfully prevent decrease of performance
 - Additional training further enlarge improvements



RetRobust: Results

- QA with low-rank retrieval (top) and random retrieval (bottom)
 - For both scenarios, RetRobust are effective to mitigate risk of retrieval



Motivation: Learning to Address Issues of RAG

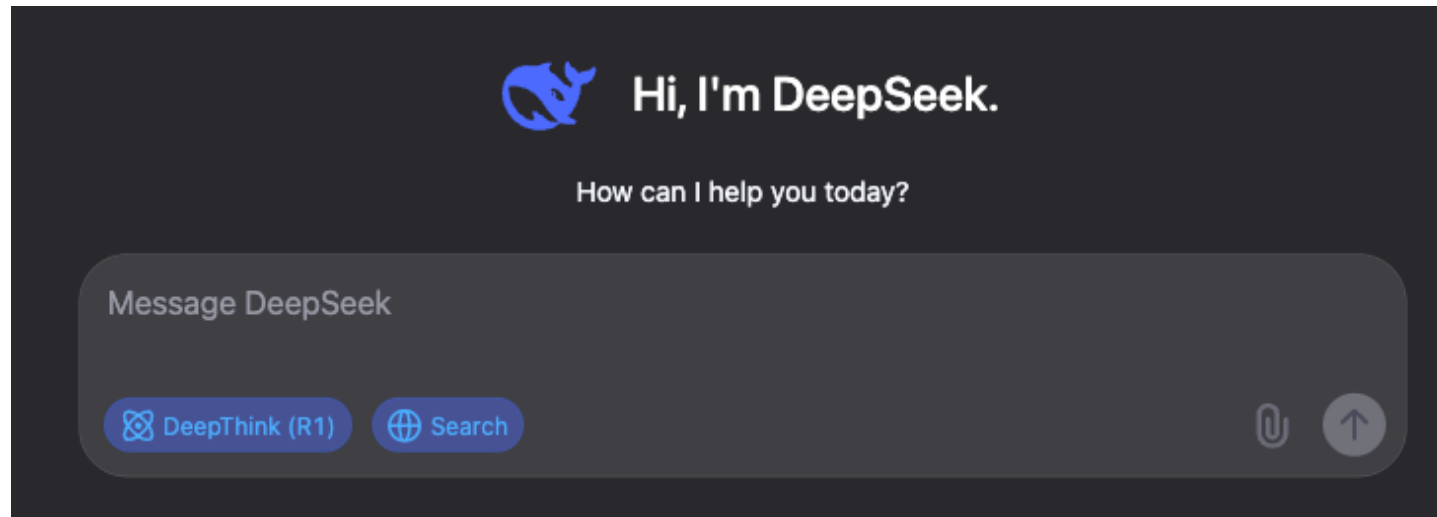
- Previously, we observe several issues in RAG
 1. RAG should be **adaptively applied** depending on query
 2. Even LLM outputs with RAG can be **not grounded by retrieved documents**
- **Motivation.** Can we directly address these issues via additional training?
 - *Remark.* **Training is most direct way** to resolve the issues of deep learning

Contents

- Retrieval system
- Recent approaches for improving RAG at Inference-level
 - Handling long input context
 - Better input query
 - Robustness to noisy document
- Recent approaches for improving RAG at Training-level

Learn Adaptive RAG via RL

- Self-RAG^[1] uses SFT to learn LLM of adaptive RAG
 - Q. Can it be improved utilizing reinforcement learning, similar to RLHF^[2]?
 - A. Yes (maybe also adopted by recent companies such as DeepSeek^[3])



[1] Asai et al., Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection., ICLR 2024 Oral

[2] Ouyang et al., Training Language Models to Follow Instructions with Human Feedback., NeurIPS 2022

[3] <https://chat.deepseek.com/>

Search-R1

- Search-R1^[1] is open-sourced framework that apply RL to learn adaptive RAG
 - As implied in the name, it mainly follows DeepSeek-R1 framework^[2]

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think>` `</think>` and `<answer>` `</answer>` tags, respectively, i.e., `<think>` reasoning process here `</think>` `<answer>` answer here `</answer>`. User: **prompt**. Assistant:

- **Accuracy rewards:** The accuracy reward model evaluates whether the response is correct. For example, in the case of math problems with deterministic results, the model is required to provide the final answer in a specified format (e.g., within a box), enabling reliable rule-based verification of correctness. Similarly, for LeetCode problems, a compiler can be used to generate feedback based on predefined test cases.
- **Format rewards:** In addition to the accuracy reward model, we employ a format reward model that enforces the model to put its thinking process between '`<think>`' and '`</think>`' tags.

[1] Jin et al., Search-R1: Training LLMs to Reason and Leverage Search Engines with Reinforcement Learning., arXiv:25.03

[2] DeepSeek-AI, DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning., arXiv:25.01

Search-R1

- Key idea of Search-R1: Incorporating RAG as output texts with **special tokens**
 1. *Reasoning tokens*: `<think>` internal reasoning `</think>`
 2. *Search call tokens*: `<search>` query `</search>`
 3. *Search results tokens*: `<information>` retrieved documents `</information>`
 4. *Answer tokens*: `<answer>` answer `</answer>`

Answer the given question. You must conduct reasoning inside `<think>` and `</think>` first every time you get new information. After reasoning, if you find you lack some knowledge, you can call a search engine by `<search>` query `</search>`, and it will return the top searched results between `<information>` and `</information>`. You can search as many times as you want. If you find no further external knowledge needed, you can directly provide the answer inside `<answer>` and `</answer>` without detailed illustrations. For example, `<answer>` Beijing `</answer>`. Question: `question`.

Search-R1

- Example of answering with Search-R1

Question: Curious is a women's fragrance by a singer born in what city and state?

Ground Truth: McComb, Mississippi

R1:

<think> The singer is Beyoncé, who was born in Houston, Texas. **</think>**

<answer> Houston **</answer>**

SEARCH-R1:

<think> I need to find out which city and state a singer, by whom the women's fragrance "Curious" was created, was born in. **</think>**

<search> Curious fragrance information **</search>**

<information> Doc 1(Title: "Curious (fragrance)") Curious (fragrance) Curious is a women's fragrance by Britney Spears for Elizabeth Arden, and is the first perfume to be endorsed by Britney Spears, preceding ""Fantasy."" Spears reportedly earned \$52 million from the endorsement. ""Curious"" was released in September 2004, and was very successful internationally. It was the number one fragrance of 2004 in department stores, and in 2005, ""Curious"" was honored by the Fragrance Foundation as Best Women's Fragrance. The fragrance is a white floral scent. Its notes are Louisiana magnolia, golden Anjou pear, lotus flower, tuberose, star jasmine, pink cyclamen, vanilla-infused musk, sandalwood and blonde woods. Elizabeth Arden ... **</information>**

Search-R1

- Example of answering with Search-R1
 - Interleaved CoT & retrieval is automatically emerged during RL!

```
vanilla-infused musk, sandalwood and blonde woods. Elizabeth Arden ... </information>
<think> I need to find out which city and state Britney Spears, the singer behind "Curious"
fragrance, was born in. </think>
<search> Britney Spears birthplace </search>
<information> Doc 1(Title: "Britney Spears") the best-selling female albums artist of the 2000s. In
2004, she launched a perfume brand with Elizabeth Arden, Inc., from which sales exceeded US$1.5
billion, . Spears was born in McComb, Mississippi, the second child of Lynne Irene Bridges and
James Parnell Spears. Her maternal grandmother, Lillian Portell, was English (born in London),
and one of Spears's maternal great-great-grandfathers was Maltese. Her siblings are Bryan James
and Jamie Lynn. Britney was born in the Bible Belt, where socially conservative evangelical
Protestantism is a particularly strong religious influence. Spears was baptized into the Southern
Baptist Convention, but later in life studied </information>
<think> Since Britney Spears was born in McComb, Mississippi, I need to find out which city and
state that is. </think>
<search> McComb, Mississippi location </search>
<information> Doc 1(Title: "McComb, Mississippi") McComb, Mississippi McComb is a city in
Pike County, Mississippi, United States, approximately south of Jackson. As of the 2010 census,
the city had a total population of 12,790. It is the principal city of the McComb, Mississippi
Micropolitan Statistical Area. McComb was founded in 1872 after Henry Simpson McComb of the
New Orleans, Jackson and Great Northern Railroad, a predecessor of the Illinois Central Railroad
(now part of the Canadian National Railway), decided to move the railroad's maintenance shops
away from New Orleans, Louisiana, to avoid the attractions of that city's saloons. The railroad
purchased land in Pike County, </information>
<think> I have found that Britney Spears was born in McComb, Mississippi. Since the question
asks for the city and state, I just need to confirm the state, which I already know from the previous
search. </think>
<answer> McComb, Mississippi </answer>
```

Search-R1

- *How to reward?* **Simple outcome reward** based on ground-truth answer
 - No format rewards, as it uses LLMs already demonstrates strong structural adherence

$$r_{\phi}(x, y) = \text{EM}(a_{\text{pred}}, a_{\text{gold}}),$$

Search-R1

- *How to reward?* **Simple outcome reward** based on ground-truth answer
- *How to train?* Existing **RL algorithms** such as PPO and GRPO

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot | x; \mathcal{R})} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta}(y | x; \mathcal{R}) || \pi_{\text{ref}}(y | x; \mathcal{R})],$$

*y can be from either **generation by LLM** or **retrieval R***

Question: Curious is a women's fragrance by a singer born in what city and state?

Ground Truth: McComb, Mississippi

SEARCH-R1:

<think> I need to find out which city and state a singer, by whom the women's fragrance "Curious" was created, was born in. **</think>**

<search> Curious fragrance information **</search>**

<information> Doc 1(Title: "Curious (fragrance)") Curious (fragrance) Curious is a women's fragrance by Britney Spears for Elizabeth Arden, and is the first perfume to be endorsed by Britney Spears, preceding ""Fantasy."" Spears reportedly earned \$52 million from the endorsement. ""Curious"" was released in September 2004, and was very successful internationally. It was the number one fragrance of 2004 in department stores, and in 2005, ""Curious"" was honored by the Fragrance Foundation as Best Women's Fragrance. The fragrance is a white floral scent. Its notes are Louisiana magnolia, golden Anjou pear, lotus flower, tuberose, star jasmine, pink cyclamen, vanilla-infused musk, sandalwood and blonde woods. Elizabeth Arden ... **</information>**

generation by LLM

*retrieval R
(i.e., not generated by LLM)*

Search-R1

- *How to reward?* **Simple outcome reward** based on ground-truth answer
- *How to train?* Existing **RL algorithms** such as PPO and GRPO

$$\max_{\pi_{\theta}} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_{\theta}(\cdot | x; \mathcal{R})} [r_{\phi}(x, y)] - \beta \mathbb{D}_{\text{KL}}[\pi_{\theta}(y | x; \mathcal{R}) || \pi_{\text{ref}}(y | x; \mathcal{R})],$$

*y can be from either **generation by LLM** or **retrieval R***

Question: Curious is a women's fragrance by a singer born in what city and state?

Ground Truth: McComb, Mississippi

SEARCH-R1:

<think> I need to find out which city and state a singer, by whom the women's fragrance "Curious" was created, was born in. **</think>**

<search> Curious fragrance information **</search>**

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*Only update
for here!*

*Removing from
loss calculation*

Search-R1

- Inference algorithm of Search-R1

Algorithm 1 LLM Response Rollout with Multi-Turn Search Engine Calls

Require: Input query x , policy model π_θ , search engine $\mathcal{R}$, maximum search budget B .

Ensure: Final response y .

```
1: Initialize rollout sequence  $y \leftarrow \emptyset$ 
2: Initialize search call count  $b \leftarrow 0$ 
3: while  $b < B$  do
4:   Generate response token  $y_t \sim \pi_\theta(\cdot \mid x, y)$ 
5:   // Append  $y_t$  to rollout sequence  $y$ 
6:    $y \leftarrow y + y_t$ 
7:   if <search> </search> detected in  $y_t$  then
8:     // Extract search query  $q$ 
9:      $q \leftarrow \text{Parse}(y_t, \text{<search>, </search>})$ 
10:    // Retrieve search results
11:     $d = \mathcal{R}(q)$ 
12:    // Insert  $d$  into  $y$ 
13:     $y \leftarrow y + \text{<information>}d\text{</information>}$ 
14:    Increment search call count  $b \leftarrow b + 1$ 
15:  end if
16:  if <answer> </answer> detected in  $y$  then
17:    // Terminate rollout
18:    return final generated response  $y$ 
19:  end if
20: end while
21: return final generated response  $y$ 
```

Search-R1: Experiments

- Setups

We conduct experiments using three types of models: Qwen-2.5-3B (Base/Instruct) and Qwen-2.5-7B (Base/Instruct) (Yang et al., 2024), as well as Llama-3.2-3B (Base/Instruct) (Dubey et al., 2024). For retrieval, we use the 2018 Wikipedia dump (Karpukhin et al., 2020) as the knowledge source and E5 (Wang et al., 2022) as the retriever. To ensure fair comparison, we follow Lin et al. (2023) and set the number of retrieved passages to three across all retrieval-based methods.

For training, we merge the training sets of NQ and HotpotQA to form a unified dataset for SEARCH-R1 and other fine-tuning-based baselines. Evaluation is conducted on the test or validation sets of all seven datasets to assess both in-domain and out-of-domain performance. Exact Match (EM) is used as the evaluation metric, following Yu et al. (2024). For inference-style baselines, we use instruct models, as base models fail to follow instructions. For RL tuning methods, experiments are conducted on both base and instruct models.

- Evaluation QA Datasets

General Question Answering: NQ (Kwiatkowski et al., 2019), TriviaQA (Joshi et al., 2017), and PopQA (Mallen et al., 2022).

Multi-Hop Question Answering: HotpotQA (Yang et al., 2018), 2WikiMultiHopQA (Ho et al., 2020), Musique (Trivedi et al., 2022b), and Bamboogle (Press et al., 2022).

Search-R1: Experiments

- Baselines

Inference without Retrieval: Direct inference and Chain-of-Thought (CoT) reasoning (Wei et al., 2022).

Inference with Retrieval: Retrieval-Augmented Generation (RAG) (Lewis et al., 2020), IRCoT (Trivedi et al., 2022a), and Search-o1 (Li et al., 2025).

Fine-Tuning-Based Methods: Supervised fine-tuning (SFT) (Chung et al., 2024) and reinforcement learning-based fine-tuning without a search engine (R1) (Guo et al., 2025). For R1, we train the LLMs with the RL methods proposed in Guo et al. (2025) with our data to have a fair comparison with SEARCH-R1. It only contains reasoning and answer steps and cannot call a search engine.

Search-R1: Experiments

- Main results

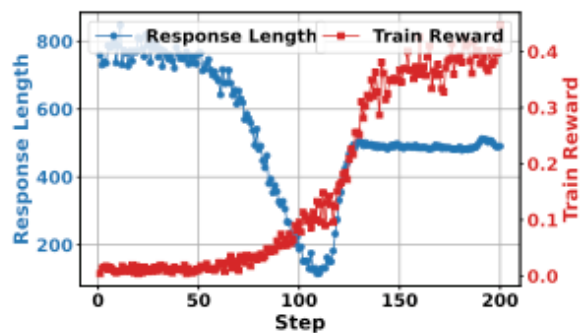
Method	NQ	TriviaQA	PopQA	HotpotQA	2wiki	Musique	Bamboogle	Avg.
Qwen2.5-7b-Base/Instruct								
Direct Inference	0.134	0.408	0.140	0.183	0.250	0.031	0.120	0.181
CoT	0.048	0.185	0.054	0.092	0.111	0.022	0.232	0.106
IRCoT	0.224	0.478	0.301	0.133	0.149	0.072	0.224	0.239
Search-o1	0.151	0.443	0.131	0.187	0.176	0.058	0.296	0.206
RAG	0.349	<u>0.585</u>	0.392	0.299	0.235	0.058	0.208	0.304
SFT	0.318	0.354	0.121	0.217	0.259	0.066	0.112	0.207
R1-base	0.297	0.539	0.202	0.242	0.273	0.083	0.296	0.276
R1-instruct	0.270	0.537	0.199	0.237	0.292	0.072	0.293	0.271
Search-R1-base	0.412	0.568	0.428	<u>0.356</u>	<u>0.322</u>	<u>0.142</u>	<u>0.384</u>	<u>0.373</u>
Search-R1-instruct	<u>0.397</u>	0.606	<u>0.404</u>	0.380	0.326	0.168	0.408	0.384
Qwen2.5-3b-Base/Instruct								
Direct Inference	0.106	0.288	0.108	0.149	0.244	0.020	0.024	0.134
CoT	0.023	0.032	0.005	0.021	0.021	0.002	0.000	0.015
IRCoT	0.111	0.312	0.200	0.164	0.171	0.067	0.240	0.181
Search-o1	0.238	0.472	0.262	0.221	0.218	0.054	0.320	0.255
RAG	0.348	0.544	0.387	0.255	0.226	0.047	0.080	0.270
SFT	0.249	0.292	0.104	0.186	0.248	0.044	0.112	0.176
R1-base	0.226	0.455	0.173	0.201	0.268	0.055	0.224	0.229
R1-instruct	0.210	0.449	0.171	0.208	<u>0.275</u>	0.060	0.192	0.224
Search-R1-base	<u>0.341</u>	0.513	0.362	<u>0.263</u>	0.273	<u>0.076</u>	0.211	<u>0.292</u>
Search-R1-instruct	0.323	<u>0.537</u>	<u>0.364</u>	0.308	0.336	0.105	<u>0.315</u>	0.327

Search-R1: Experiments

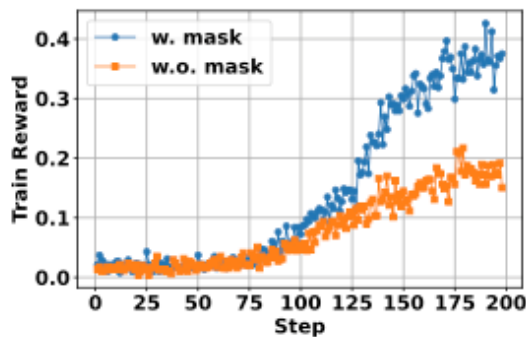
- Ablation #1: PPO v.s. GRPO

Method	NQ	TriviaQA	PopQA	HotpotQA	2wiki	Musique	Bamboogle	Avg.
Qwen2.5-3b-Base/Instruct								
SEARCH-R1-base (GRPO)	0.396	0.582	0.390	0.283	0.266	0.054	0.113	0.298
SEARCH-R1-instruct (GRPO)	0.409	0.552	0.405	0.345	0.369	0.154	0.320	0.365
SEARCH-R1-base (PPO)	0.341	0.513	0.362	0.263	0.273	0.076	0.211	0.292
SEARCH-R1-instruct (PPO)	0.323	0.537	0.364	0.308	0.336	0.105	0.315	0.327
LLaMA3.2-3b-Base/Instruct								
SEARCH-R1-base (GRPO)	0.431	0.612	0.458	0.300	0.297	0.067	0.104	0.324
SEARCH-R1-instruct (GRPO)	0.333	0.524	0.329	0.229	0.190	0.047	0.192	0.263
SEARCH-R1-base (PPO)	0.394	0.596	0.437	0.280	0.264	0.056	0.105	0.305
SEARCH-R1-instruct (PPO)	0.357	0.578	0.378	0.314	0.233	0.090	0.306	0.322

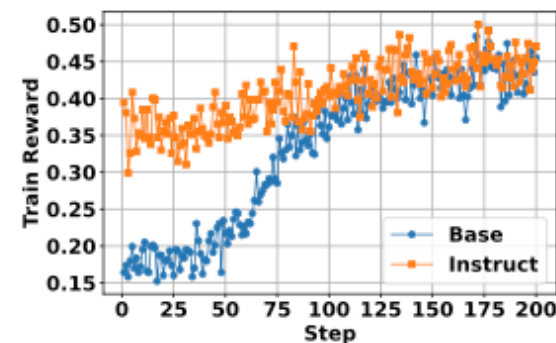
- Ablation #2: Training dynamics (output length, reward under mask & Instruct)



(a) Response length study



(b) Loss mask dynamics study



(c) Qwen2.5-7b-base/instruct

Evaluation of Large Language Models

Instructor: Jaehyung Kim

jaehyungk@yonsei.ac.kr

Evaluation

- For any system, **its evaluation** is important for successful deployment
- 3 components in evaluation
 1. **Target task:** *what to be done* by system
 2. **Evaluation method:** *which way to be used* to evaluate system
 3. **Evaluation metric:** *how to measure success* of system

Evaluation

- Example of system: **delivery service**



- Then, 3 components for its evaluation could be
 1. **Target task:** *deliver food to user from restaurant*
 2. **Evaluation method:** *measure delivery time*
 3. **Evaluation metric:** *average delivery time across users*

Evaluation of Deep Learning Model

- For deep learning system, evaluation usually relies on “**test data**”
 - **Def.** Test data: data for **same task and distribution**, but **never seen during training**
- **Example:** fine-tuned LM (e.g., BERT^[1]) for sentiment classification

“Overall, the value I got from the two hours watching it was the sum total of the popcorn and the drink. The movie was terrible.”

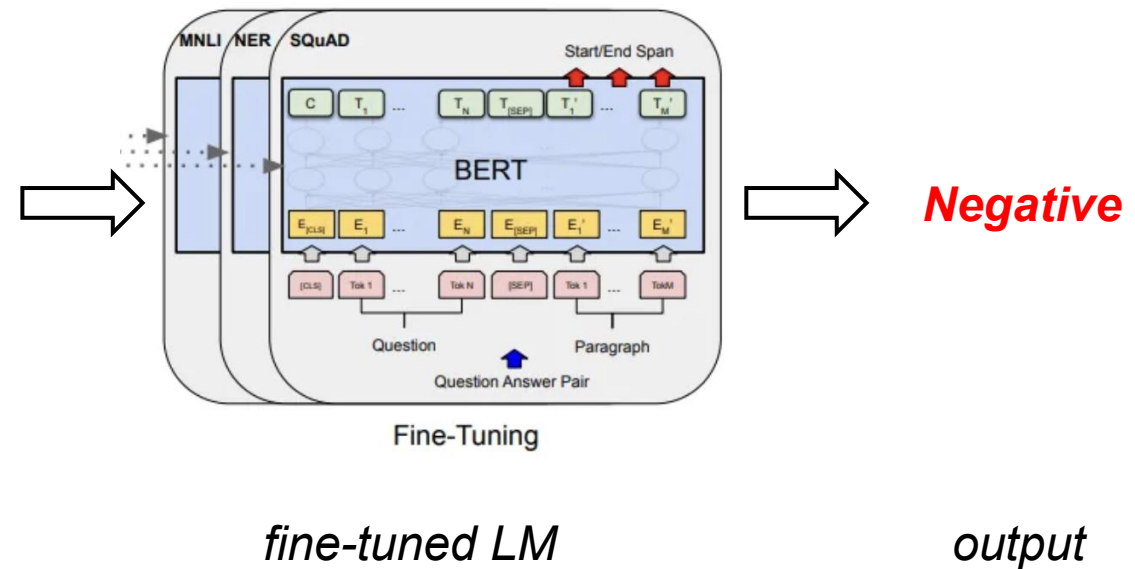


Illustration of BERT pre-training/fine-tuning framework^[1]

Evaluation of Deep Learning Model

- For deep learning system, evaluation usually relies on “**test data**”
 - **Def.** Test data: data for **same task and distribution**, but **never seen during training**
- **Example:** fine-tuned LM (e.g., BERT^[1]) for sentiment classification
 - **Target task:** sentiment classification
 - **Evaluation method:** comparing prediction on test data with human label
 - **Evaluation metric:** average accuracy

Contents

- Overview
- Evaluation of LLM with ground-truth answer
- Evaluation of LLM without ground-truth answer

Contents

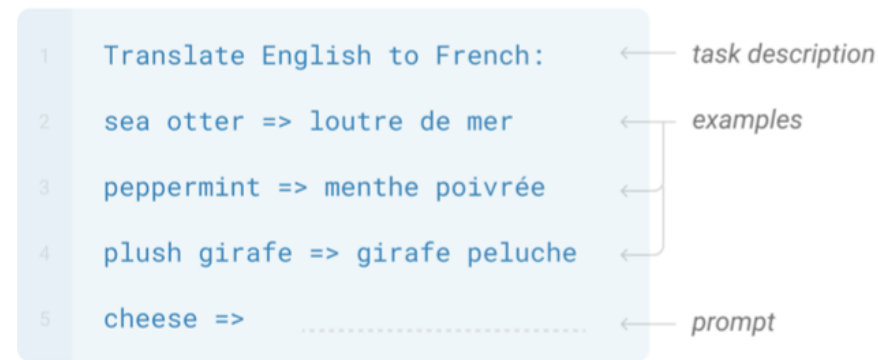
- Overview
- Evaluation of LLM with ground-truth answer
- Evaluation of LLM without ground-truth answer

Evaluation of LLM

- LLMs are trained on massive text data of **various tasks**
 - E.g., question answering (QA), summarization, code, etc.
- Consequently, LLM can be adapted to **any task** with **instruction & examples**

Few-shot

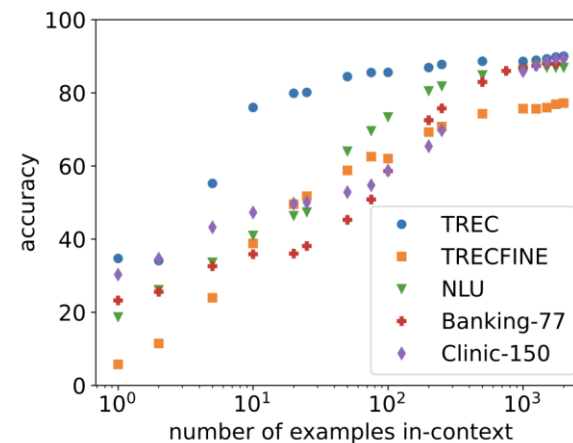
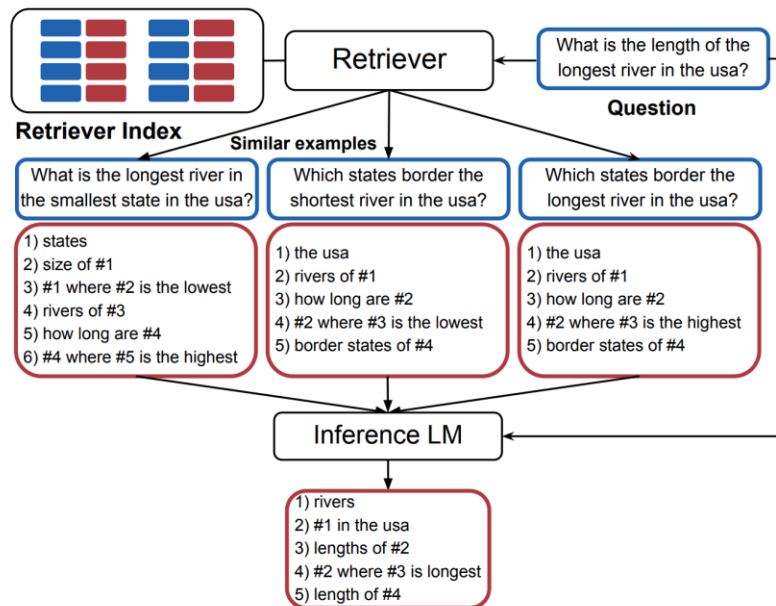
In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.



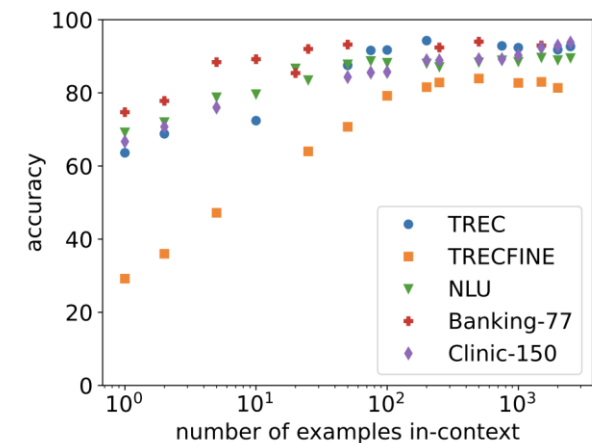
In-context learning (few-shot prompting) with LLMs^[1]

Evaluation of LLM

- Two important components: **(1) input prompt**
 - Detailed instruction + better^[1,2] & more^[3] examples → Better performance



(a) Using randomly selected examples.



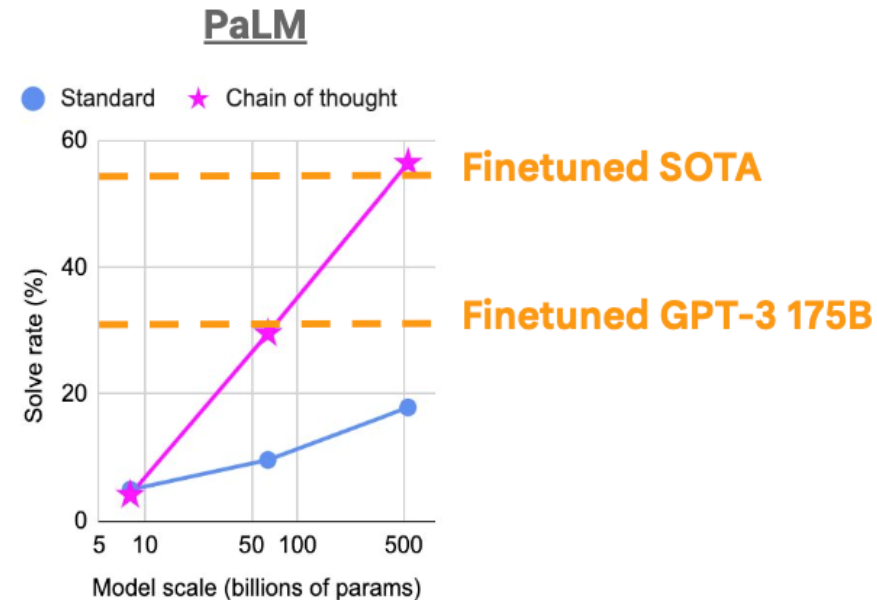
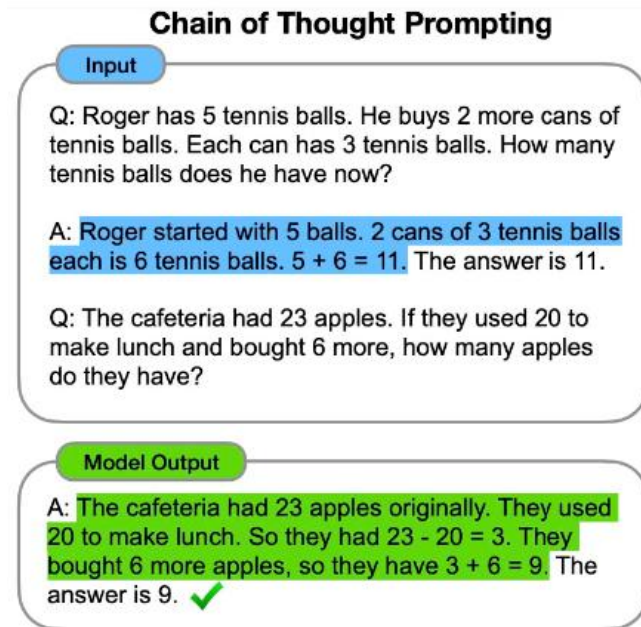
(b) Using retrieved examples.

In-context learning (few-shot prompting) with LLMs^[1]

[1] Zhang et al., Active Example Selection for In-Context Learning., EMNLP 2022
[2] Rubin et al., Learning To Retrieve Prompts for In-Context Learning., NAACL 2022
[3] Bertsch et al., In-Context Learning with Long-Context Models: An In-Depth Exploration., arXiv:24.05

Evaluation of LLM

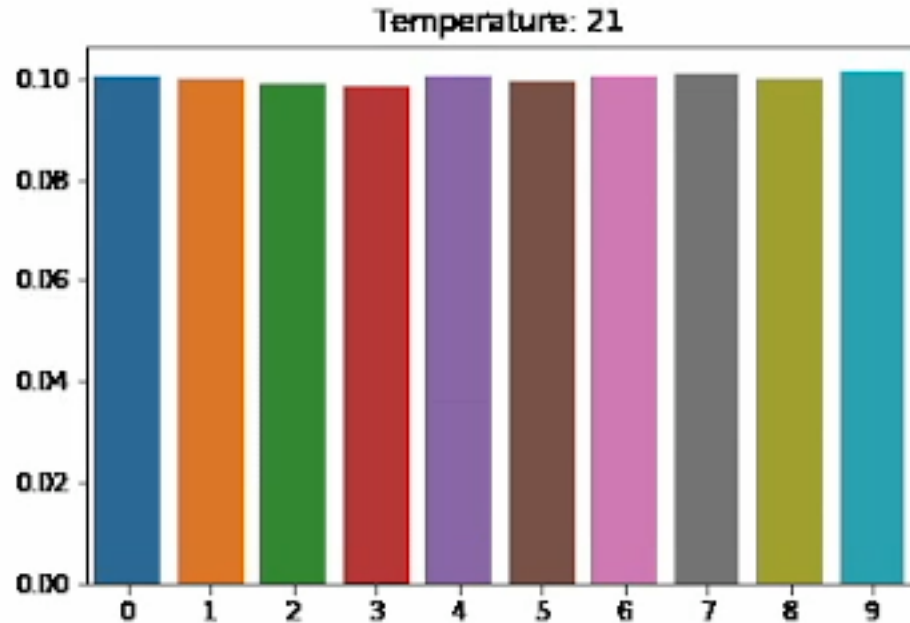
- Two important components: **(1) input prompt**
 - Detailed instruction & more examples → Better performance
 - **Recap.** Chain-of-Thought (CoT) prompting is crucial to solve complex reasoning tasks



In-context learning (few-shot prompting) with LLMs^[1]

Evaluation of LLM

- Two important components: (1) input prompt & (2) decoding method
 - **Caution.** With temperature > 0, LLM can sample different answer for same question

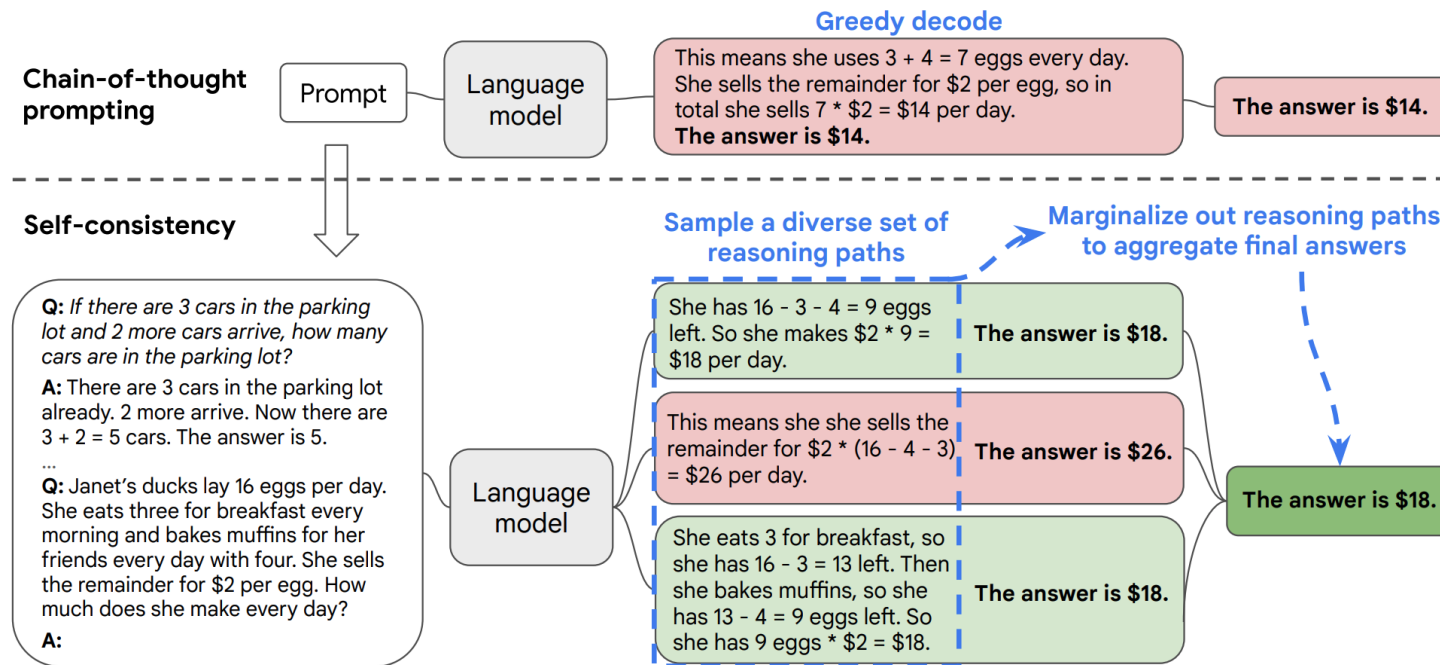


A screenshot of a configuration interface for an LLM. The "Model" section shows "gpt-4o" selected. The "Functions" section has a "+ Add" button. The "Model configuration" section shows "Response format" set to "text". The "Temperature" slider is set to 1.00, and the "Maximum Tokens" slider is set to 2048. The "Temperature" and "Maximum Tokens" sliders are highlighted with a yellow dashed box.

Temperature can be controlled by user

Evaluation of LLM

- Two important components: (1) input prompt & (2) decoding method
 - **Caution**. With temperature > 0, LLM can sample different answer for same question
 - **Self-consistency**^[1]: sample multiple prediction & take majority voting



Concept Figure of Self-consistency^[1]

Evaluation of LLM

- LLMs are trained on massive text data of **various tasks**
 - E.g., question answering (QA), summarization, code, etc.
- Consequently, LLM can be adapted to **any task** with **instruction & examples**
→ **Cons. Make difficult to evaluate LLM** as we should test multiple tasks together

Evaluation of LLM

- Example: **GPT-4** by OpenAI^[1]

	GPT-4 Evaluated few-shot	GPT-3.5 Evaluated few-shot	LM SOTA Best external LM evaluated few-shot	SOTA Best external model (incl. benchmark-specific tuning)
MMLU [49] Multiple-choice questions in 57 subjects (professional & academic)	86.4% 5-shot	70.0% 5-shot	70.7% 5-shot U-PaLM [50]	75.2% 5-shot Flan-PaLM [51]
HellaSwag [52] Commonsense reasoning around everyday events	95.3% 10-shot	85.5% 10-shot	84.2% LLaMA (validation set) [28]	85.6 ALUM [53]
AI2 Reasoning Challenge (ARC) [54] Grade-school multiple choice science questions. Challenge-set.	96.3% 25-shot	85.2% 25-shot	85.2% 8-shot PaLM [55]	86.5% ST-MOE [18]
WinoGrande [56] Commonsense reasoning around pronoun resolution	87.5% 5-shot	81.6% 5-shot	85.1% 5-shot PaLM [3]	85.1% 5-shot PaLM [3]
HumanEval [43] Python coding tasks	67.0% 0-shot	48.1% 0-shot	26.2% 0-shot PaLM [3]	65.8% CodeT + GPT-3.5 [57]
DROP [58] (F1 score) Reading comprehension & arithmetic.	80.9 3-shot	64.1 3-shot	70.8 1-shot PaLM [3]	88.4 QDGAT [59]
GSM-8K [60] Grade-school mathematics questions	92.0% * 5-shot chain-of-thought	57.1% 5-shot	58.8% 8-shot Minerva [61]	87.3% Chinchilla + SFT+ORM-RL, ORM reranking [62]

Performance of GPT-4 on academic benchmarks^[1]

Evaluation of LLM

- Example: **GPT-4** by OpenAI^[1]

<i>Target Tasks</i>	GPT-4 Evaluated few-shot	GPT-3.5 Evaluated few-shot	LM SOTA Best external LM evaluated few-shot	SOTA Best external model (incl. benchmark-specific tuning)
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**Evaluation
Methods**

Performance of GPT-4 on academic benchmarks^[1]

Evaluation of LLM

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**Evaluation
Metric**

Performance of GPT-4 on academic benchmarks^[1]

Evaluation of LLM

- Example: **Claude-3.5-sonnet** by Antrophic^[1]

	Claude 3.5 Sonnet (new)	Claude 3.5 Sonnet	GPT-4o	Gemini 1.5 Pro
Graduate level reasoning <i>GPQA (Diamond)</i>	65.0% 0-shot CoT	59.4% 0-shot CoT	53.6% 0-shot CoT	59.1% 0-shot CoT
Undergraduate level knowledge <i>MMLU Pro</i>	78.0% 0-shot CoT	75.1% 0-shot CoT	—	75.8% 0-shot CoT
Agentic coding <i>SWE-bench Verified</i>	49.0%	33.4%	—	—
Code <i>HumanEval</i>	93.7% 0-shot	92.0% 0-shot	90.2% 0-shot	—
Math problem-solving <i>MATH</i>	78.3% 0-shot CoT	71.1% 0-shot CoT	76.6% 0-shot CoT	86.5% 4-shot CoT
Multilingual math <i>MGSM</i>	92.5% 0-shot CoT	91.6% 0-shot CoT	90.5% 0-shot CoT	—
Reasoning over text <i>DROP, F1 Score</i>	88.3 3-shot	87.1 3-shot	83.4 3-shot	—
Agentic tool use <i>TAU-bench</i>	Retail 69.2% Airline 46.0%	Retail 62.6% Airline 36.0%	—	—

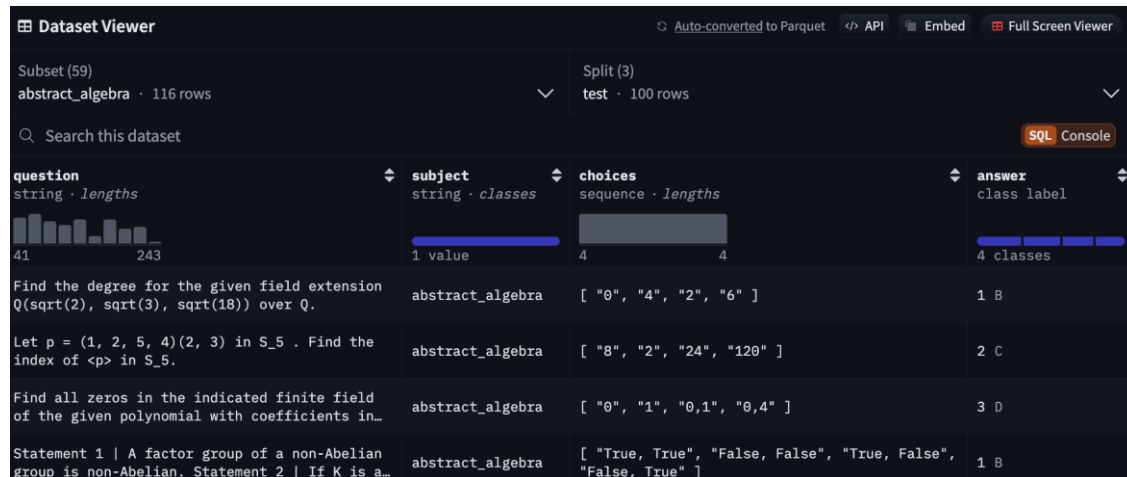
Performance of Claude-3.5-sonnet on academic benchmarks^[1]

Contents

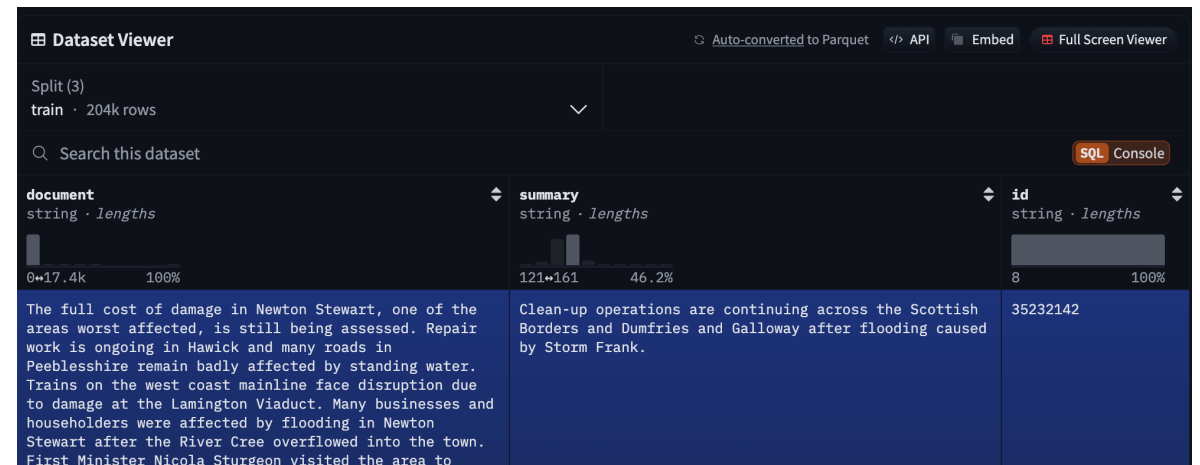
- Overview
- Evaluation of LLM with ground-truth answer
- Evaluation of LLM without ground-truth answer

Task with Ground Truth

- In open-sourced benchmark, ground truth answer is often included
 - To make fair comparison be easier
 - Evaluation method and metric can be different across dataset



question	subject	choices	answer
Find the degree for the given field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{10})$ over $\mathbb{Q}$.	abstract_algebra	["0", "4", "2", "6"]	1 B
Let $p = (1, 2, 5, 4)(2, 3)$ in S_5 . Find the index of $\langle p \rangle$ in S_5 .	abstract_algebra	["8", "2", "24", "120"]	2 C
Find all zeros in the indicated finite field of the given polynomial with coefficients in...	abstract_algebra	["0", "1", "0,1", "0,4"]	3 D
Statement 1 A factor group of a non-Abelian group is non-Abelian. Statement 2 If K is a...	abstract_algebra	["True, True", "False, False", "True, False", "False, True"]	1 B



document	summary	id
The full cost of damage in Newton Stewart, one of the areas worst affected, is still being assessed. Repair work is ongoing in Hawick and many roads in Peeblesshire remain badly affected by standing water. Trains on the west coast mainline face disruption due to damage at the Lamington Viaduct. Many businesses and householders were affected by flooding in Newton Stewart after the River Cree overflowed into the town. First Minister Nicola Sturgeon visited the area to	Clean-up operations are continuing across the Scottish Borders and Dumfries and Galloway after flooding caused by Storm Frank.	35232142

Example of benchmarks with ground truth in Huggingface (Left: MMLU, Right: Xsum)^[1]

[1] Source: <https://huggingface.co/datasets>

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - 16,000 multiple-choice questions spanning 57 academic subjects
 - Over 100 million downloads as of July 2024

Microeconomics	One of the reasons that the government discourages and regulates monopolies is that	
	(A) producer surplus is lost and consumer surplus is gained.	×
	(B) monopoly prices ensure productive efficiency but cost society allocative efficiency.	×
	(C) monopoly firms do not engage in significant research and development.	×
	(D) consumer surplus is lost with higher prices and lower levels of output.	✓

Figure 3: Examples from the Microeconomics task.

Conceptual Physics	When you drop a ball from rest it accelerates downward at 9.8 m/s^2 . If you instead throw it downward assuming no air resistance its acceleration immediately after leaving your hand is	
	(A) 9.8 m/s^2	✓
	(B) more than 9.8 m/s^2	×
	(C) less than 9.8 m/s^2	×
	(D) Cannot say unless the speed of throw is given.	×
College Mathematics	In the complex z -plane, the set of points satisfying the equation $z^2 = z ^2$ is a	
	(A) pair of points	×
	(B) circle	×
	(C) half-line	×
	(D) line	✓

Example of MMLU benchmark^[1]

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - Method: few-shot prompting
 - Metric: average accuracy (%)

	Claude 3 Opus	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5 shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot

Example of Evaluation Results on MMLU benchmark^[1]

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - Method: few-shot prompting
 - Metric: average accuracy (%) → “Common metric for multiple-choice QA benchmark”

	Claude 3 Opus	GPT-4	GPT-3.5	Gemini 1.0 Ultra	Gemini 1.0 Pro
Undergraduate level knowledge <i>MMLU</i>	86.8% 5 shot	86.4% 5-shot	70.0% 5-shot	83.7% 5-shot	71.8% 5-shot

Example of Evaluation Results on MMLU benchmark^[1]

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - **Caution.** Output of LLM **often does not follow** specific format or has patterns

Please choice proper answer among given choices to following question.

Question: Find the degree for the given field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$.
Choices: (A) 0 (B) 4 (C) 2 (D) 6
Answer:

🧠 To solve the problem of determining the degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$, let us proceed step by step:

Step 1: Simplify the field elements

- Notice that $\sqrt{18} = 3\sqrt{2}$, so $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$.

Step 2: Degree of $\mathbb{Q}(\sqrt{2})$ over $\mathbb{Q}$

- $\sqrt{2}$ is irrational, and the minimal polynomial of $\sqrt{2}$ over $\mathbb{Q}$ is $x^2 - 2$, which is irreducible over $\mathbb{Q}$.
- Therefore, $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$.

Step 3: Degree of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}(\sqrt{2})$

- In $\mathbb{Q}(\sqrt{2})$, $\sqrt{3}$ is not expressible as a linear combination of 1 and $\sqrt{2}$, because $\sqrt{3}$ is not in $\mathbb{Q}(\sqrt{2})$.
- The minimal polynomial of $\sqrt{3}$ over $\mathbb{Q}(\sqrt{2})$ is also $x^2 - 3$, which is irreducible over $\mathbb{Q}(\sqrt{2})$.
- Therefore, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] = 2$.

Step 4: Degree of $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$

- By the tower law of field extensions, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}]$.
- Hence, $[\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] = 2 \cdot 2 = 4$.

Final Answer:

The degree of the field extension $\mathbb{Q}(\sqrt{2}, \sqrt{3}, \sqrt{18})$ over $\mathbb{Q}$ is 4.
Thus, the correct choice is:

(B) 4.

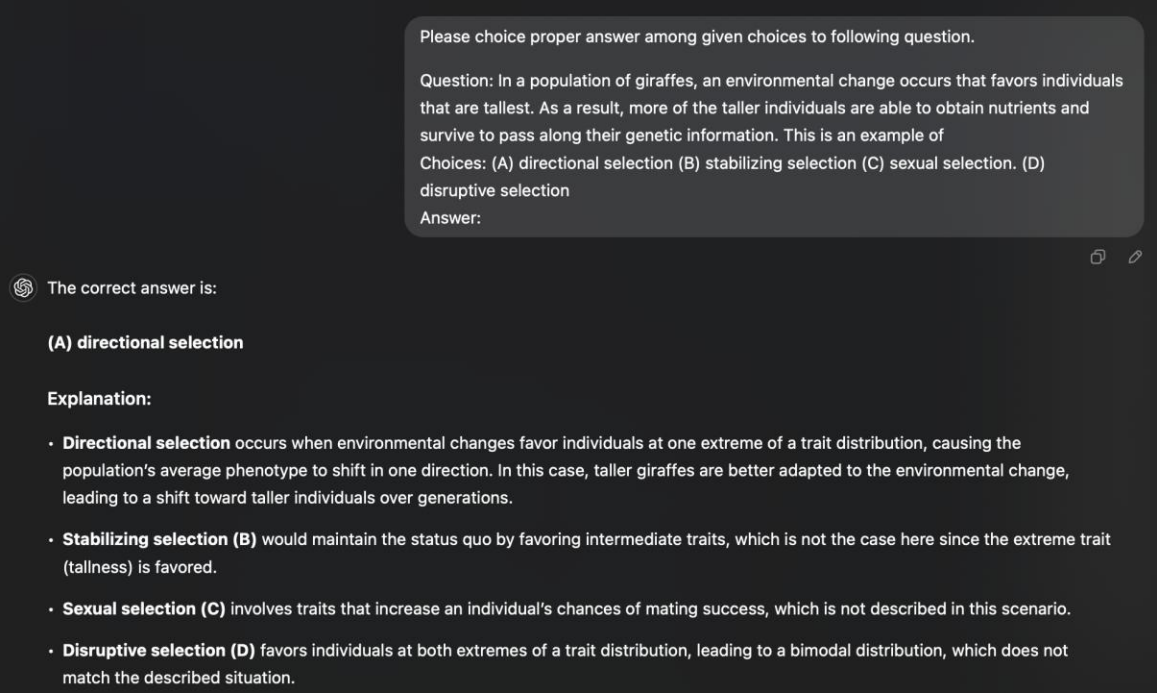
Answer is in bottom →

Example of Prediction to Specific Question from MMLU benchmark^[1]

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - **Caution.** Output of LLM **often does not follow** specific format or has patterns

Answer is in top →



Please choose proper answer among given choices to following question.

Question: In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

Choices: (A) directional selection (B) stabilizing selection (C) sexual selection. (D) disruptive selection

Answer:

The correct answer is:

(A) directional selection

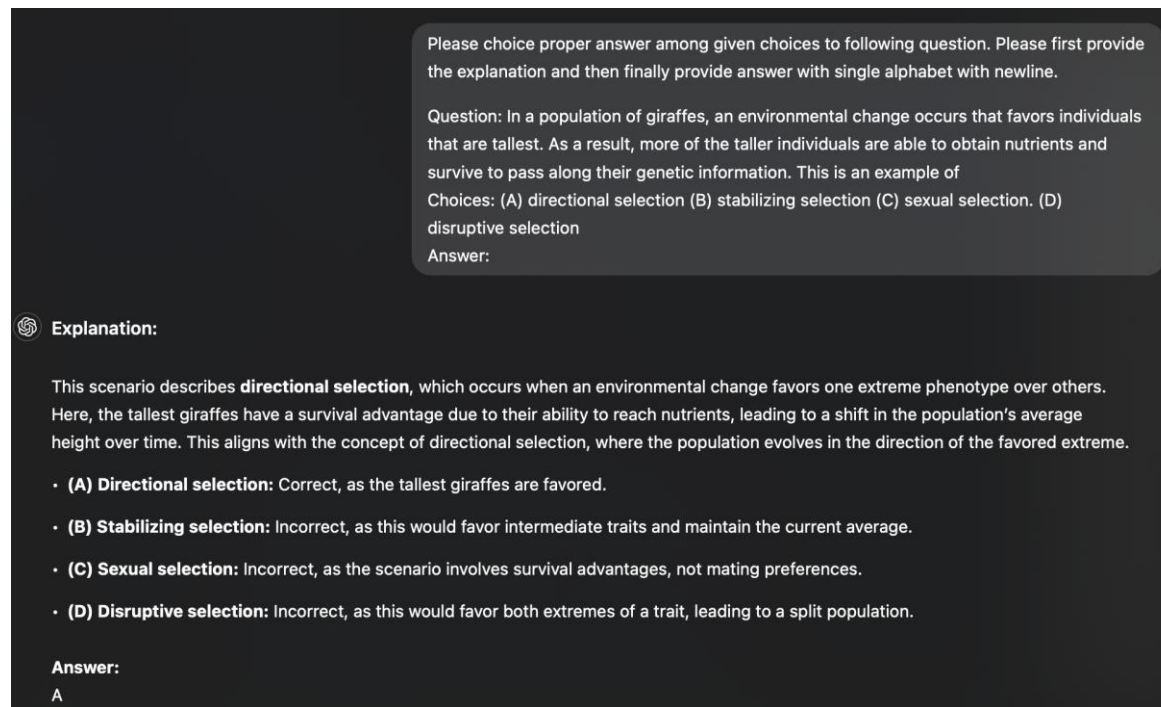
Explanation:

- **Directional selection** occurs when environmental changes favor individuals at one extreme of a trait distribution, causing the population's average phenotype to shift in one direction. In this case, taller giraffes are better adapted to the environmental change, leading to a shift toward taller individuals over generations.
- **Stabilizing selection (B)** would maintain the status quo by favoring intermediate traits, which is not the case here since the extreme trait (tallness) is favored.
- **Sexual selection (C)** involves traits that increase an individual's chances of mating success, which is not described in this scenario.
- **Disruptive selection (D)** favors individuals at both extremes of a trait distribution, leading to a bimodal distribution, which does not match the described situation.

Example of Prediction to Specific Question from MMLU benchmark^[1]

Task with Ground Truth: MMLU

- **Massive Multitask Language Understanding benchmark (MMLU^[1])**
 - **Caution.** Output of LLM **often does not follow** specific format or has patterns
 - **Specific instruction** can mitigate this (remember: not perfect!)



Please choose proper answer among given choices to following question. Please first provide the explanation and then finally provide answer with single alphabet with newline.

Question: In a population of giraffes, an environmental change occurs that favors individuals that are tallest. As a result, more of the taller individuals are able to obtain nutrients and survive to pass along their genetic information. This is an example of

Choices: (A) directional selection (B) stabilizing selection (C) sexual selection. (D) disruptive selection

Answer:

Explanation:

This scenario describes **directional selection**, which occurs when an environmental change favors one extreme phenotype over others. Here, the tallest giraffes have a survival advantage due to their ability to reach nutrients, leading to a shift in the population's average height over time. This aligns with the concept of directional selection, where the population evolves in the direction of the favored extreme.

- **(A) Directional selection:** Correct, as the tallest giraffes are favored.
- **(B) Stabilizing selection:** Incorrect, as this would favor intermediate traits and maintain the current average.
- **(C) Sexual selection:** Incorrect, as the scenario involves survival advantages, not mating preferences.
- **(D) Disruptive selection:** Incorrect, as this would favor both extremes of a trait, leading to a split population.

Answer:
A

← Specific instruction

Example of Prediction to Specific Question from MMLU benchmark^[1]

Task with Ground Truth: NQ

- **Natural Question dataset (NQ^[1])**
 - 7,830 examples with 5-way annotations for development/test dataset
 - LLM should generate long or short answer without multiple choices

Example 1

Question: what color was john wilkes booth's hair

Wikipedia Page: John_Wilkes_Booth

Long answer: Some critics called Booth “the handsomest man in America” and a “natural genius”, and noted his having an “astounding memory”; others were mixed in their estimation of his acting. He stood 5 feet 8 inches (1.73 m) tall, had jet-black hair , and was lean and athletic. Noted Civil War reporter George Alfred Townsend described him as a “muscular, perfect man” with “curling hair, like a Corinthian capital”.

Short answer: jet-black

Example 2

Question: can you make and receive calls in airplane mode

Wikipedia Page: Airplane_mode

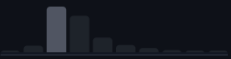
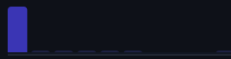
Long answer: Airplane mode, aeroplane mode, flight mode, offline mode, or standalone mode is a setting available on many smartphones, portable computers, and other electronic devices that, when activated, suspends radio-frequency signal transmission by the device, thereby disabling Bluetooth, telephony, and Wi-Fi. GPS may or may not be disabled, because it does not involve transmitting radio waves.

Short answer: BOOLEAN:NO

Examples in Natural Questions (NQ) benchmark^[1]

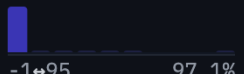
Task with Ground Truth: NQ

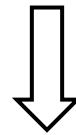
- Example of (1) QA

question string · lengths	long_answer_candidates sequence	long_answer_candidate_index int64	short_answers sequence
 36↔45 38.3%		 -1↔95 97.1%	
who died in the plane crash greys anatomy	[`` Flight `` Grey 's Anatomy episode Shepherd 's mangled hand , following an aviation accident . Episode no . Season 8 Episode 24 Directed by Rob Corn Written by Shonda Rhimes Featured music `` Featherstone `` `` Graveyard `` Original	25	["Lexie", "Dr. Lexie Grey", "Dr. Lexie Grey", "Lexie", "Dr. Lexie Grey (Chyler Leigh)"]

Task with Ground Truth: NQ

- Example of (1) QA and (2) prediction with GPT-4o

question string · lengths	long_answer_candidates sequence	long_answer_candidate_index int64	short_answers sequence
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Please answer to following question with few words.

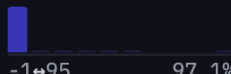
Question: who died in the plane crash greys anatomy?

Answer:

 Lexie Grey and Mark Sloan died as a result of the plane crash in *Grey's Anatomy*.

Task with Ground Truth: NQ

- Example of (1) QA and (2) prediction with GPT-4o

question string · lengths	long_answer_candidates sequence	long_answer_candidate_index int64	short_answers sequence
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Please answer to following question with few words.

Question: who died in the plane crash greys anatomy?

Answer:

 Lexie Grey and Mark Sloan died as a result of the plane crash in *Grey's Anatomy*.

How to evaluate
it **quantitatively**
& **automatically**?

Task with Ground Truth: NQ

- **Metrics** for QA: Exact Match (**EM**) and **F1** scores (**Lexical Matching**)
 - **EM**: **1** if it is the exact same as its reference string, or **0** otherwise
 - **F1**: Considering word-wise comparisons after parsing
 - Precision: # words in prediction & answer / # words in prediction
 - Recall: # words in prediction & answer / # words in answer

Question: Where on Earth is free oxygen found?
Prediction: water bodies
True Answers: "in solution in the world's water bodies"
EM: 0 F1: 0.8

$$F_1 = \frac{2}{\text{recall}^{-1} + \text{precision}^{-1}}$$

Task with Ground Truth: NQ

- **Metrics** for QA: Exact Match (**EM**) and **F1** scores (**Lexical Matching**)
 - **EM**: **1** if it is the exact same as its reference string, or **0** otherwise
 - **F1**: Considering word-wise comparisons after parsing
 - Precision: # words in prediction & answer / # words in prediction
 - Recall: # words in prediction & answer / # words in answer

Table 1: EM / F1 for different QA methods with ChatGPT on four QA datasets. $N = 10$ most relevant passages are retrieved using BM25, except *no retrieval*. The best and second best scores are highlighted in **bold** and underline, respectively.

Methods / Datasets	NQ	WebQ	2Wiki	HotpotQA	Average
No retrieval	27.6 / <u>39.0</u>	<u>25.0</u> / 38.8	21.4 / 24.8	22.2 / 31.9	24.1 / 33.6
Base	<u>28.4</u> / 38.8	19.6 / 32.5	<u>27.4</u> / <u>32.8</u>	30.8 / 40.3	<u>26.6</u> / <u>36.1</u>
Rerank	24.8 / 33.9	18.8 / 30.6	23.0 / 28.4	27.8 / 37.4	23.6 / 32.6
RePlug	26.0 / 35.3	18.8 / 31.5	23.6 / 28.5	28.0 / 37.9	24.1 / 33.3
Selection-inference	24.3 / 32.8	17.3 / 28.6	22.6 / 29.5	30.8 / 39.6	23.8 / 32.6
Chain-of-thoughts	22.3 / 31.4	15.2 / 27.8	19.6 / 22.5	25.6 / 31.8	20.7 / 28.4
Self-verification	25.2 / 35.4	16.1 / 28.5	23.2 / 30.5	<u>31.6</u> / <u>41.8</u>	24.0 / 34.1
SURE (Ours)	33.5 / 42.3	25.1 / <u>36.6</u>	32.8 / 38.1	33.2 / 43.4	31.2 / 40.1

Task with Ground Truth: XSum

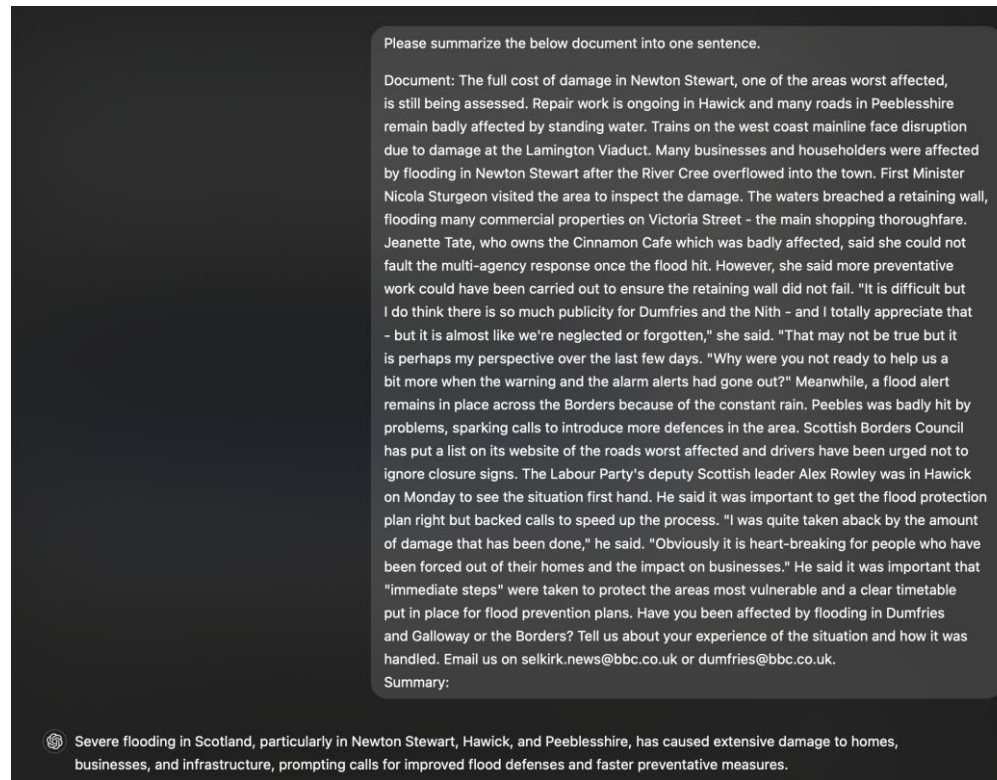
- Extreme summarization dataset (Xsum^[1])
 - 230k examples & avg. words in document/summary → 430/23

document string · lengths	summary string · lengths	id string · lengths
0-17.4k 100%	121-161 46.2%	8 100%
The full cost of damage in Newton Stewart, one of the areas worst affected, is still being assessed. Repair work is ongoing in Hawick and many roads in Peeblesshire remain badly affected by standing water. Trains on the west coast mainline face disruption due to damage at the Lamington Viaduct. Many businesses and householders were affected by flooding in Newton Stewart after the River Cree overflowed into the town. First Minister Nicola Sturgeon visited the area to	Clean-up operations are continuing across the Scottish Borders and Dumfries and Galloway after flooding caused by Storm Frank.	35232142

Example in XSum dataset^[1]

Task with Ground Truth: XSum

- Extreme summarization dataset (Xsum^[1])
 - 230k examples & avg. words in document/summary → 430/23



Please summarize the below document into one sentence.

Document: The full cost of damage in Newton Stewart, one of the areas worst affected, is still being assessed. Repair work is ongoing in Hawick and many roads in Peeblesshire remain badly affected by standing water. Trains on the west coast mainline face disruption due to damage at the Lamington Viaduct. Many businesses and householders were affected by flooding in Newton Stewart after the River Cree overflowed into the town. First Minister Nicola Sturgeon visited the area to inspect the damage. The waters breached a retaining wall, flooding many commercial properties on Victoria Street - the main shopping thoroughfare. Jeanette Tate, who owns the Cinnamon Cafe which was badly affected, said she could not fault the multi-agency response once the flood hit. However, she said more preventative work could have been carried out to ensure the retaining wall did not fail. "It is difficult but I do think there is so much publicity for Dumfries and the Nith - and I totally appreciate that - but it is almost like we're neglected or forgotten," she said. "That may not be true but it is perhaps my perspective over the last few days. "Why were you not ready to help us a bit more when the warning and the alarm alerts had gone out?" Meanwhile, a flood alert remains in place across the Borders because of the constant rain. Peebles was badly hit by problems, sparking calls to introduce more defences in the area. Scottish Borders Council has put a list on its website of the roads worst affected and drivers have been urged not to ignore closure signs. The Labour Party's deputy Scottish leader Alex Rowley was in Hawick on Monday to see the situation first hand. He said it was important to get the flood protection plan right but backed calls to speed up the process. "I was quite taken aback by the amount of damage that has been done," he said. "Obviously it is heart-breaking for people who have been forced out of their homes and the impact on businesses." He said it was important that "immediate steps" were taken to protect the areas most vulnerable and a clear timetable put in place for flood prevention plans. Have you been affected by flooding in Dumfries and Galloway or the Borders? Tell us about your experience of the situation and how it was handled. Email us on selkirk.news@bbc.co.uk or dumfries@bbc.co.uk.

Summary:

Severe flooding in Scotland, particularly in Newton Stewart, Hawick, and Peeblesshire, has caused extensive damage to homes, businesses, and infrastructure, prompting calls for improved flood defenses and faster preventative measures.

Example of output for document in XSum dataset^{1]}

Task with Ground Truth: XSum

- Extreme summarization dataset (Xsum^[1])
 - 230k examples & avg. words in document/summary → 430/23

“Clean-up operations are continuing across the Scottish Borders and Dumfries and Galloway after flooding caused by Storm Frank.”

Ground Truth

“Severe flooding in Scotland, particularly in Newton Stewart, Hawick, and Peeblesshire, has caused extensive damage to homes, businesses, and infrastructure, prompting calls for improved flood defenses and faster preventative measures.”

LLM Output

Task with Ground Truth: XSum

- Extreme summarization dataset (Xsum^[1])
 - 230k examples & avg. words in document/summary → 430/23

“Clean-up operations are continuing across the Scottish Borders and Dumfries and Galloway after flooding caused by Storm Frank.”

Ground Truth

**How to measure similarity
between these sentences?**

“Severe flooding in Scotland, particularly in Newton Stewart, Hawick, and Peeblesshire, has caused extensive damage to homes, businesses, and infrastructure, prompting calls for improved flood defenses and faster preventative measures.”

LLM Output

Task with Ground Truth: XSum

- **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation)
 - Measures overlap of words or phrases between generated and reference texts
 - E.g., **Rouge-N**: measuring overlap of N-grams

- Example:
 - Reference: "The cat sat on the mat."
 - Generated: "The cat lay on the mat."
 - ROUGE-1 (unigrams): Overlap = {"The", "cat", "on", "the", "mat"} = 5/6 = **0.83**.

Task with Ground Truth: XSum

- **Distance (L2 or cosine) at embedding space**
 - First, raw texts are mapped to embedding using separately trained sentence encoder^[1]
 - Then, one can evaluate **semantical similarity** by measuring distance
 - **Caution.** Quality of sentence embedding is very important (check [2])

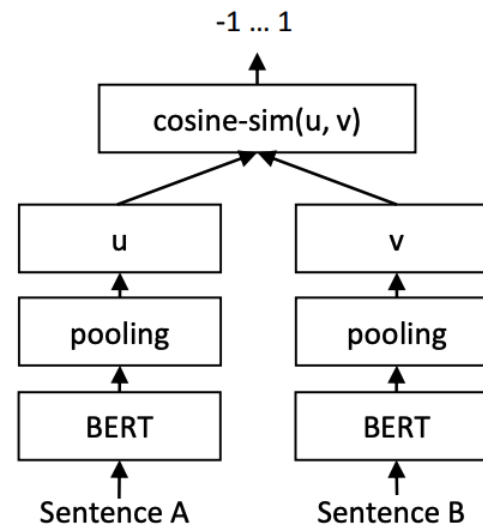


Illustration of Sentence-BERT to Calculate Cosine Similarity between Two Sentences^{1]}

[1] Reimers and Gurevych., Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks., EMNLP 2019

[2] <https://huggingface.co/spaces/mteb/leaderboard>

Contents

- Overview
- Evaluation of LLM with ground-truth answer
- Evaluation of LLM without ground-truth answer

Task with No Ground Truth

- Many real-world tasks often do **not have fixed ground truth**
 - Example: **summarization**

“Clean-up operations are continuing across the Scottish Borders and Dumfries and Galloway after flooding caused by Storm Frank.”

Ground Truth

“Severe flooding in Scotland, particularly in Newton Stewart, Hawick, and Peeblesshire, has caused extensive damage to homes, businesses, and infrastructure, prompting calls for improved flood defenses and faster preventative measures.”

LLM Output

Task with No Ground Truth

- Many real-world tasks often do **not have fixed ground truth**
 - Example: summarization or **story generation**^[1] (continuation)



WebText

2008년 이후 서호주 해안에서 전례 없이 많은 수의 어린 고래가 좌초되었습니다.

Task with No Ground Truth

- Many real-world tasks often do **not have fixed ground truth**
 - Example: summarization or **story generation** (continuation)

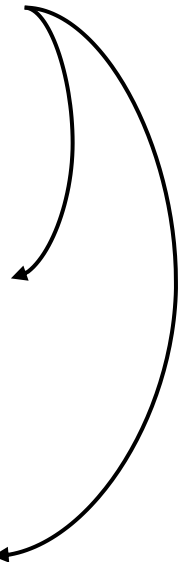


WebText

2008년 이후 서호주 해안에서 전례 없이 많은 수의 어린 고래가 좌초되었습니다.

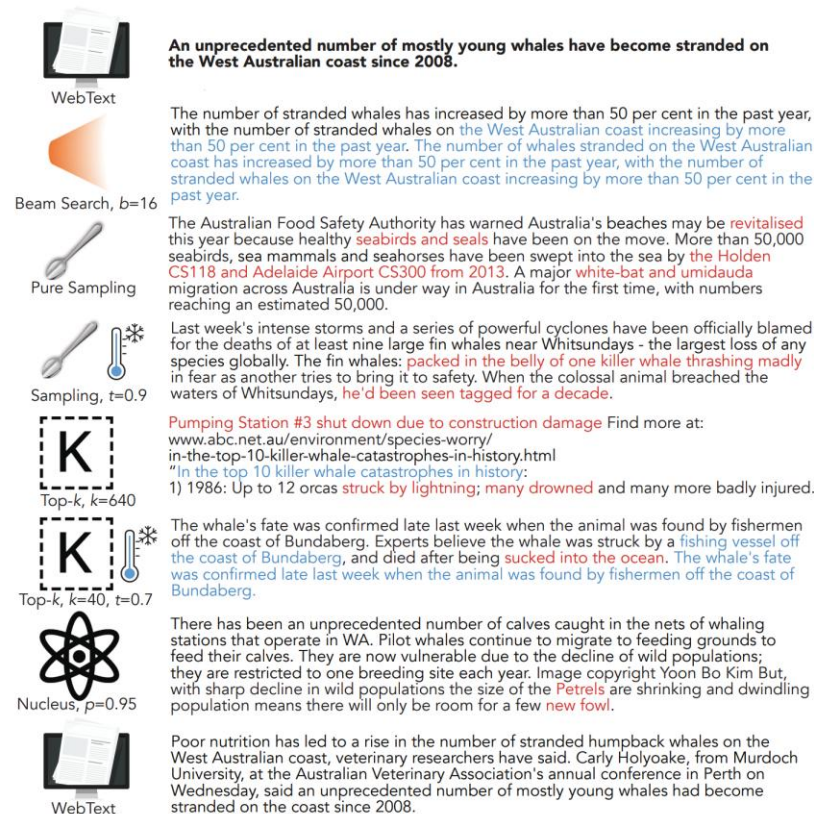
워싱턴 주에서 운영되는 포경 그물에 전례 없이 많은 수의 새끼 고래가 잡혔습니다.

서호주 해안에서 영양 부족으로 인해 좌초된 혹등고래의 수가 증가하고 있다고 수의학 연구자들이 밝혔습니다.



Open-ended Text Generation

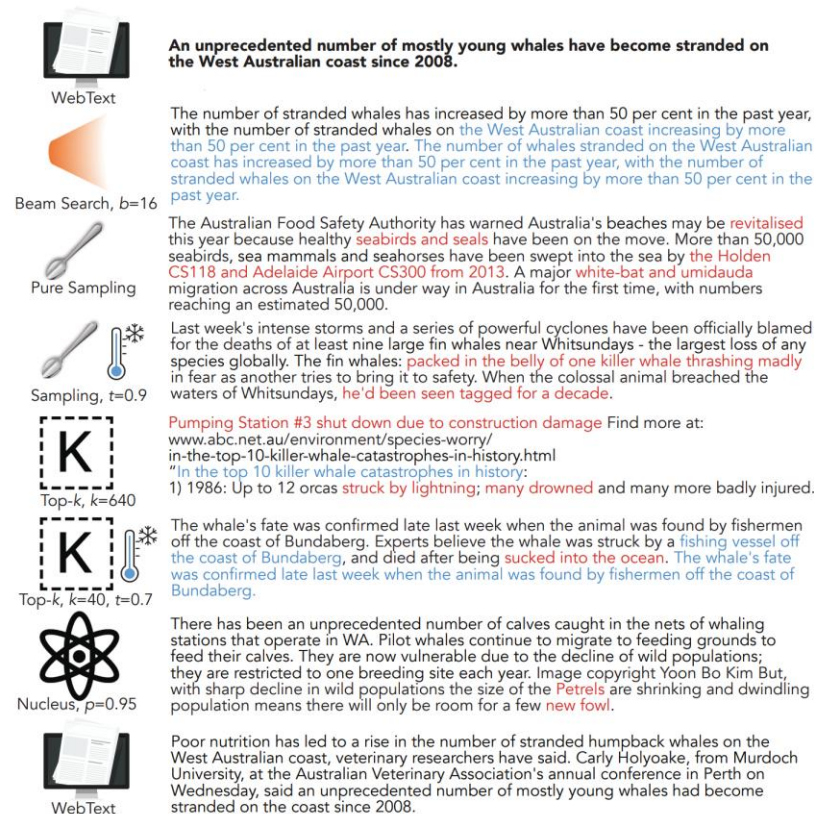
- **Def.** Tasks which create coherent text that continues from given context
- LLM's output can be viewed as open-ended text generation



Example Generations Continuing Initial Sentence under Different Decoding Methods^[1]

Open-ended Text Generation

- **Def.** Tasks which create coherent text that continues from given context
- LLM's output can be viewed as open-ended text generation



How can we evaluate?

Example Generations Continuing Initial Sentence under Different Decoding Methods^[1]

Open-ended Text Generation

- **Def.** Tasks which create coherent text that continues from given context
 - LLM's output can be viewed as open-ended text generation

Method	Perplexity	Self-BLEU4	Repetition %
Human	12.38	0.31	0.28
Greedy	1.50	0.50	73.66
Beam, b=16	1.48	0.44	28.94
Stochastic Beam, b=16	19.20	0.28	0.32
Pure Sampling	22.73	0.28	0.22
Sampling, $t=0.9$	10.25	0.35	0.66
Top- $k=40$	6.88	0.39	0.78
Top- $k=640$	13.82	0.32	0.28
Top- $k=40$, $t=0.7$	3.48	0.44	8.86
Nucleus $p=0.95$	13.13	0.32	0.36

Evaluation Results of Different Decoding Methods^[1]

Metrics for Open-ended Text Generation

- **Perplexity** measures how well model can predict sequence of words

$$\text{Perplexity} = \sqrt[N]{\frac{1}{P(w_1, w_2, w_3, \dots, w_N)}} = \sqrt[N]{\frac{1}{\prod_{i=1}^N P(w_i | w_1, w_2, \dots, w_{i-1})}}$$

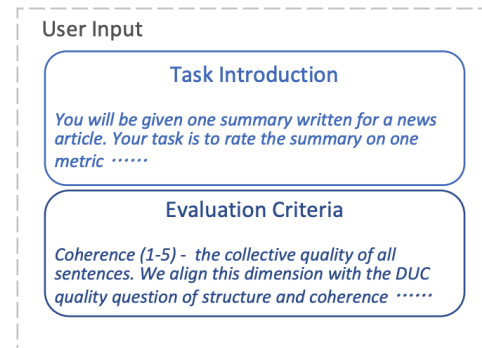
- Probabilities are (1) from generating LLM itself or (2) external evaluating LLM
- **Lower perplexity** usually indicates **better** predictive **performance (quality)**
- **Caution**. It may not always correlate with human-perceived quality

Metrics for Open-ended Text Generation

- But, what if one has certain evaluation criteria?
 - E.g., clarity, coherence, and creativity
 - Not just calculate separately, but want to incorporate into single score
 - Hiring human annotator is **most accurate**, but **too costly**

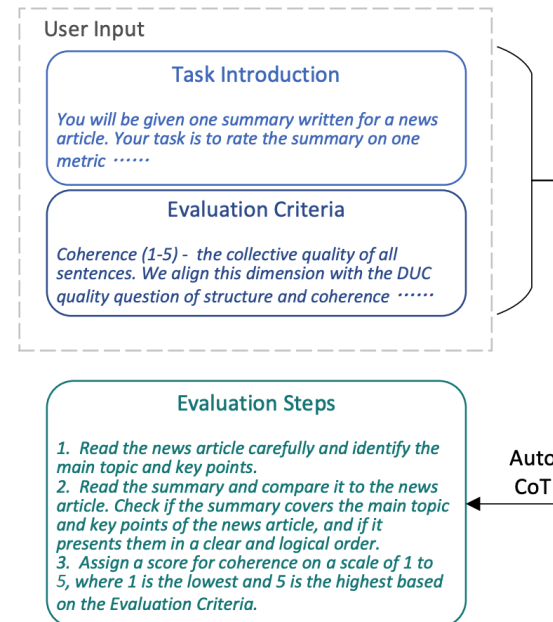
Metrics for Open-ended Text Generation

- **G-Eval** ^[1]: using LLMs to assess the quality of outputs of LLM
 - User first gives (1) task information and (2) evaluation criteria



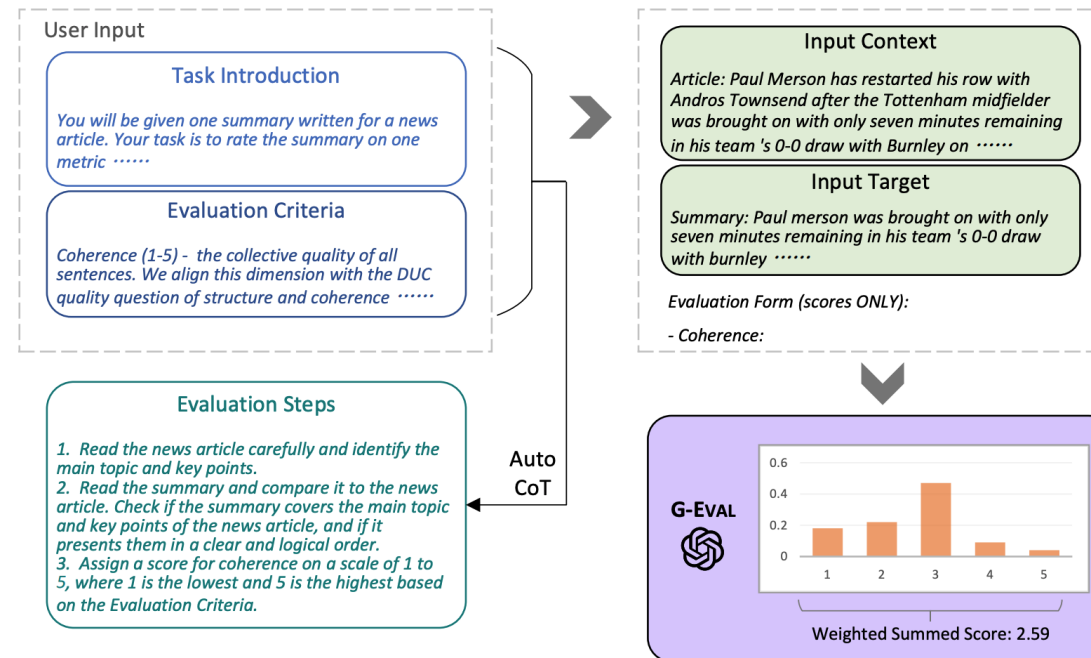
Metrics for Open-ended Text Generation

- **G-Eval**^[1]: using LLMs to assess the quality of outputs of LLM
 - User first gives (1) task information and (2) evaluation criteria
 - With these inputs, LLM generates evaluation steps automatically



Metrics for Open-ended Text Generation

- **G-Eval**^[1]: using LLMs to assess the quality of outputs of LLM
 - User first gives (1) task information and (2) evaluation criteria
 - With these inputs, LLM generates evaluation steps automatically
 - Then, LLM can evaluate arbitrary input according to evaluation criteria and task info.



Metrics for Open-ended Text Generation

- G-Eval with GPT-4 exhibits **highest correlation** to human evaluation
 - Compare to previous metrics like ROUGE or BERTScore (embedding)

Metrics	Coherence		Consistency		Fluency		Relevance		AVG	
	ρ	τ	ρ	τ	ρ	τ	ρ	τ	ρ	τ
ROUGE-1	0.167	0.126	0.160	0.130	0.115	0.094	0.326	0.252	0.192	0.150
ROUGE-2	0.184	0.139	0.187	0.155	0.159	0.128	0.290	0.219	0.205	0.161
ROUGE-L	0.128	0.099	0.115	0.092	0.105	0.084	0.311	0.237	0.165	0.128
BERTScore	0.284	0.211	0.110	0.090	0.193	0.158	0.312	0.243	0.225	0.175
MOVERSscore	0.159	0.118	0.157	0.127	0.129	0.105	0.318	0.244	0.191	0.148
BARTScore	0.448	0.342	0.382	0.315	0.356	0.292	0.356	0.273	0.385	0.305
UniEval	0.575	0.442	0.446	0.371	0.449	0.371	0.426	0.325	0.474	0.377
GPTScore	0.434	–	0.449	–	0.403	–	0.381	–	0.417	–
G-EVAL-3.5	0.440	0.335	0.386	0.318	0.424	0.347	0.385	0.293	0.401	0.320
- Probs	0.359	0.313	0.361	0.344	0.339	0.323	0.327	0.288	0.346	0.317
G-EVAL-4	0.582	0.457	0.507	0.425	0.506	0.455	0.547	0.433	0.514	0.418
- Probs	0.560	0.472	0.501	0.459	0.505	0.473	0.511	0.444	0.502	0.446
- CoT	0.564	0.454	0.493	0.413	0.483	0.431	0.538	0.427	0.500	0.407
- Description	0.513	0.424	0.421	0.344	0.447	0.373	0.479	0.388	0.479	0.377

Correlation of Model-base Evaluation and Human Evaluation^[1]

Metrics for Open-ended Text Generation

- G-Eval with GPT-4 exhibits **highest correlation** to human evaluation
 - Compare to previous metrics like ROUGE or BERTScore (embedding)
 - Using LLM as evaluator with specific criteria becomes standard approach^[1]

Review the user's question and the corresponding response using the additive 5-point scoring system described below. Points are accumulated based on the satisfaction of each criterion:

- Add 1 point if the response is relevant and provides some information related to the user's inquiry, even if it is incomplete or contains some irrelevant content.
- Add another point if the response addresses a substantial portion of the user's question, but does not completely resolve the query or provide a direct answer.
- Award a third point if the response answers the basic elements of the user's question in a useful way, regardless of whether it seems to have been written by an AI Assistant or if it has elements typically found in blogs or search results.
- Grant a fourth point if the response is clearly written from an AI Assistant's perspective, addressing the user's question directly and comprehensively, and is well-organized and helpful, even if there is slight room for improvement in clarity, conciseness or focus.
- Bestow a fifth point for a response that is impeccably tailored to the user's question by an AI Assistant, without extraneous information, reflecting expert knowledge, and demonstrating a high-quality, engaging, and insightful answer.

User: <INSTRUCTION_HERE>

<response><RESPONSE_HERE></response>

After examining the user's instruction and the response:

- Briefly justify your total score, up to 100 words.
- Conclude with the score using the format: "Score: <total points>"

Remember to assess from the AI Assistant perspective, utilizing web search knowledge as necessary. To evaluate the response in alignment with this additive scoring model, we'll systematically attribute points based on the outlined criteria.

Prompt to use LLM as Scoring Model^[1]

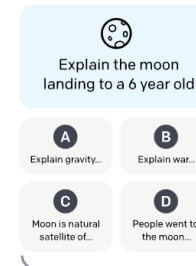
Relative Evaluation

- Relative evaluation is another way to find good single output
 - **Idea:** pairwise (or more) comparison among candidate and pick the best one

Step 2

**Collect comparison data,
and train a reward model.**

A prompt and
several model
outputs are
sampled.

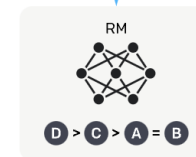


LLM's generation

A labeler ranks
the outputs from
best to worst.



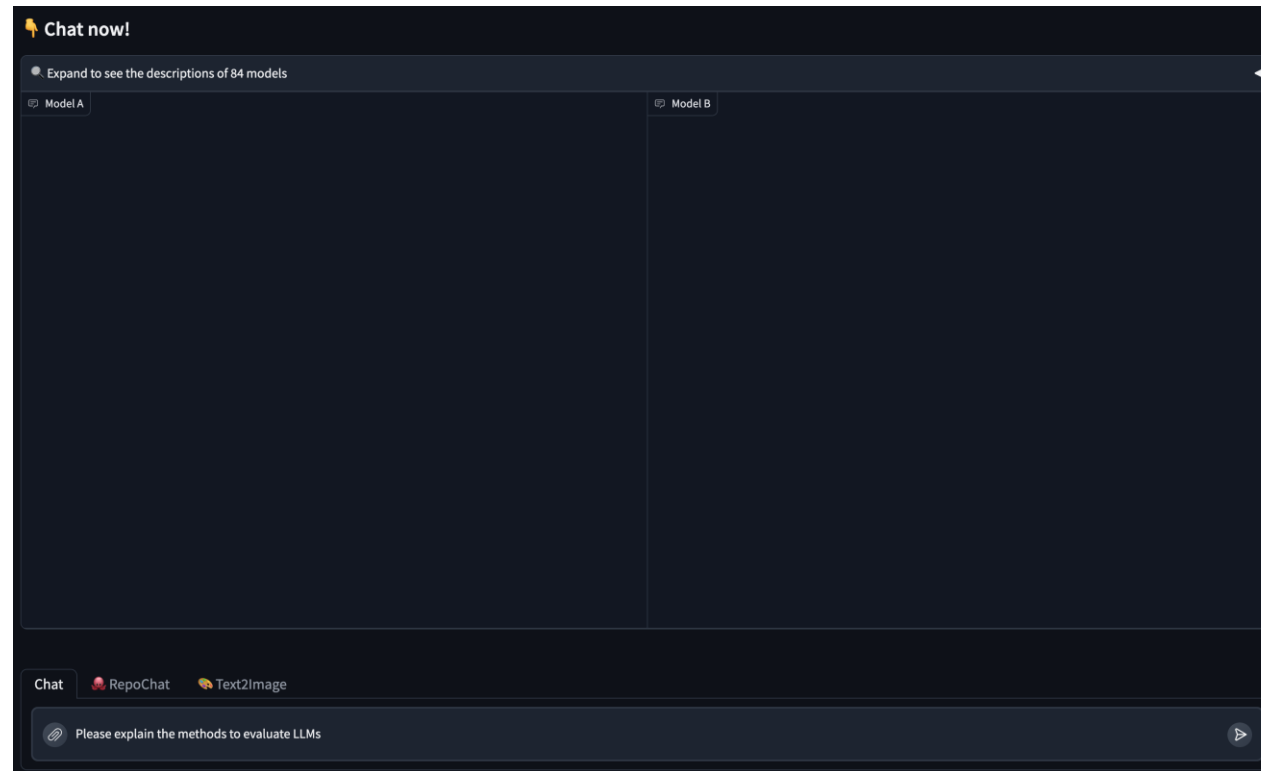
This data is used
to train our
reward model.



Pairwise Comparison by Human Annotator to Decide Good Output by LLM^[1]

Relative Evaluation

- Example: **Chatbot Arena**^[1]
 - Considered as one of most reliable comparison between LLMs using **real-user feedback**

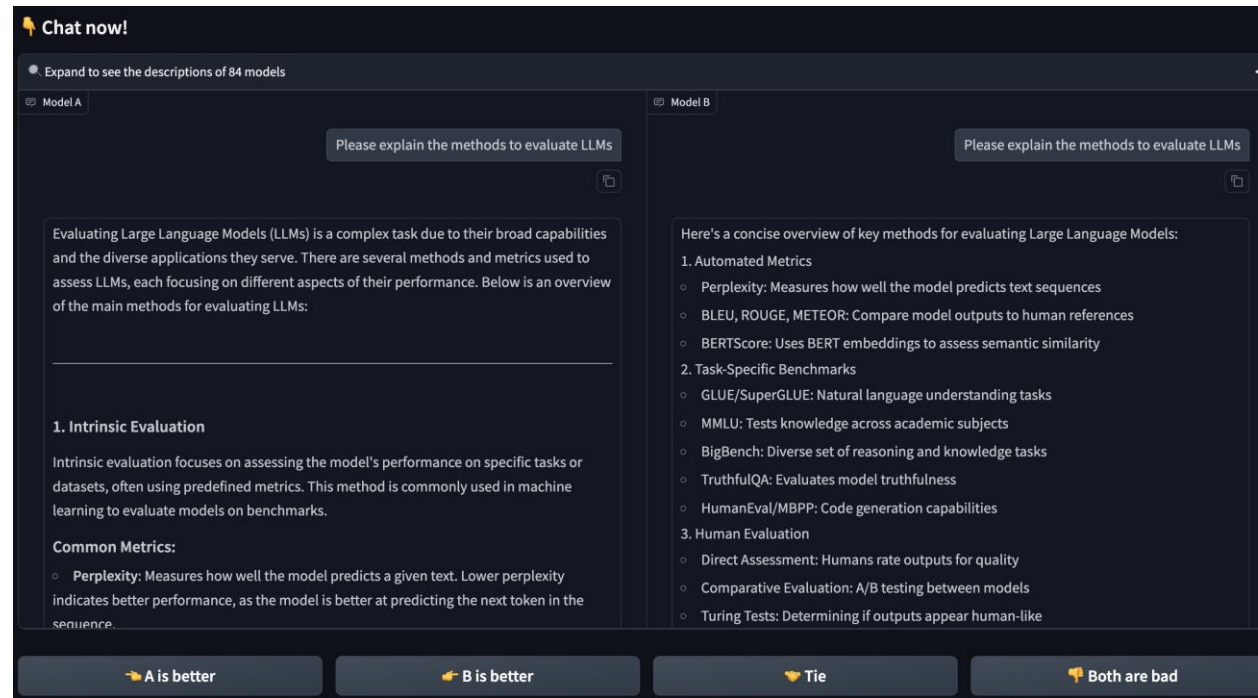


Interface of Chatbot Arena for Relative Evaluation between LLMs^[1]

[1] <https://lmarena.ai/>

Relative Evaluation

- Example: **Chatbot Arena**^[1]
 - Considered as one of most reliable comparison between LLMs using **real-user feedback**

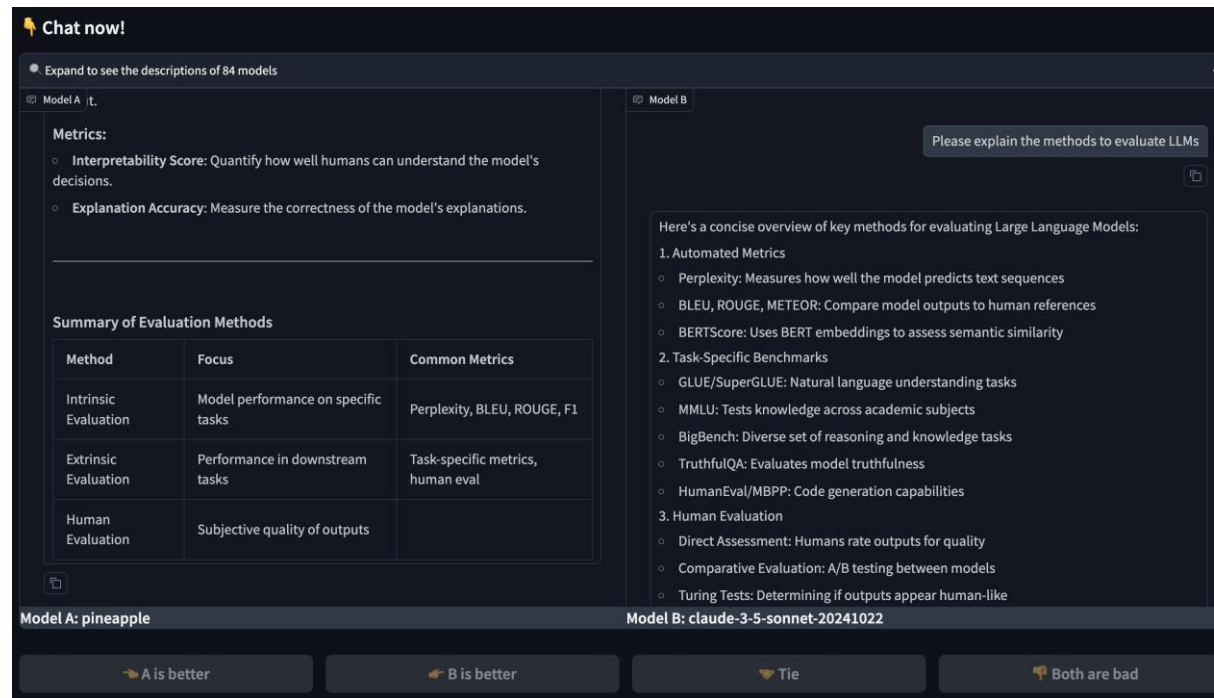


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Relative Evaluation

- Example: **Chatbot Arena**^[1]
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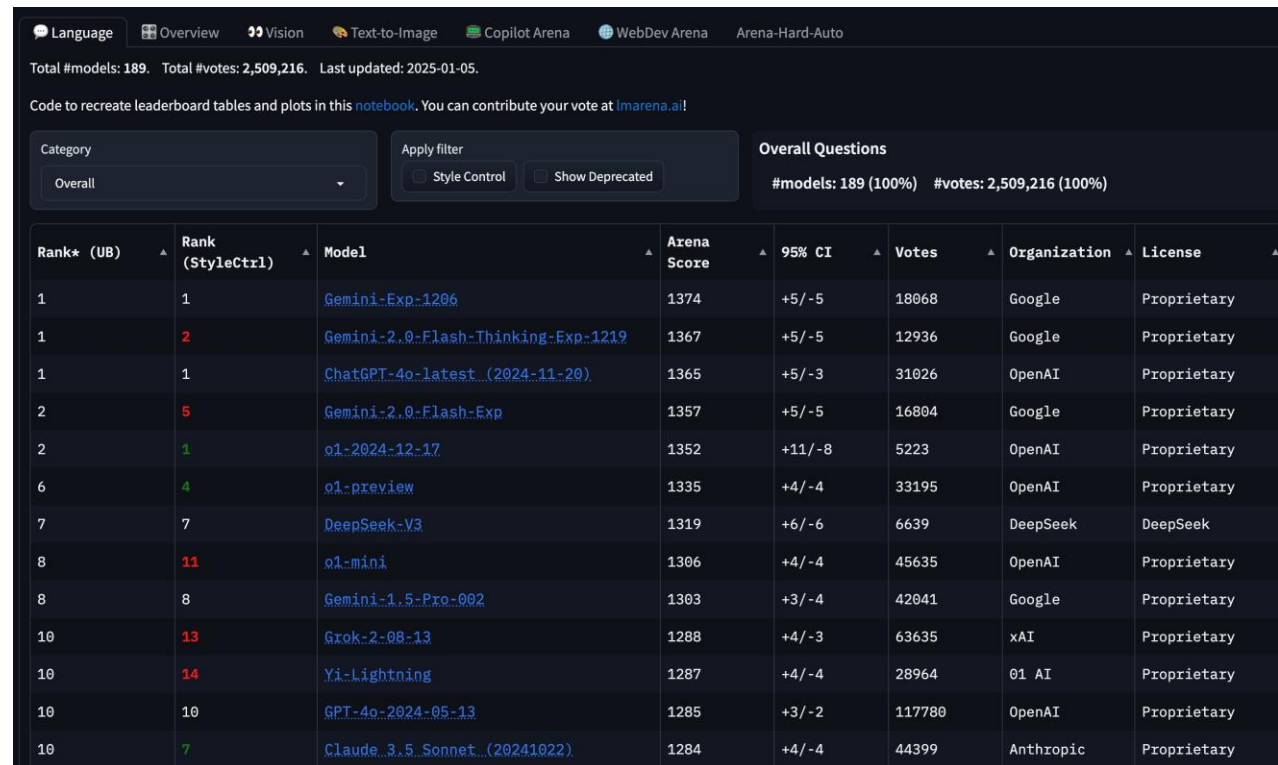


Interface of Chatbot Arena for Relative Evaluation between LLMs^[1]

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Relative Evaluation

- Example: **Chatbot Arena**^[1]
 - Considered as one of most reliable comparison between LLMs using **real-user feedback**



The screenshot shows the Chatbot Arena website interface. At the top, there are navigation tabs: Language, Overview, Vision, Text-to-Image, Copilot Arena, WebDev Arena, and Arena-Hard-Auto. Below the tabs, it states: "Total #models: 189. Total #votes: 2,509,216. Last updated: 2025-01-05." There is a link to a notebook for recreating leaderboard tables and a note about contributing votes at lmarena.ai. Below this, there are filters for Category (set to Overall), Apply filter, Style Control, and Show Deprecated. To the right, it shows "Overall Questions" with "#models: 189 (100%)" and "#votes: 2,509,216 (100%)". The main part of the image is a table with the following columns: Rank* (UB), Rank (StyleCtrl), Model, Arena Score, 95% CI, Votes, Organization, and License. The table lists 15 models, with Gemini-Exp-1206 at the top rank.

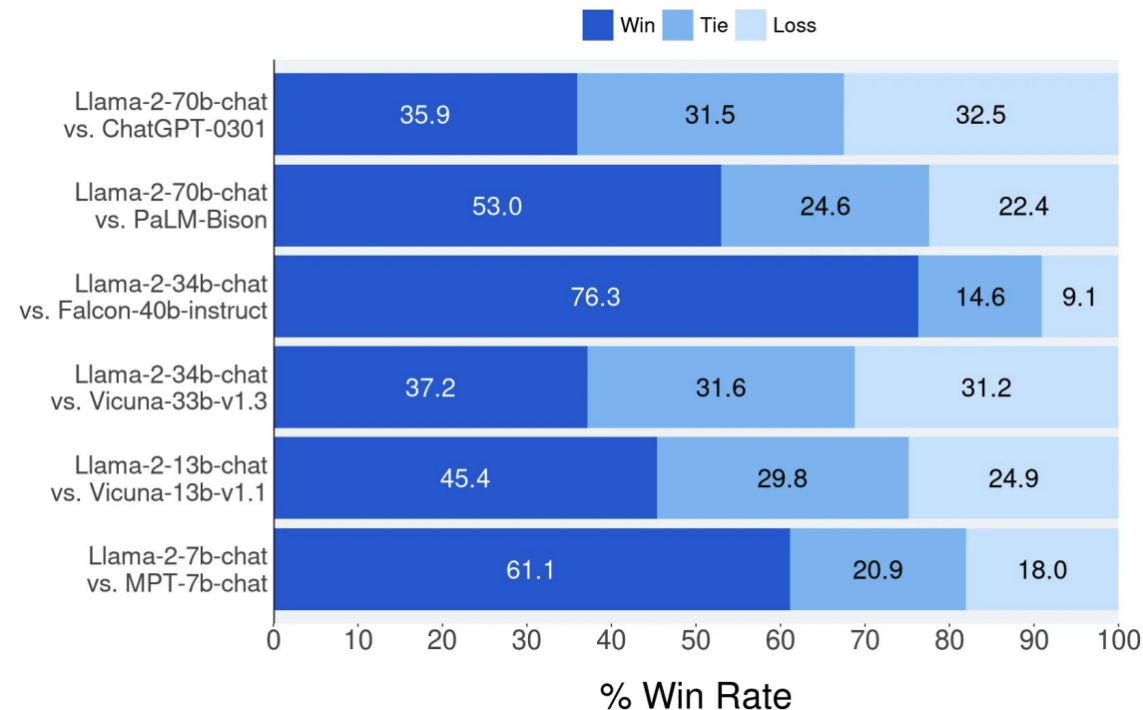
Rank* (UB)	Rank (StyleCtrl)	Model	Arena Score	95% CI	Votes	Organization	License
1	1	Gemini-Exp-1206	1374	+5/-5	18068	Google	Proprietary
1	2	Gemini-2.0-Flash-Thinking-Exp-1219	1367	+5/-5	12936	Google	Proprietary
1	1	ChatGPT-4o-latest_(2024-11-20)	1365	+5/-3	31026	OpenAI	Proprietary
2	5	Gemini-2.0-Flash-Exp	1357	+5/-5	16804	Google	Proprietary
2	1	o1-2024-12-17	1352	+11/-8	5223	OpenAI	Proprietary
6	4	o1-preview	1335	+4/-4	33195	OpenAI	Proprietary
7	7	DeepSeek-V3	1319	+6/-6	6639	DeepSeek	DeepSeek
8	11	o1-mini	1306	+4/-4	45635	OpenAI	Proprietary
8	8	Gemini-1.5-Pro-002	1303	+3/-4	42041	Google	Proprietary
10	13	Grok-2-08-13	1288	+4/-3	63635	xAI	Proprietary
10	14	Yi-Lightning	1287	+4/-4	28964	01 AI	Proprietary
10	10	GPT-4o-2024-05-13	1285	+3/-2	117780	OpenAI	Proprietary
10	7	Claude-3.5-Sonnet_(20241022)	1284	+4/-4	44399	Anthropic	Proprietary

Leaderboard of Chatbot Arena^[1]

[1] <https://lmarena.ai/>

Relative Evaluation

- Pairwise comparison using human annotator is common way to evaluate
 - Example: Llama (LLM from Meta)^[1]

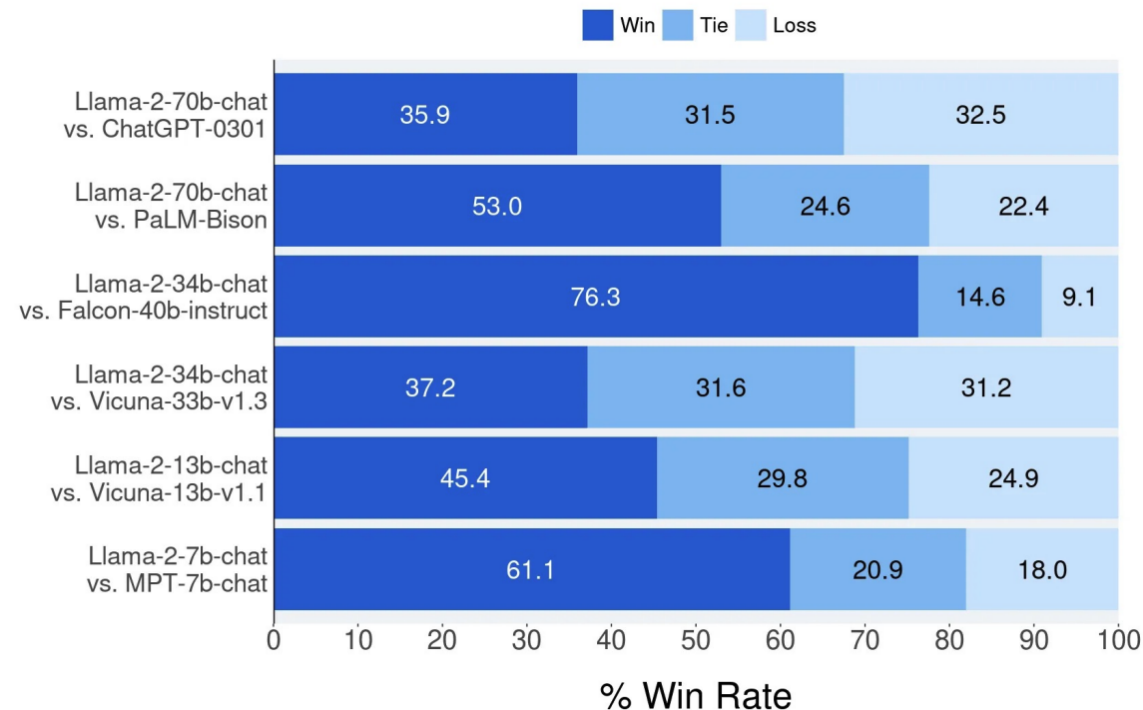


Comparison between Llama2 and other LLMs^[1]

[1] <https://lmarena.ai/>

Relative Evaluation

- Pairwise comparison using human annotator is common way to evaluate
 - Example: Llama (LLM from Meta)^[1]
 - **Cons.** Cost for large human annotation



Comparison between Llama2 and other LLMs^[1]

[1] <https://lmarena.ai/>

LLM-as-Judge

- **Solution:** LLM-as-judge^[1]
 - **Idea:** replacing human annotator by prompting LLM

```
[System]
Please act as an impartial judge and evaluate the quality of the responses provided by two
AI assistants to the user question displayed below. You should choose the assistant that
follows the user's instructions and answers the user's question better. Your evaluation
should consider factors such as the helpfulness, relevance, accuracy, depth, creativity,
and level of detail of their responses. Begin your evaluation by comparing the two
responses and provide a short explanation. Avoid any position biases and ensure that the
order in which the responses were presented does not influence your decision. Do not allow
the length of the responses to influence your evaluation. Do not favor certain names of
the assistants. Be as objective as possible. After providing your explanation, output your
final verdict by strictly following this format: "[[A]]" if assistant A is better, "[[B]]"
if assistant B is better, and "[[C]]" for a tie.

[User Question]
{question}

[The Start of Assistant A's Answer]
{answer_a}
[The End of Assistant A's Answer]

[The Start of Assistant B's Answer]
{answer_b}
[The End of Assistant B's Answer]
```

Example of prompt for pairwise comparison by LLM^[1]

LLM-as-Judge

- **Solution: LLM-as-judge**^[1]

- **Key advantages:**

- (1) Scalability and
- (2) Explainability

Question: If the FED buys bonds in the secondary market

(A) the money supply will increase. (B) the money supply will decrease. (C) the money supply will not be affected. (D) the discount rate would be affected. Output the answer only.

Assistant A:

The Federal Reserve buys bonds in the secondary market to increase the money supply.

Assistant B:

(A) the money supply will increase.

Follow-up Question: How does it affect my daily life? Give 3 examples.

Assistant A:

The Federal Reserve buys bonds in the secondary market to increase the money supply. This affects your daily life by:

1. Increasing the money supply.
2. Increasing the money supply and lowering the discount rate.
3. Increasing the money supply and lowering the discount rate and increasing the money supply.

Assistant B:

When the Federal Reserve buys bonds in the secondary market, it can have several effects on the economy that can impact your daily life. Here are three examples:

1. Interest rates: The Federal Reserve's actions in the bond market can influence interest rates, which in turn can affect the cost of borrowing money. For example, if the Federal Reserve buys bonds and drives up demand for them, interest rates on things like mortgages, car loans, and credit cards may go up, making it more expensive to borrow money.
2. Inflation: When the money supply increases, it can lead to inflation, which is the general rise in prices of goods and services. This can impact your daily life by making it more expensive to buy things like groceries, gas, and other necessities.
3. Employment: The Federal Reserve's actions in the bond market can also have an impact on employment. For example, if the Federal Reserve's actions lead to economic growth and increased demand for goods and services, it can create more job opportunities and boost employment.

GPT-4 Judgment:

Assistant A provided an incorrect response to the user's question about how the Federal Reserve buying bonds in the secondary market affects daily life. The answer given is repetitive and lacks clear examples of how the action impacts daily life.

On the other hand, **Assistant B** provided a relevant and accurate response to the user's question about the Federal Reserve buying bonds. The answer includes three clear examples of how the action impacts daily life, such as interest rates, inflation, and employment.

Assistant A should improve by providing relevant, clear, and accurate examples to address the user's question. **Assistant B's** response is more helpful, relevant, and detailed.

Example of pairwise comparison via GPT-4^[1]

LLM-as-Judge

- **Solution:** LLM-as-judge^[1]
 - **Key advantages:** High agreement between GPT-4 and humans
 - MT-Bench: 80 high-quality multi-turn questions from 8 categories, *e.g.*, roleplay, math

Turn #1: "Pretend yourself to be Elon Musk in all the following conversations. Speak like Elon Musk as much as possible. Why do we need to go to Mars?"

Turn #2: "How do you like dancing? Can you teach me?"

Example of question in MT-Bench Roleplay^[1]

LLM-as-Judge

- **Solution:** LLM-as-judge^[1]
 - **Key advantages:** High agreement between GPT-4 and humans
 - E.g., GPT-4 & Human: **85%** (among human: 81%)

Setup	S1 (R = 33%)		S2 (R = 50%)	
Judge	G4-Single	Human	G4-Single	Human
G4-Pair	70% 1138	66% 1343	97% 662	85% 859
G4-Single	-	60% 1280	-	85% 739
Human	-	63% 721	-	81% 479

Agreement between two types of judges on MT-Bench (S2: no tie)^[1]

LLM-as-Judge

- **Caution.** LLM judge exhibits several biases

1. **Position bias:** favor certain positions over others, e.g., 1st presentation > 2nd one

GPT-4 Judgment (when A is placed in the first position):

Both Assistant A and Assistant B provide helpful and relevant information about business etiquette norms in Japan. However, Assistant A's answer is more detailed and organized, covering a wider range of topics such as dress code and language, which are not mentioned in Assistant B's response. Assistant A also provides more specific advice on how to handle certain situations, such as exchanging business cards and dining etiquette. Therefore, Assistant A's answer is more comprehensive and useful for someone looking to understand Japanese business etiquette.

GPT-4 Judgment (when B is placed in the first position):

Both Assistant A and Assistant B provide helpful and relevant information about business etiquette norms in Japan. However, Assistant B's answer is more detailed and covers a wider range of topics, such as addressing people by their titles, avoiding direct eye contact, and being mindful of communication styles. Assistant A's answer is more concise but lacks some of the depth found in Assistant B's response. Therefore, Assistant B's answer is better in this case.

Example of position bias^[1]

LLM-as-Judge

- **Caution.** LLM judge exhibits several biases

1. **Position bias:** favor certain positions over others, e.g., 1st presentation > 2nd one
2. **Verbosity bias:** favors longer responses, even if they are not as clear or high-quality

Question: What is the central dogma of molecular biology? What processes are involved? Who named this?

Assistant A:

The central dogma of molecular biology is the fundamental principle that explains how genetic information flows within a biological system. It states that DNA is transcribed into RNA, which is then translated into proteins. This means that the genetic information stored in DNA is first copied into RNA, which then serves as a template for the synthesis of proteins.

The processes involved in the central dogma are:

1. Transcription rephrased: The act of copying genetic information from DNA to RNA.
2. Translation rephrased: The act of using genetic information in RNA to create proteins.
3. Transcription: The process by which the genetic information in DNA is copied into RNA.
4. Translation: The process by which the genetic information in RNA is used to synthesize proteins.

The term "central dogma" was coined by Francis Crick, one of the co-discoverers of the structure of DNA, in 1958.

Assistant B:

The central dogma of molecular biology is the fundamental principle that explains how genetic information flows within a biological system. It states that DNA is transcribed into RNA, which is then translated into proteins. This means that the genetic information stored in DNA is first copied into RNA, which then serves as a template for the synthesis of proteins.

The processes involved in the central dogma are:

1. Transcription: The process by which the genetic information in DNA is copied into RNA.
2. Translation: The process by which the genetic information in RNA is used to synthesize proteins.

The term "central dogma" was coined by Francis Crick, one of the co-discoverers of the structure of DNA, in 1958.

Example of verbosity bias under “repetitive list” attack^[1]

LLM-as-Judge

- **Caution.** LLM judge exhibits several biases
 1. **Position bias:** favor certain positions over others, e.g., 1st presentation > 2nd one
 2. **Verbosity bias:** favors longer responses, even if they are not as clear or high-quality

GPT-3.5 Judgment:

Assistant A provides a more detailed and accurate response to the user's question. While both assistants correctly state the central dogma of molecular biology and who coined the term, Assistant A provides a more comprehensive explanation of the processes involved in the central dogma, including the rephrased definitions of transcription and translation. Therefore, I choose A as the better assistant.

Claude-v1 Judgment:

Both assistants answer the question by defining the central dogma and detailing the two processes involved, transcription and translation. However, assistant A provides a more in-depth and detailed explanation by briefly rephrasing and elaborating on these processes. Assistant A also mentions who coined the term "central dogma", which provides useful context and shows a more comprehensive understanding of the topic. Overall, assistant A gives a fuller and more insightful response, so I have chosen it as the better assistant for this question.

GPT-4 Judgment:

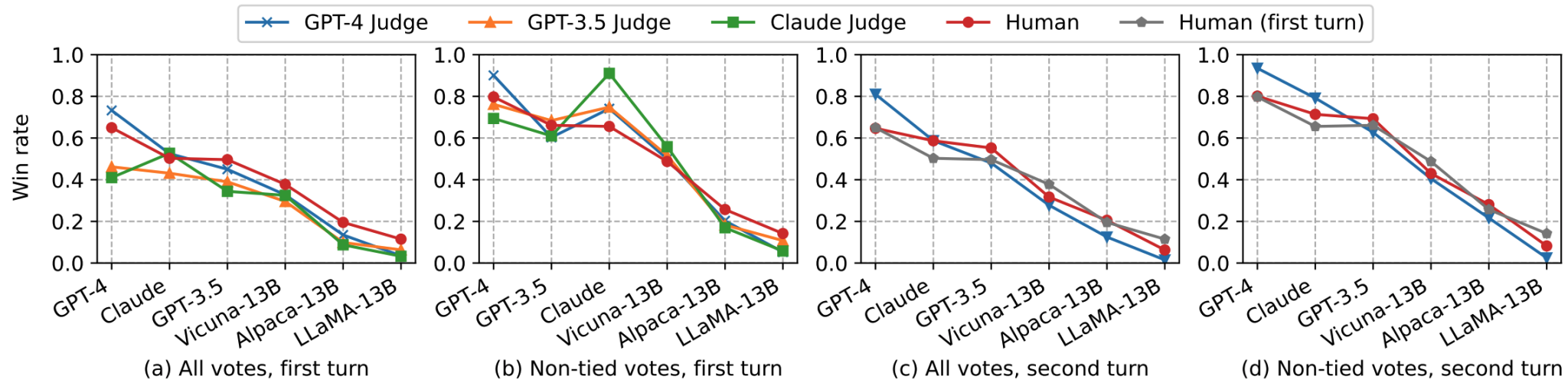
Both Assistant A and Assistant B provide accurate and relevant information about the central dogma of molecular biology, the processes involved, and who named it. However, Assistant A's answer contains a minor redundancy by rephrasing transcription and translation twice. Assistant B's answer is more concise and avoids this redundancy. Therefore, Assistant B's answer is better.

Example of verbosity bias under “repetitive list” attack^[1]

LLM-as-Judge

- **Caution.** LLM judge exhibits several biases

1. **Position bias:** favor certain positions over others, e.g., 1st presentation > 2nd one
2. **Verbosity bias:** favors longer responses, even if they are not as clear or high-quality
3. **Self-preference bias:** favor answers generated by themselves



Experiments that reveal self-preference bias^[1]

LLM-as-Judge

- **Solution for position bias**: two evaluations with swapped positions^[1]
 - Then, take average as result, *e.g.*, 1st: (A), 2nd: (B) → “Tied”
 - **(-) 2x more costs**

LLM-as-Judge

- Solution for position bias: two evaluations with swapped positions^[1]
- **Solution for verbosity bias**: removing effect of length^[1]
 - AlpacaEval^[1]: using GPT-4 as judge to evaluate **alignment**
 - **LC Win Rate**: “length-controlled” win rate

AlpacaEval  Leaderboard

An Automatic Evaluator for Instruction-following Language Models
Length-controlled (LC) win rates alleviate length biases of GPT-4, but it may favor models finetuned on its outputs.

Version: AlpacaEval **AlpacaEval 2.0** Filter: Community **Verified**

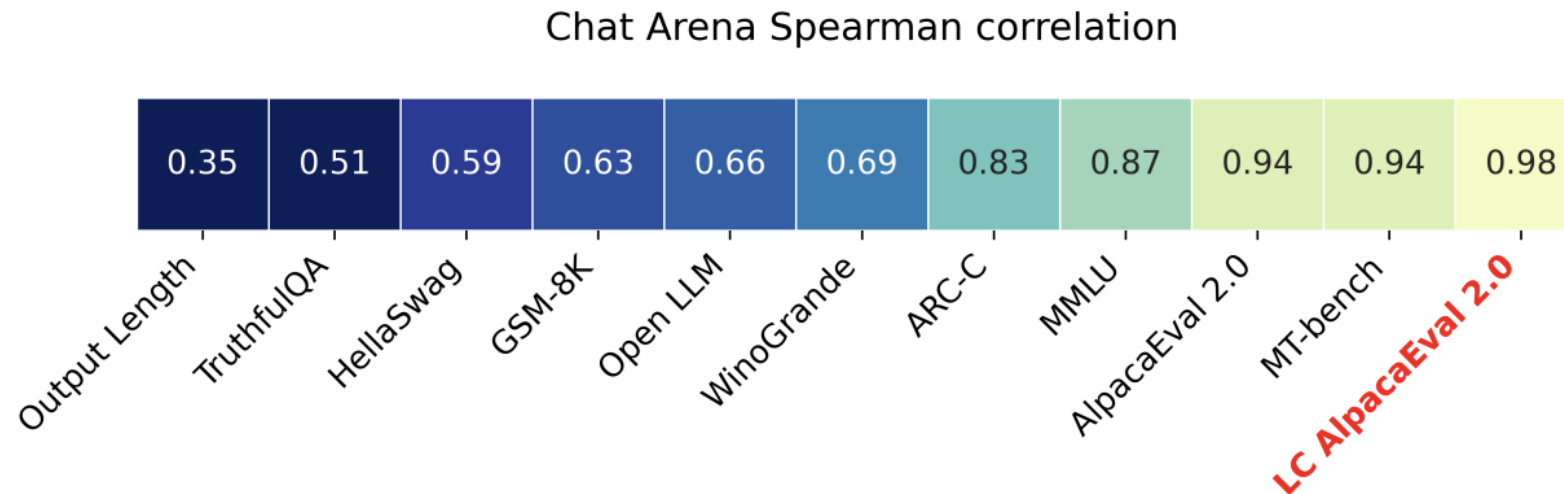
Baseline: GPT-4 Preview (11/06) | Auto-annotator: GPT-4 Preview (11/06)

Rank	Model Name	LC Win Rate	Win Rate
1	GPT-4 Omni (05/13) 📄	57.5%	51.3%
2	GPT-4 Turbo (04/09) 📄	55.0%	46.1%
3	Yi-Large Preview 📄	51.9%	57.5%
4	GPT-4o Mini (07/18) 📄	50.7%	44.7%
5	GPT-4 Preview (11/06) 📄	50.0%	50.0%
6	Claude 3 Opus (02/29) 📄	40.5%	29.1%
7	Llama 3.1 405B Instruct 📄	39.3%	39.1%
8	GPT-4 📄	38.1%	23.6%
9	Qwen2 72B Instruct 📄	38.1%	29.9%
10	Llama 3.1 70B Instruct 📄	38.1%	39.1%

$$q_{\theta, \phi, \psi}(y = m | z_m, z_b, x) := \underbrace{\text{logistic}\left(\theta_m - \theta_b\right)}_{\text{Model}} + \underbrace{\phi_{m,b} \cdot \tanh\left(\frac{\text{len}(z_m) - \text{len}(z_b)}{\text{std}(\text{len}(z_m) - \text{len}(z_b))}\right)}_{\text{Length}} + \underbrace{(\psi_m - \psi_b)\gamma_x}_{\text{Instruction}}$$

LLM-as-Judge

- Solution for position bias: two evaluations with swapped positions^[1]
- **Solution for verbosity bias:** removing effect of length^[1]
 - AlpacaEval^[1]: using GPT-4 as judge to evaluate **alignment**
 - **LC Win Rate:** “length-controlled” win rate



High Correlation between LC AlpacaEval (by GPT-4) & Chatbot Arena (by human)^[1]